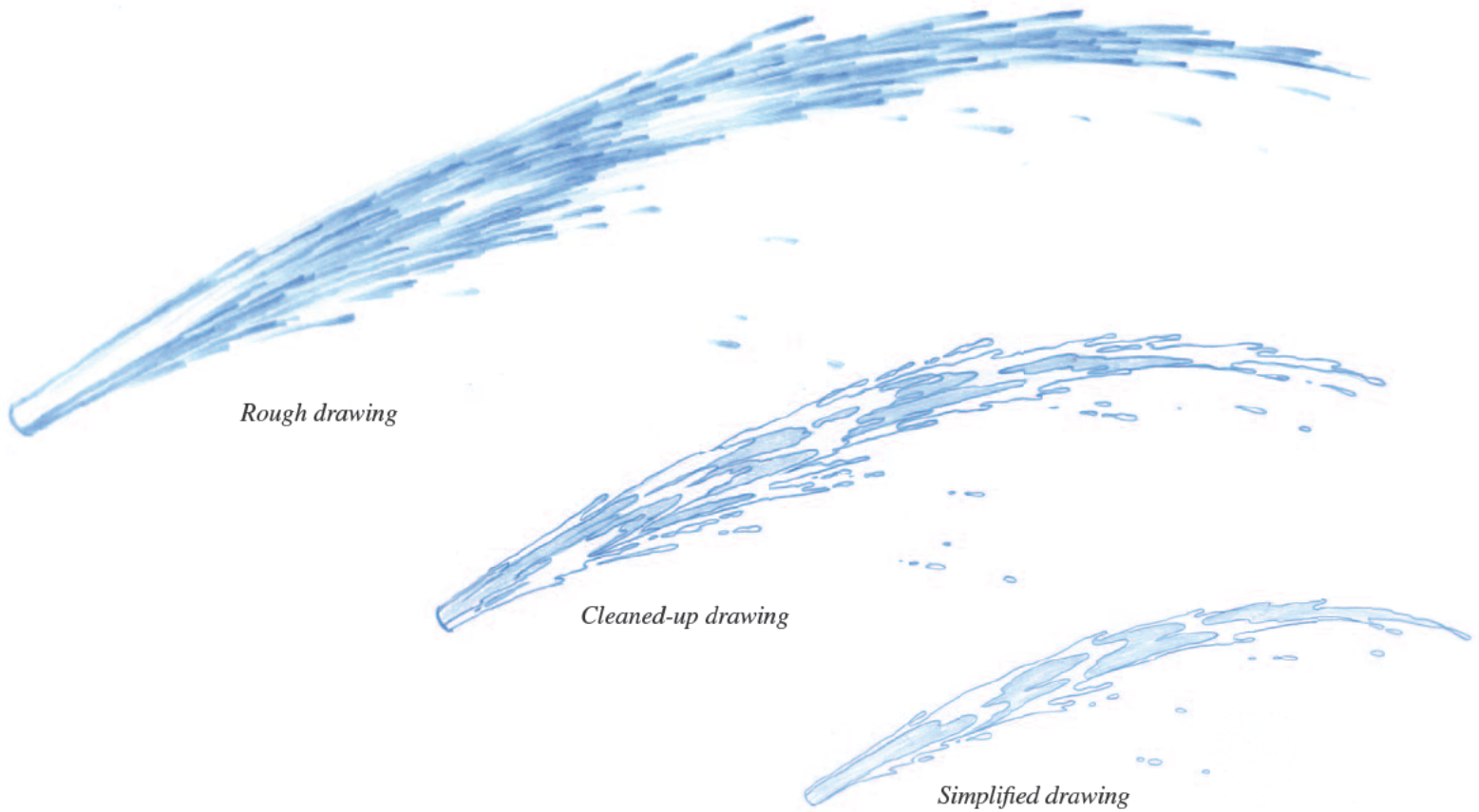
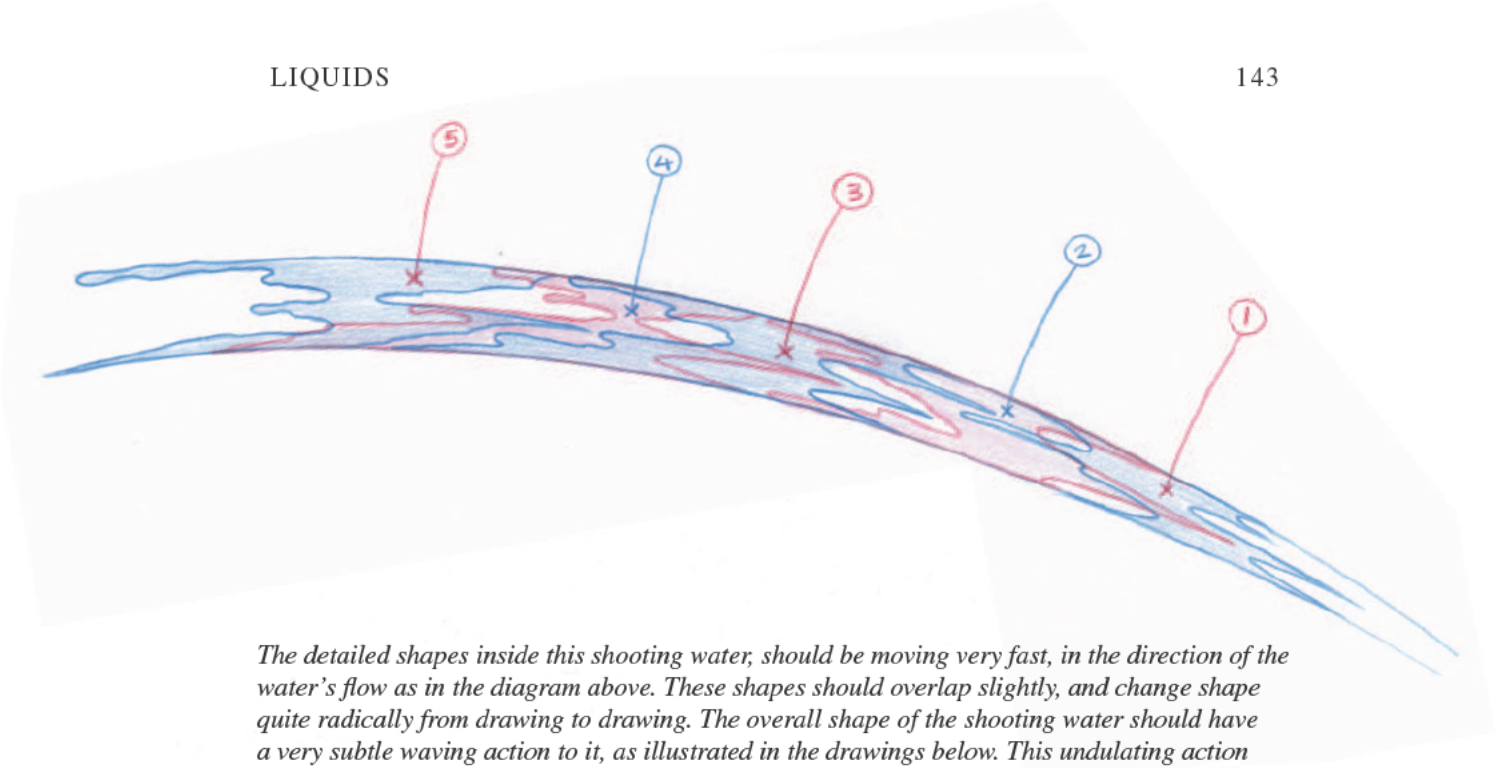
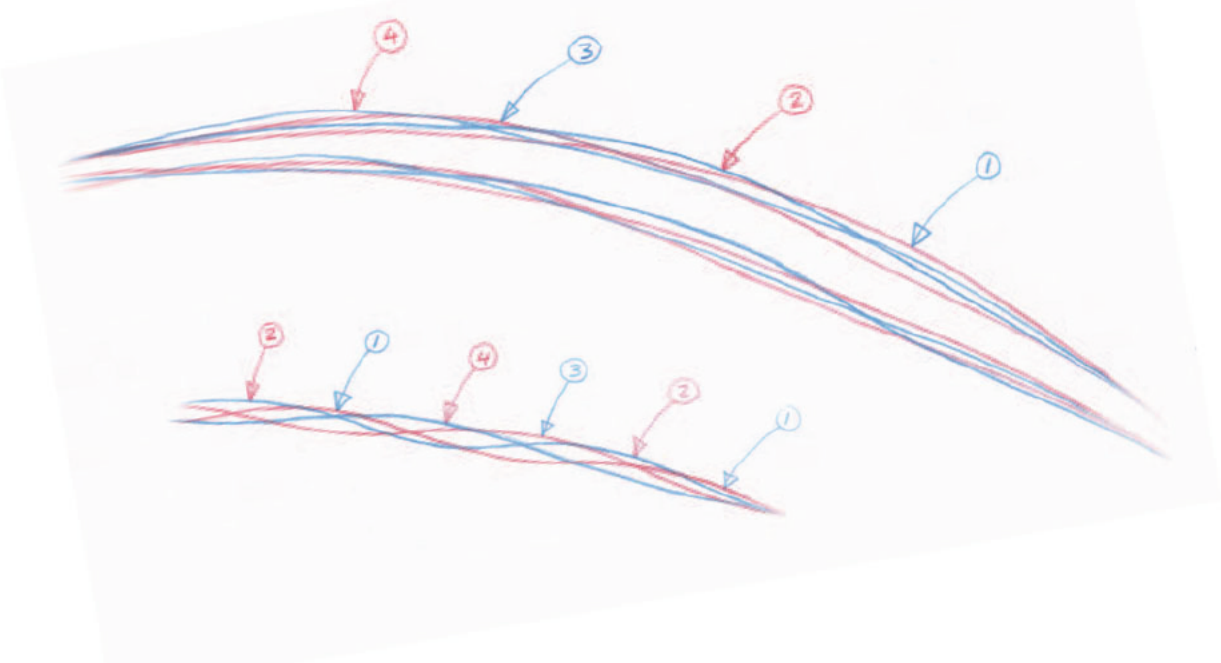


Water shooting out of a hose, should have a really strong directional energy flow to it. When drawing it rough, use fast, energetic strokes to create good directional movement to each drawing. If the rough drawings have energy in them, the shooting water will retain that when the drawings are cleaned up. Even if it is a very simple cartoon style, with far less detail, as in the third example drawing here, if the rough has good energy so does the cleaned-up drawing.

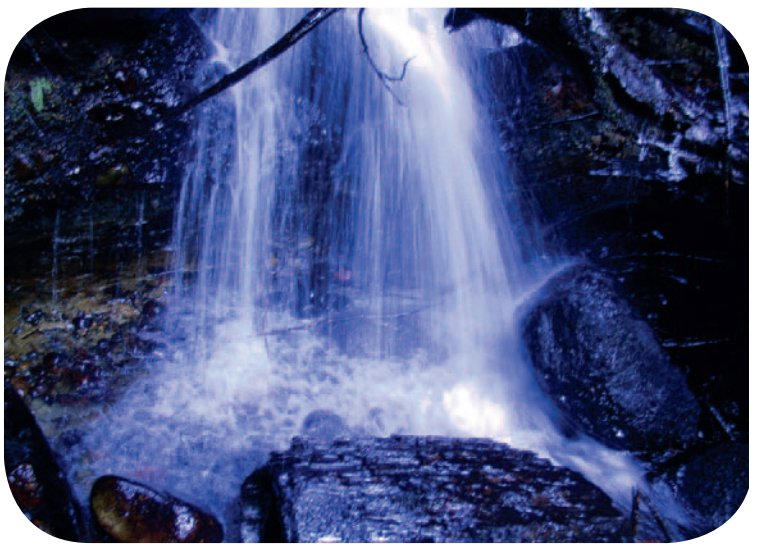


The detailed shapes inside this shooting water, should be moving very fast, in the direction of the water's flow as in the diagram above. These shapes should overlap slightly, and change shape quite radically from drawing to drawing. The overall shape of the shooting water should have a very subtle waving action to it, as illustrated in the drawings below. This undulating action should be very fast as well, and it can also be somewhat erratic, as water shooting out at high speed does have a very active randomness to its movement. These principles can be applied to many water elements, such as waterfalls, fountains, or water running out of a tap.



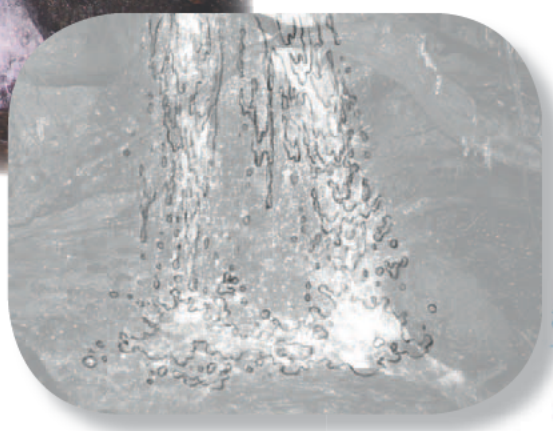
Animating waterfalls by hand can be a daunting task, given that even a very small waterfall, like the one in this photo, is an extremely complex element to illustrate. It is imperative to find a more simplified, stylized design in order to make the level of detail manageable to draw. This photo above was taken with a very fast shutter speed, so there is no motion blur whatsoever.

When designing a hand-drawn waterfall, we should add a certain amount of directional energy and motion blur, which not only helps to simplify the animation process, but can also add the all-important energy which I emphasize so much as an integral part of all effects animation. The photo of the exact same waterfall on the right was taken with a much slower shutter speed, so you can see the directional flow of the waterfall far better.



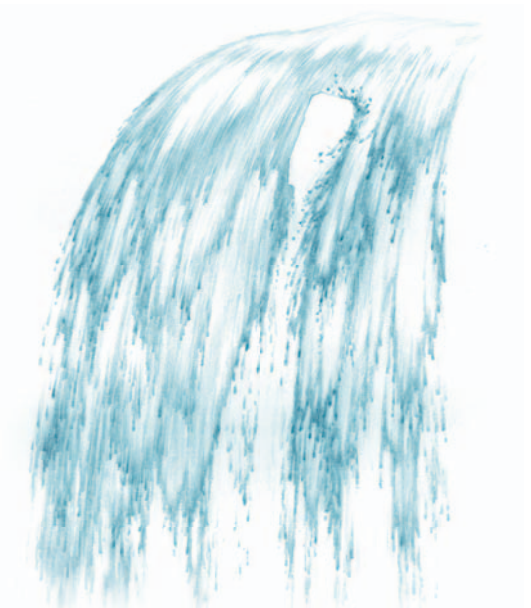
Simplifying something as complex as even a very small, relatively simple waterfall is no easy task. There are so many overlapping shapes happening at once to describe the water's path of action. My drawing here is still extremely complex, even though I have reduced the amount of detail at least 50% compared to the photo.

Taking the first waterfall photo from the preceding pages, I have illustrated how, by simplifying the shapes and stretching them out, we can illustrate the directional energy of the waterfall at the same time that we simplify it enough so that we can animate it without too much detail.



Even with a great deal of simplification, the drawing on the right is still a very detailed and difficult design to animate! Animating waterfalls is another area of special effects animation that can be much more "effectively" done with the aid of digital technology, and still retain a hand-drawn look.

When drawing and animating a waterfall, the details can get way out of hand. To consider animating each little droplet and water shape as it descends is out of the question. If one must animate such a thing, keep in mind that if the general flow of the shapes is moving in the right direction, the illusion that every drop is moving in that direction will succeed. Some of the details can be followed through, but it is not necessary to animate all of them! Keep it as simple as possible! When deciding which details to

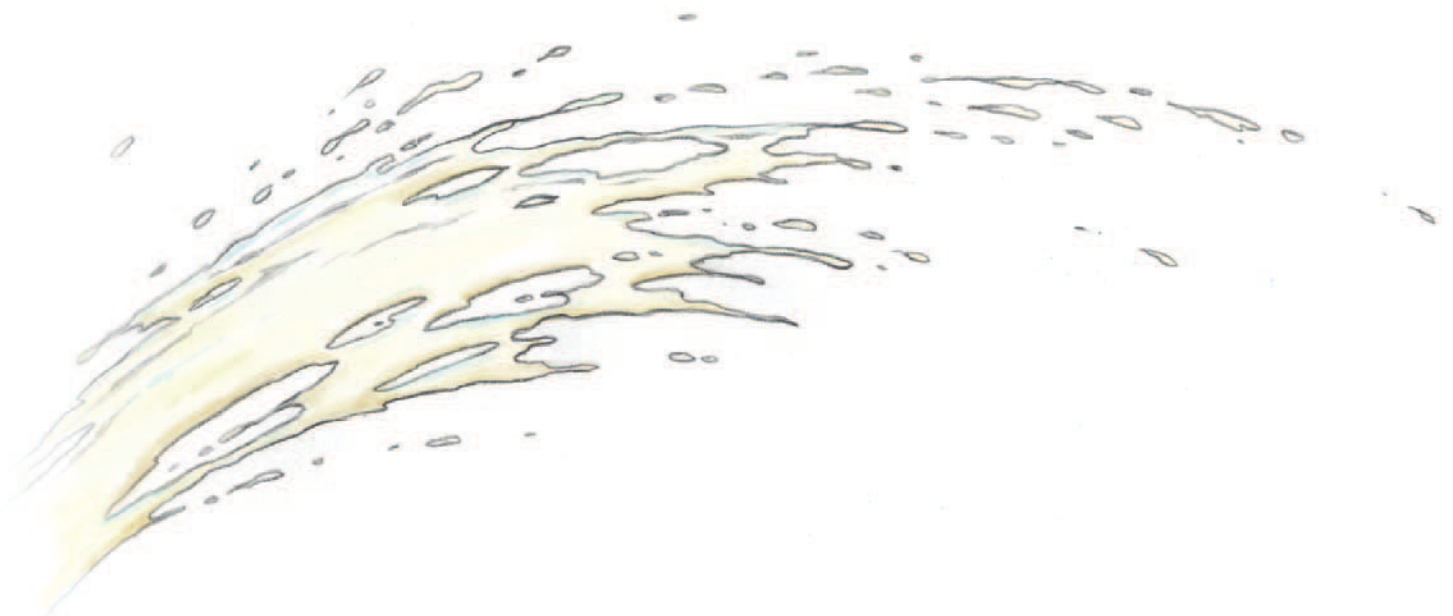


focus your efforts on, choose the parts of the effects design that make the strongest silhouette. These are worth detailing as the viewer's eyes will see these shapes more easily than shapes that are inferior to the effect. Choose your battles carefully, and you will save much time and your effects will be more powerful. I must emphasize once again, that waterfalls are one effects element that can be beautifully done digitally, even in a 2D environment, so considering their inherent complexity, animating them by hand is not always the best way to go!

Of course not all liquids are water. In the course of my career as an effects animator I have had to animate alcohol, oil, salad dressing, mucus, lava, mud, blood, gasoline, milk, oatmeal, and other liquids, and I handled every one slightly differently. Most important is to establish the viscosity, or thickness of the liquid we are attempting to animate. The higher the viscosity, the thicker the liquid. Suppose water is our standard for average viscosity as it is the most commonly animated liquid.

If we are animating motor oil, or cooking oil, which has a far greater viscosity than water, our shapes need to be more blobby and rounded, and our animation has to be somewhat slower as well. Greater viscosity means the molecules of the substance cling together more stubbornly, thus it is thicker and moves more slowly.

Thus gasoline has a very low viscosity, it is thinner than water, as are paint thinner and pure alcohol. These liquids will displace themselves more quickly and break up into beads much more easily than water. On the slightly thicker, higher viscosity side, something like milk, which is only slightly thicker than water will displace and break up slightly slower than water. Thicker still are all types of oils, thicker even more are mud, lava, tar, molasses, etc.



If we are animating gasoline, paint thinner, or kerosene, the opposite principles apply, as these liquids have less viscosity, and therefore are thinner and break up more easily than water. So keeping that in mind, gasoline will break up into thin sheets and tiny droplets much quicker than water does. In a puddle it would spread much faster as well.

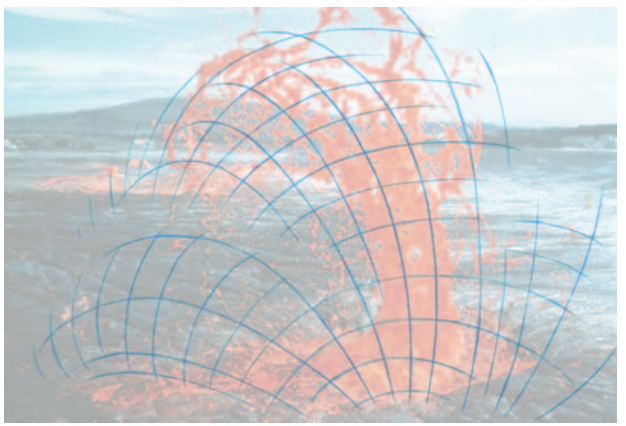
As viscosity increases, our animation must slow down appropriately, and our designs become more rounded and blobby, as these thicker liquids hold together much better than water, and resist the forces that move them a lot more as well. Research and observation are very important when tackling these liquid elements. Many have been beautifully animated in films like Fantasia, and observing the design and animation timing in these films is extremely valuable, especially for the beginner.

If we are animating lava, which has a far greater viscosity than water, oil or gasoline, our shapes need to be even more blobby and rounded than our oil, and our animation has to be slower although it depends on how quickly lava is being ejected from a volcanoe, it can be moving very fast. Lava is very challenging to animate, as it is usually cooling quickly, and the cooler lava is thicker, therefore we need to deal with different viscosities at the same time.

This lava splashing up out of a fissure is a beautiful example of how the initial force behind the movement of any liquid creates a flowering, spreading design that has a logical path of action. As this lava is thrust upward, it is also spreading and rotating as it stretches and breaks up into smaller and smaller pieces. In the illustrations on the following pages, I have found the essential energetic design of this spewing lava and drawn it clearly. This flowering, fractal design can be seen in all dynamically moving liquids, as well as in fire and smoke, dust, dirt, gas, and vapor, etc. It is the cosmic design behind everything.

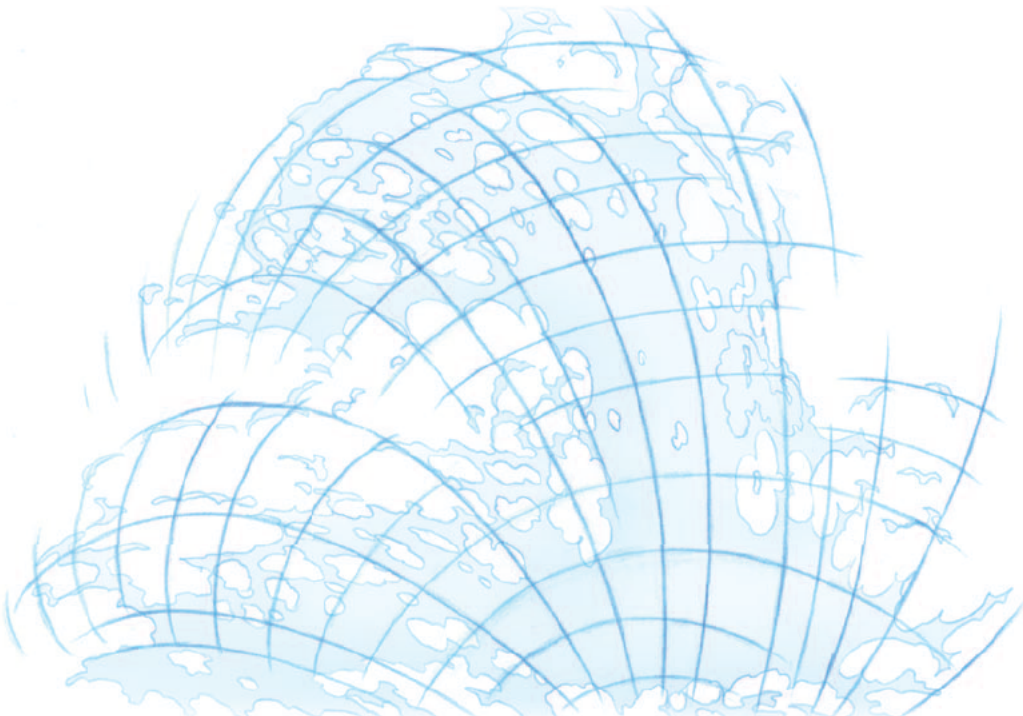
Spectacular lava fountain, Hawaii Volcanoes National Park.

This photo of spewing lava really caught my eye, because it illustrates beautifully, the directional energy and the arcing and sweeping lines of its trajectory. As it spews upwards, it also fans out and away from its source, a direct result of the pressure below, being released into the air above. It is like a blossom opening up! Pure energy unleashed.



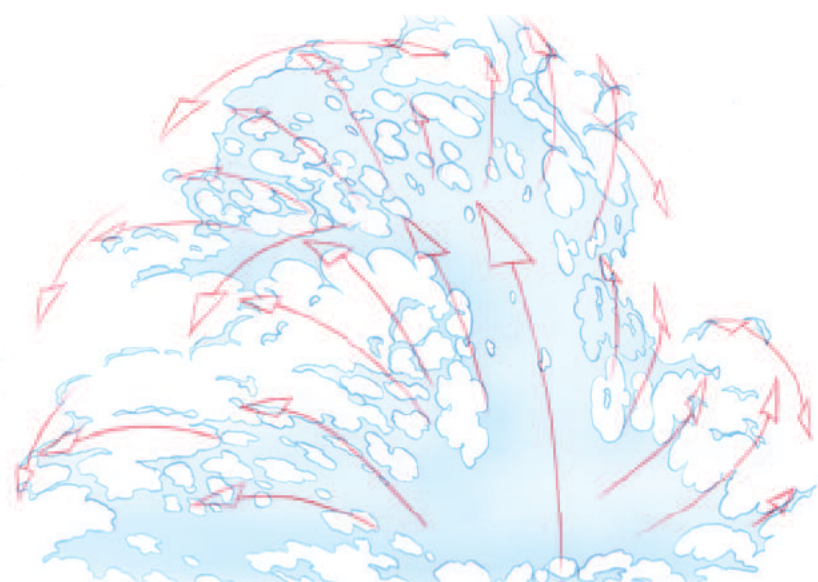
First I have drawn a very technical-looking grid over the splashing lava, that reflects the path of action and underlying structure of the splash. This kind of grid can help us to see the structure that every little droplet and detail of the lava is attached to as it spews, seemingly into abstract shapes, up and out of the volcano below.

Here is a simplified, stylized, slightly "cartoony" looking drawing of the splashing lava. I have drawn it with very clear elliptical shapes in order to point out a very important aspect of special effects design. Frequently I see attempts to draw this sort of effect, having a very unorganized and non-energetic feel to them, and that is because they are missing this all important principle. These shapes and lines may appear to the untrained eye to be mostly abstract in nature, but they are in fact extremely structured. Let's look at how and why.

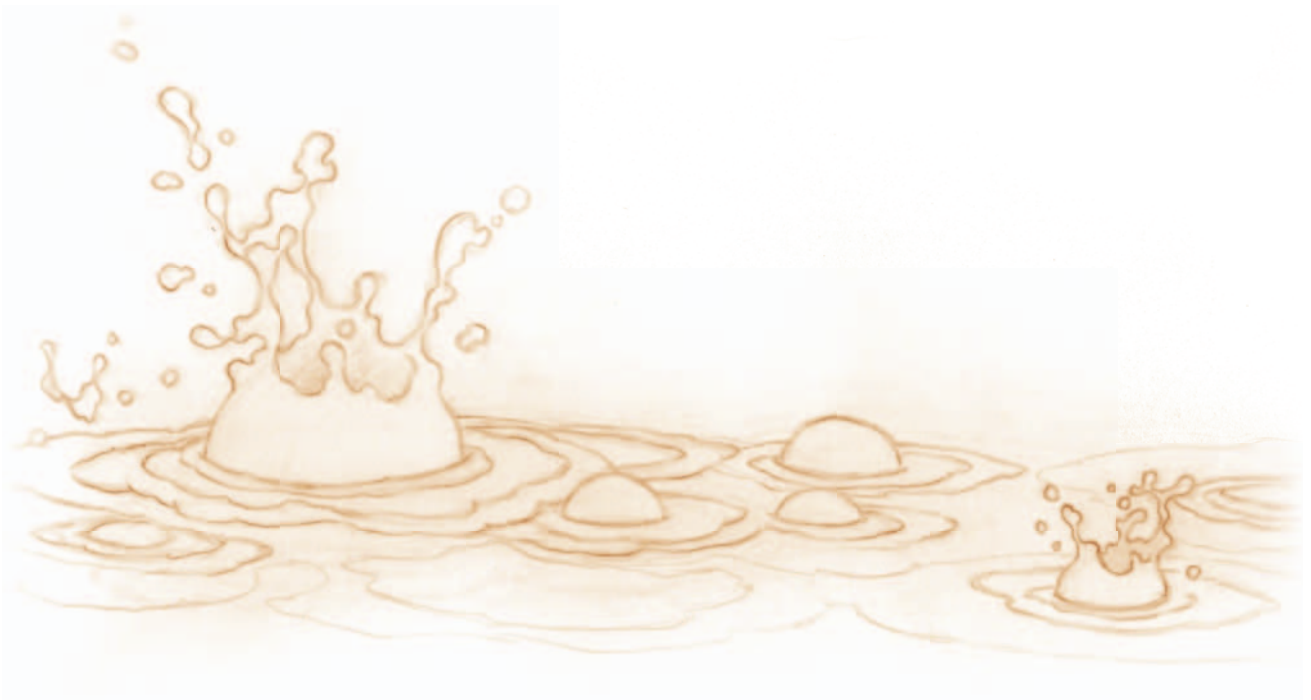


When we superimpose the stylized drawing over the rigid, technical-looking grid that I drew, we see how structured the cartoony looking splash really is. Note that every single ellipse "fits" into the grid, much like the simple ellipses I drew earlier in this chaper to describe how ripples adhere to a perspective grid. This is what makes good special effects of any kind look fantastic and filled with energy. The blossoming energy that has initiated and drives the element's action, informs every aspect of its design and movement. This is an essential principle underlying all special effects animation.

To further illustrate this principle, here are sets of arrows describing both the upwards energy of the splash and the outward fanning-out motion of it. There are many subsets of energy in a splash as energetic and complex as this one, and I could draw hundreds of little arrows to describe them. Most importantly, they are all connected to the main driving force behind this splash. Think of the energy of a flower or a tree, the veins in our arms, a fireworks display, or a river delta fanning out into the ocean. Driven by a force of nature, or by artificial intervention, all elements move with this universal energy.



Experimentation with the actual substance we are trying to animate and shooting reference photos or footage can be enormously helpful, and is also a great deal of fun. Remember to have fun whenever you get the chance, because the sheer volume of work involved in animating liquids can be daunting, and it's not all fun and games. Play when you get the chance!



When animating mud, the viscosity is somewhere in-between that of oil and lava, and like lava, the viscosity can vary depending on the temperature or water content of the mud. As mud dries it may become brittle and form chunks and strangely shaped blobs. It can be a lot of fun to animate, as it may have an extremely wide variety of thicknesses, or viscosities, from 100% watery, to brittle dry clay.

Digital Waves

How do all of these classical effects animation principles come into play when we are approaching animating water using today's C.G.I. technology, which is more often than not the way animation is being created these days? Well, if we want our water to look and feel really fantastic, understanding what makes water animation look great from a hands-on approach can help us create far better CGI water effects.

Let's consider the design principles that we have discussed in this chapter, and in Chapter 2 as well. Using what we know about strong classical design, we always strive for striking silhouettes in our animation, an absence of repetitive shapes, or too much symmetry, and in our action we also strive to overlap and offset our timing as much as possible, exaggerating these principles to make our animation more dynamic and alive.

This approach to animating water digitally, can immediately set our work a cut above the rest. This is particularly important in the world of CGI cartoons. Do the same, photorealistic effects simulations that we see in big budget science fiction thrillers really belong in our animated films? The answer is an emphatic no! Animation is a world of imagination, where *all* of the elements should have even more life, character, pizazz, and zest than they do in the "real" world.

That is what sets animation apart from other mediums, our ability to exaggerate reality, to push it farther, into a more fantastic, and ultimately entertaining version of reality.

CGI effects animator John Thornton at Blue Sky Studios has taken 3D effects to a whole new level, using innovative CGI rigs to animate water effects.

The majority of 3D programs have preset effects simulations, which we can of course alter and manipulate in a variety of ways. Whether it is an undulating water surface, or an enormous splash that we are animating, if we approach our computer with the knowledge that natural elemental energies contain within them a great deal of randomness and chaos, we can, with patience and technical know-how, imbue our digital effects with that fantastic aliveness that sets great effects animation apart! If we are content to set up a fully simulated effect, one that runs on its own, flowing with preset behaviors such as weight, velocity, viscosity, collision parameters etc., what we will end up with may be a

convincing enough-looking liquid simulation, but in the context of animation, is that all we are looking for? Can we not, armed with our imaginations and the knowledge of the elemental magic at work in the universe, add something to our computer animation that will honor the art form, and bring it to a higher level of artistry?

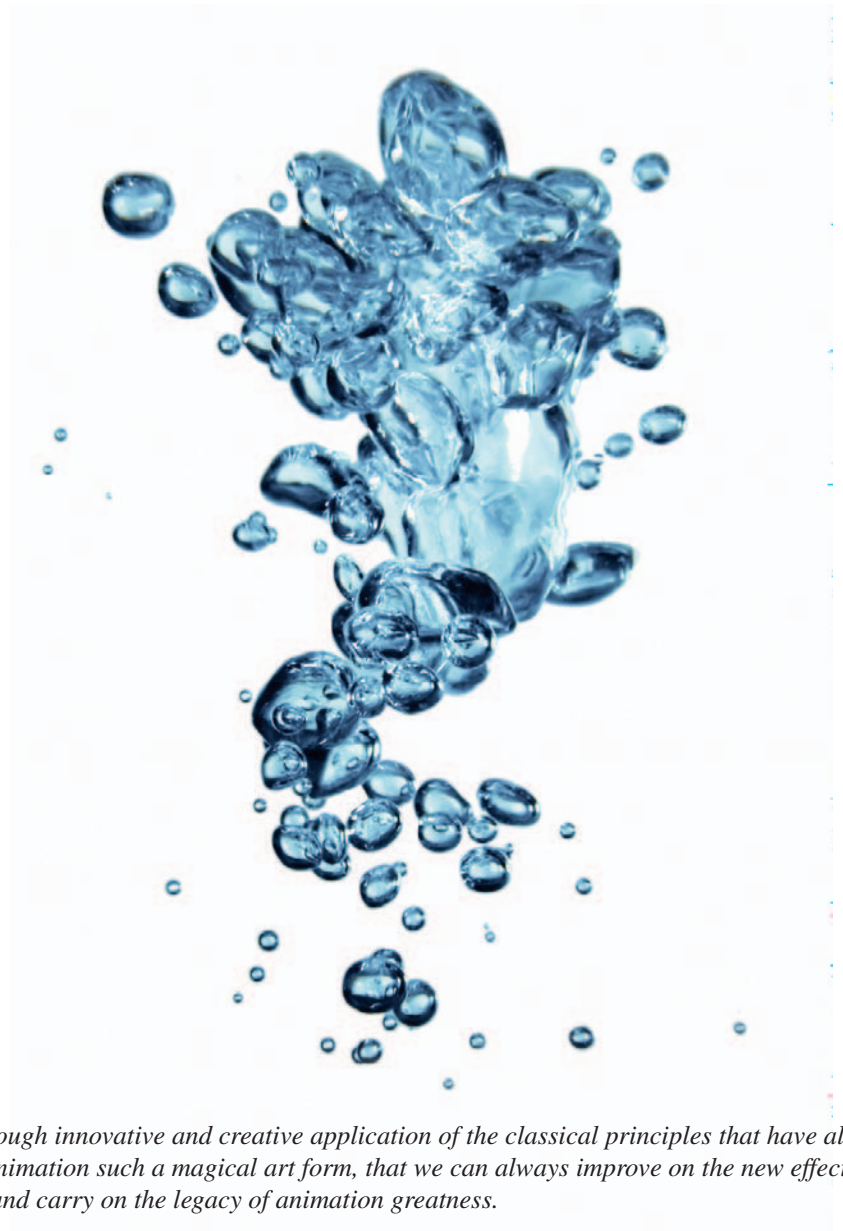
Approaching the same CGI simulation I described above, if we consider the "intent" of the effect, and the design of the film, particularly if it is a CGI animated feature, we should immediately see that a mere computer simulation lacks character and design, compared to the fanciful characters and environments that populate today's CGI animated features.

The CGI water special effects in Blue Sky's "Ice Age, The Meltdown" have a fantastic, 'stylized' look. Effects artists there are able to create dynamic shapes with their 3D effects rigs, and to manipulate the effects manually to give them more life.

The same wild, imaginative design principles that drive the rest of the elements that make up an animated film, must be applied to our CGI effects animation as well. And too often, that is simply not the case.

There are, here and there in the industry, effects artists pushing the envelope of digital effects animation in a direction that honors the unique ability of animation to be more entertaining than reality. Not coincidentally, many of these artists have extensive classical animation backgrounds. At studios like Sony and Blue Sky, old-school effects animators are rolling up their sleeves and working with savvy technical directors to create digital special effects that really stand apart from the vast majority of bland computer-generated effects that we too often see in animated films today. Whether it is a massive curling wave threatening to devour a hapless surfer, or just a small, whimsical little splash created by a tiny character jumping in the water just for the fun of it, special effects animation should be just that. *Special!*

So to each and every one of you special effects animators out there, regardless of the tools you choose to use, or need to use in order to create your effects—remember the spirit of animation. Learn how to *feel* what a good effect *feels* like. Remember that is is our responsibility to give the audience more than just enough. We can make children's eyes grow with wonder, and their parents jaws drop in awe, if we inject our effects animation with that awe and that wonder!



It is through innovative and creative application of the classical principles that have always made animation such a magical art form, that we can always improve on the new effects we create and carry on the legacy of animation greatness.

Animating water is like trying to catch the wind. Try this someday. Just sit down by a pond or a lake or the ocean, and pay attention to the way the shapes undulate and morph into each other wildly and unpredictably. Find a few stones and throw them into the water, and observe the shapes of the splashes they create. Although each splash may resemble the next, each is more different than we can possibly imagine. There is an entire universe of uniqueness and natural perfection in every drop, every bubble, every ripple, every wave.

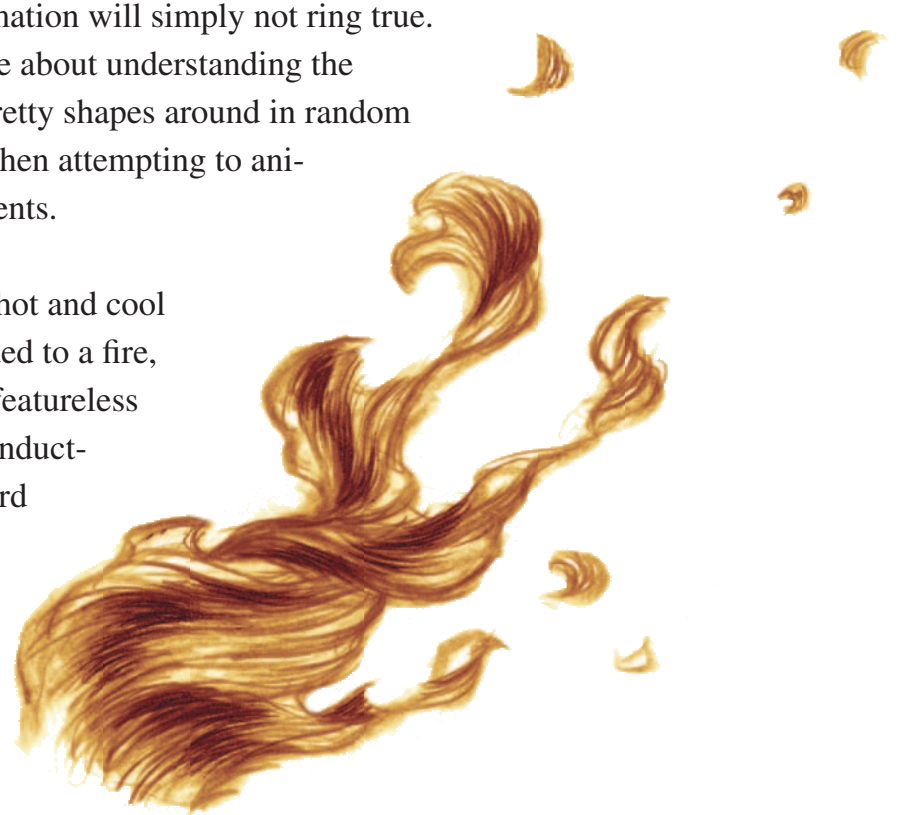
Water reacts to every force
exerted upon it, with the absolutely
perfect defense mechanism of displacement.
If you push it here, it expands over there. If you rob it of
its space, it seeps into the adjacent one. If you heat it up, it flies aways, if you throw a rock at it,
it swallows it up instantly and seals the hole as if nothing ever happened. If you scream at it,
it absorbs your sound waves. If you thrash at it wildly, it just moves out of the way and
regroups elsewhere. Hurricanes can move it into mountains that shape our
shorelines, but after even the most violent storm, it simply returns
to its flat, calm, perfect self. Water is perfect design,
the fluid of life in motion, the very
building blocks of
our essence.

Being drawn to the water,
be it a small waterfall in the woods
or a raging Hawaiian seashore, I find myself
completely mesmerized by its continuous restless movement, and its
indescribable perfection. To attempt to know it, and capture the magic of its movement,
has been a lifelong learning experience for me. And it will continue to be. Every
time I attempt to animate a splash, it draws me into the perfection
of nature's cosmic design. As it catches light like a mirror,
and reflects everything back at me, it swallows my
imagination like an enormous blue whale,
and swimming in its darkness,
I touch infinity.

Chapter 5

Fire, Smoke, and Explosions

When attempting to animate fire or smoke, the importance of understanding the underlying forces of energy cannot be understated. Without this core energy, fire and smoke animation will simply not ring true. Animating fire and smoke is far more about understanding the energy that is in play than moving pretty shapes around in random ways, as we so often see artists do when attempting to animate these sublimely beautiful elements.



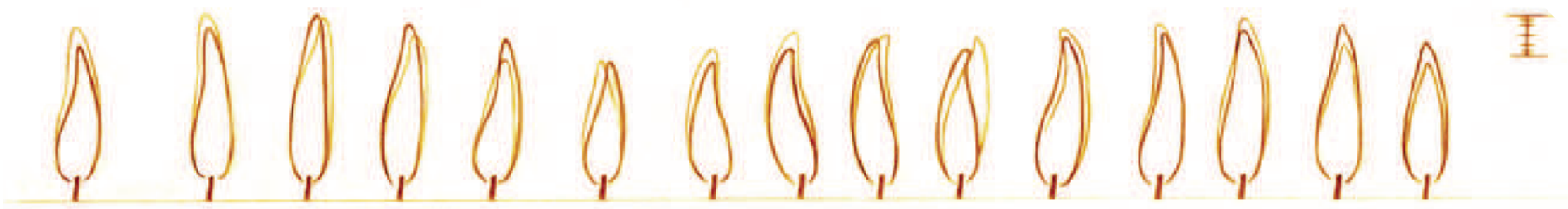
Without the forces of gravity, wind, hot and cool air currents, or excess fuel being added to a fire, a fire will actually burn as a perfect featureless sphere! (As was discovered when conducting experiments in zero gravity aboard a space shuttle.) When we look at a fire's flame, we are actually seeing luminous gases being released and rising quickly from their former state into the air as a given material is heated to its point of combustion.

Fire can be drawn in any number of styles. There is no right way to draw fire. What is absolutely essential though is that the drawing contain the right *energy* and *flow*!

The fire on the left is drawn in a highly detailed style, slightly stylized, with a combination of smooth flowing lines and angular corners combined. The fire above is drawn with a very smooth and liquid style. Next to that we have an extremely stylized angular style of drawing fire. Finally on the right is a very smooth, swooping style of drawing fire. All of them are perfectly valid styles, depending on the style of the animated film you are working on. Anything goes, as long as the energy is in the drawing!

First let us analyze a small flame like we might find on a candle or match. Its shape is a simple tapering affair, fatter at the base and tapering off to a dull point. In a sheltered setting, with little or no wind, its movement will be a subtle stretching and squashing.

There should be no extra shapes emanating off of or breaking away from a flame this small and simple. To add detail would suggest a much larger fire. In a relatively calm setting, the small flame throbs and bobs gently as it subtly gains on and then loses against the cooler air surrounding it.



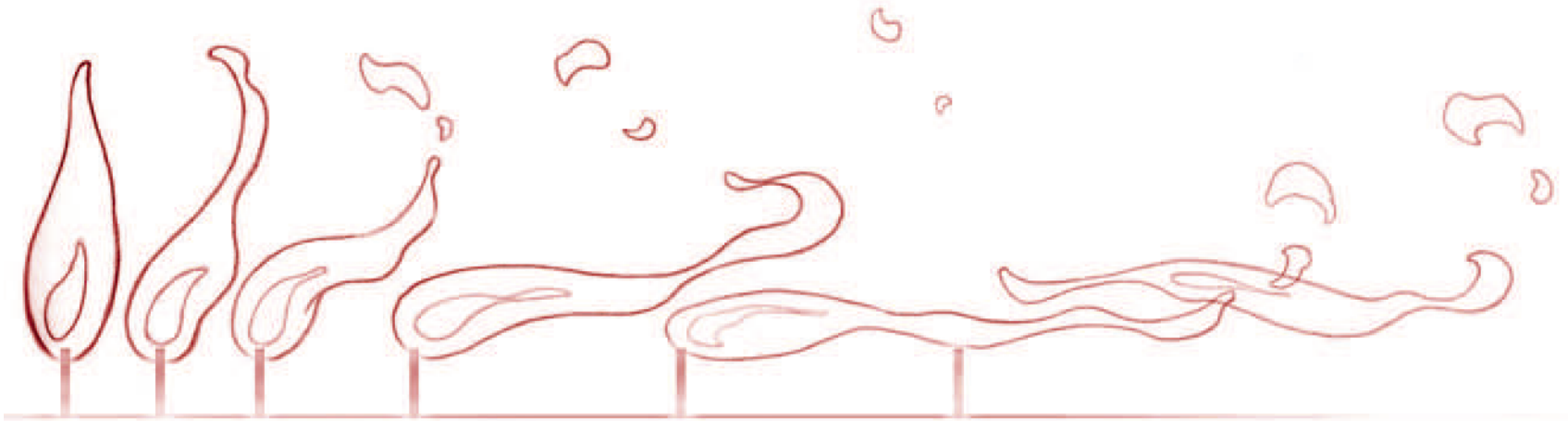
In the series of drawings above, I have animated a candle flame behaving as if it is in a room with very little turbulence in the air. The flame simply grows taller, and then shorter, weaving with an extremely subtle wave action as it stretches and contracts. The simple graph on the right shows the highest and lowest points of the tip of the flame. There should be a slowing-in cushion as it reaches its highest and lowest points, with the drawings being much closer together. The lighter part of the drawing is the subsequent position, to illustrate the movement more clearly.

I am always on the lookout with my camera for good special effects reference material. It is one of the greatest ways to observe and keep engaged with the effects learning experience.

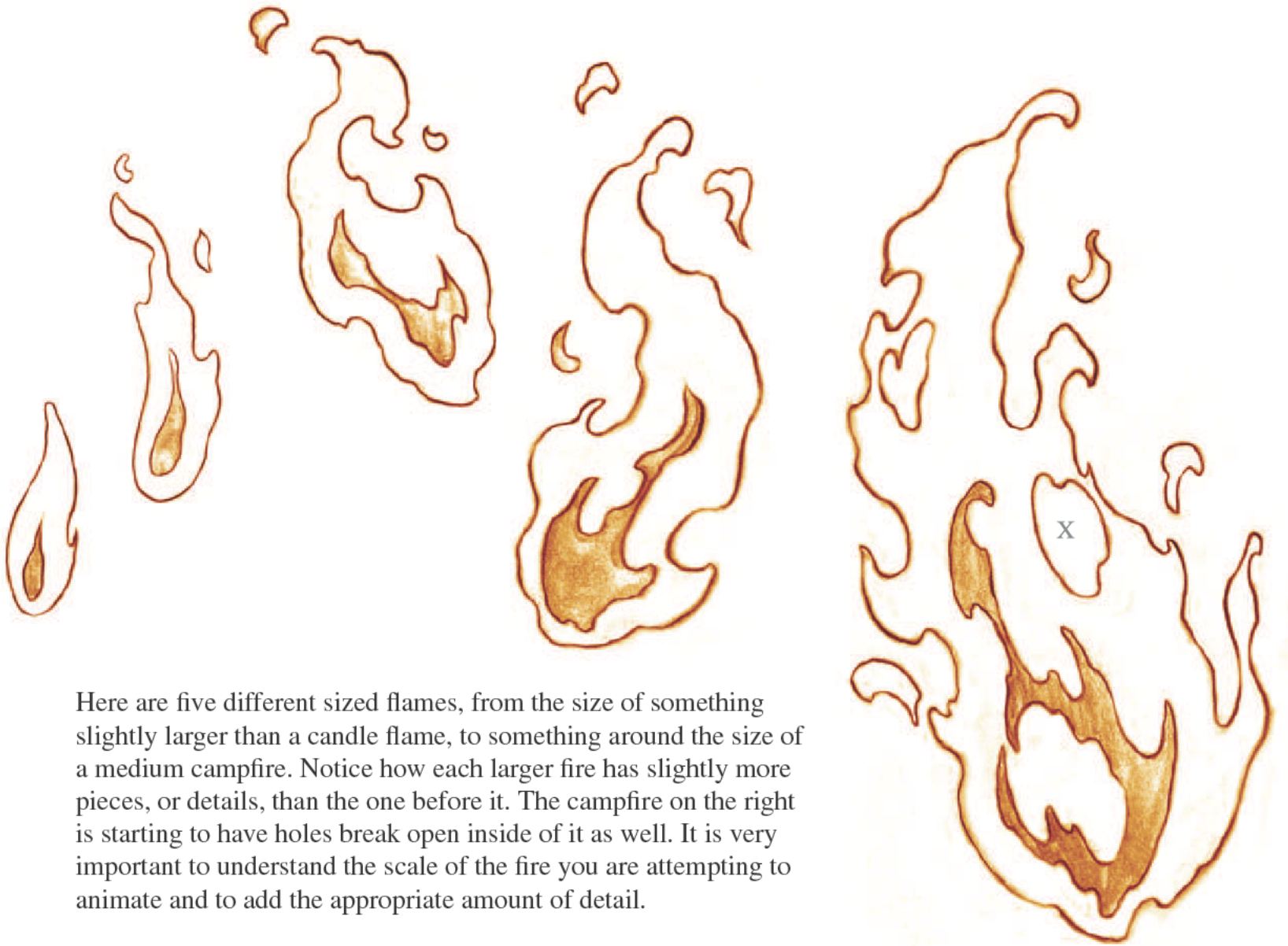
If we add a small amount of wind to the scenario, which also adds oxygen, or fuel to the fire, our candle flame will stretch out in the direction of the wind, and pieces may break off at this point. It becomes slightly more violent and reactionary to the outside forces. Extremely subtle turbulence occurs, introducing opposing curves, billows and flickers.

If the wind gets any stronger, a candle flame is quite easily extinguished. In its last dying breath, the flame may momentarily flare out with a burst of oxygen, become larger for one or two frames, stretching out to maybe three times its normal length, and then snapping off a little flicker before ultimately dying. This will usually be followed by a small puff of linear smoke, which appears two to six frames after the flame is extinguished, and is generally very short-lived, maybe a second or two at the most.

Remember those imitation flames made out of flimsy fabric in Chapter 2 with a fan blowing straight up from underneath them? This is a perfect example of how fire animates. Hot and cool air colliding causes the wind and turbulence that creates the shapes we see in a real fire, which is why this illusion works so well.

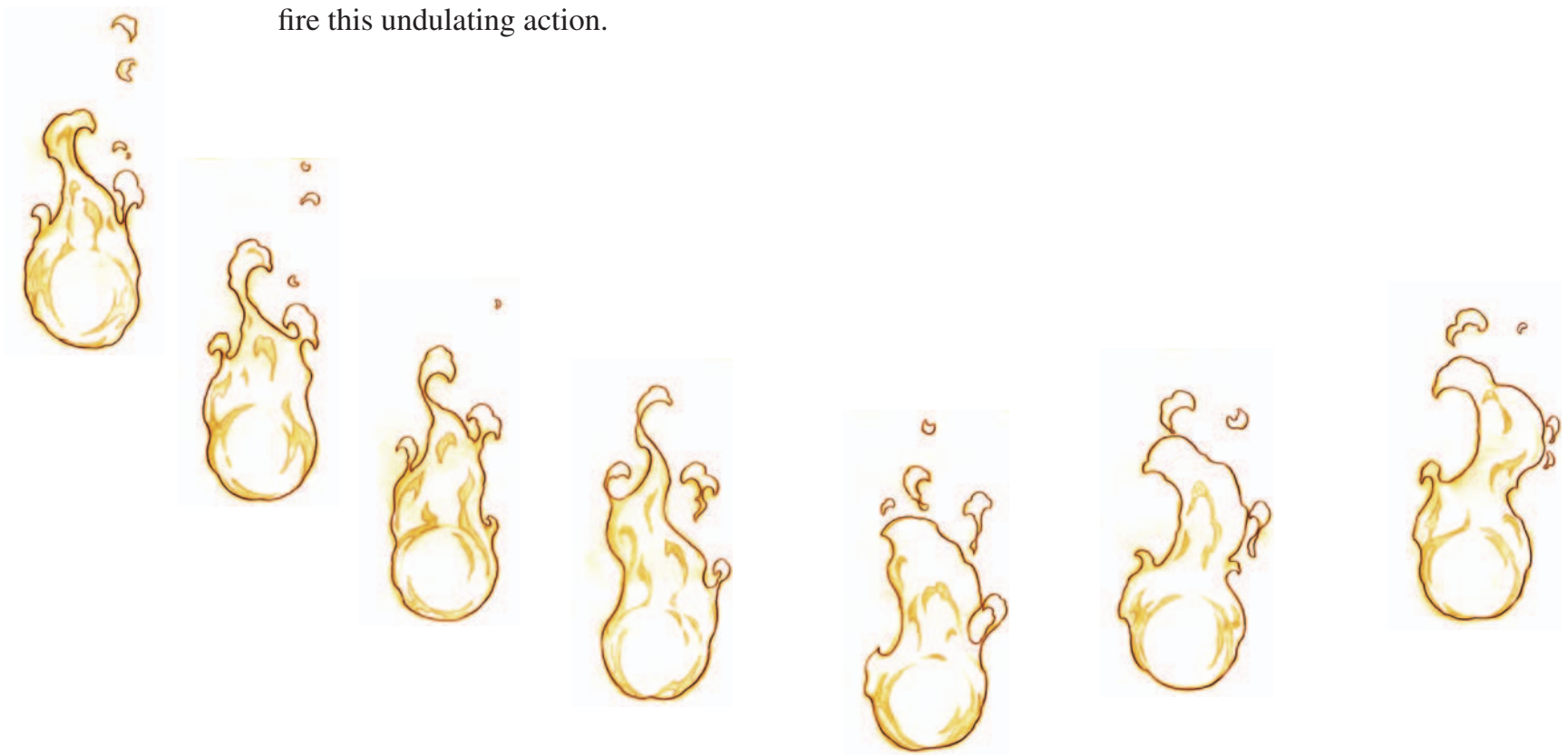


In this sequence of drawings, we see a candle flame being blown out. It doesn't take a very big breeze, or blow, to extinguish a candle flame, except of course when it's your birthday and you're trying to blow out a lot of them at the same time! Interestingly, before actually getting blown out, a flame will actually expand in volume a little bit, fueled by the oxygen in the wind, as we can see in the second, third and forth drawings. When animating a flame being blown out, it is a good idea to stretch it much further than you might expect it to actually stretch in reality. Remember, exaggeration is our friend, and it makes our animation more punchy and believable! It would be feasible to add one more drawing perhaps of the dissipating pieces of the flame, but it works well if it simply vanishes quickly!

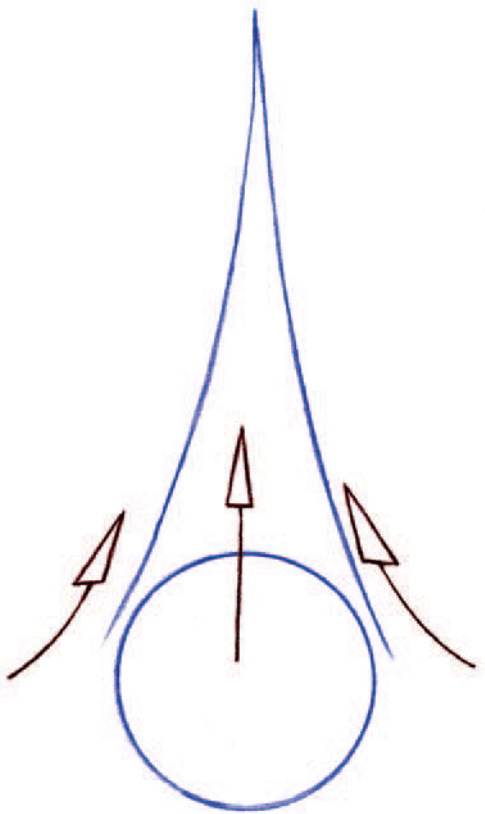


Here are five different sized flames, from the size of something slightly larger than a candle flame, to something around the size of a medium campfire. Notice how each larger fire has slightly more pieces, or details, than the one before it. The campfire on the right is starting to have holes break open inside of it as well. It is very important to understand the scale of the fire you are attempting to animate and to add the appropriate amount of detail.

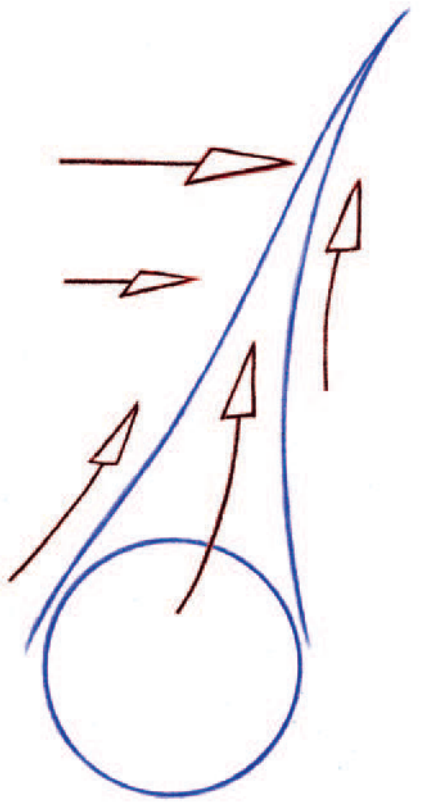
As we advance to a larger fire, as in a campfire or a burning torch, the bobbing motion will become more pronounced, as will the reversing internal arcs, which undulate from side-to-side. This motion is caused by one side of the fire cooling faster than the other; the hotter side will then rise faster, rolling upward and over, causing a flame shape to break off of the main silhouette. This series of events repeats itself as forces interact with it, especially wind—even an imperceptible current can give fire this undulating action.



Before any interior details can be considered, it is best to construct the basic flow, or gesture, of the fire. On the second pass, the primary interior shapes can be conceived as sections of spherical masses moving upwards, which then diminish in size. We build the silhouette with sharper edges, connecting these spherical masses. The over arching idea of a fire is a triangular shape, or a series of triangular shapes feeding one another. These quickly unite into larger silhouettes, depending on the severity and frequency of the forces (wind, gravity, pressure) and the amount of available fuel from the material that has been ignited (paper, straw, wood, gasoline, etc.).

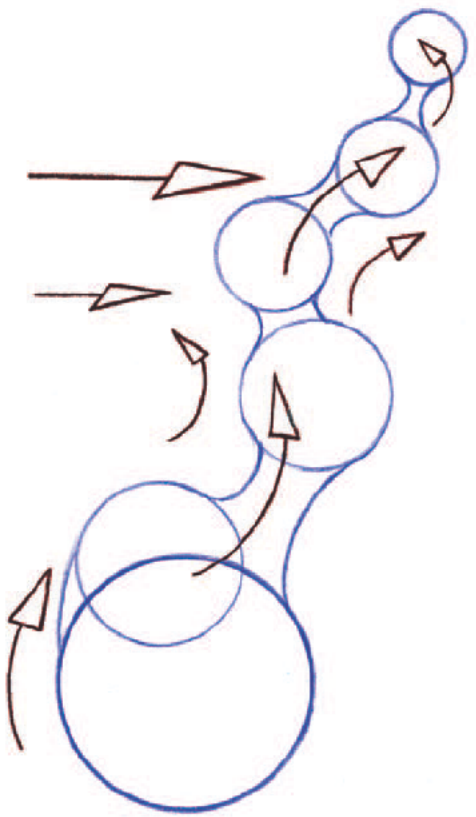


1. Hot air rises from a fuel source, drawing in cooler air in from the bottom.



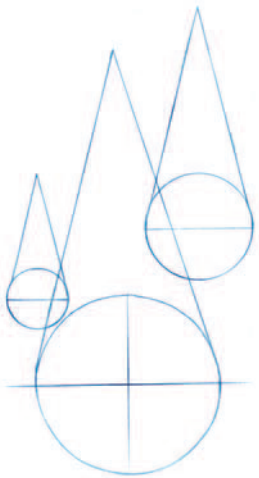
2. Add a small breeze, pushing the flame to the side. It begins to arc.

This simple yet effective formula for animating a medium-size fire was originally designed by a colleague of mine at Walt Disney, David Tidgwell. I have never found a better explanation!



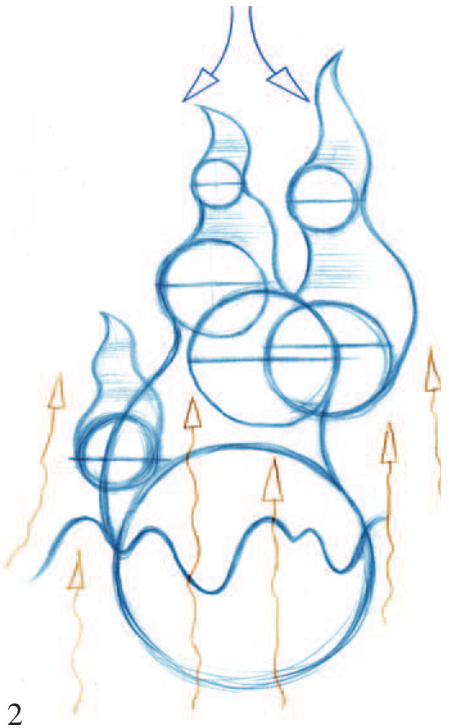
3. Add turbulence. As the cooler air interacts with the rising hot air, it creates alternating eddies of swirling air.

4. Add details. The energy pattern is simply and clearly described within the details.



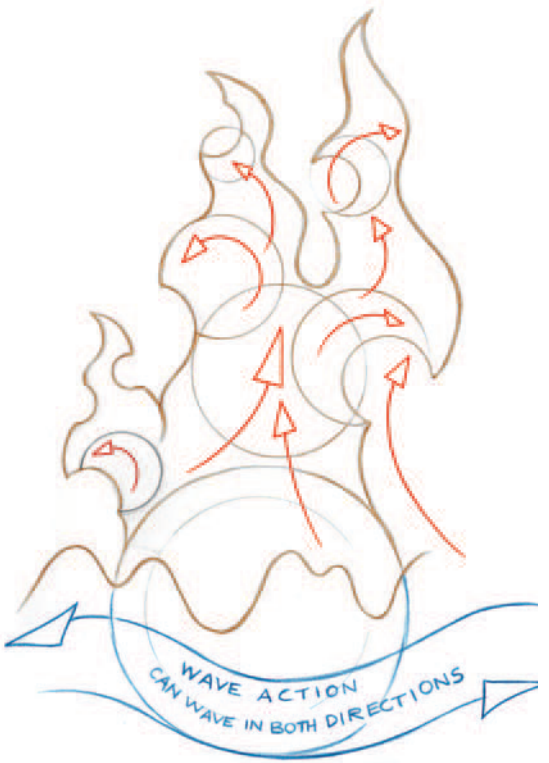
1

Here is another approach to "building" a fire. I have started with basic triangles—note that they are different sizes. Because of the repetitive nature of fire shapes, it is important to always vary your shape sizes.



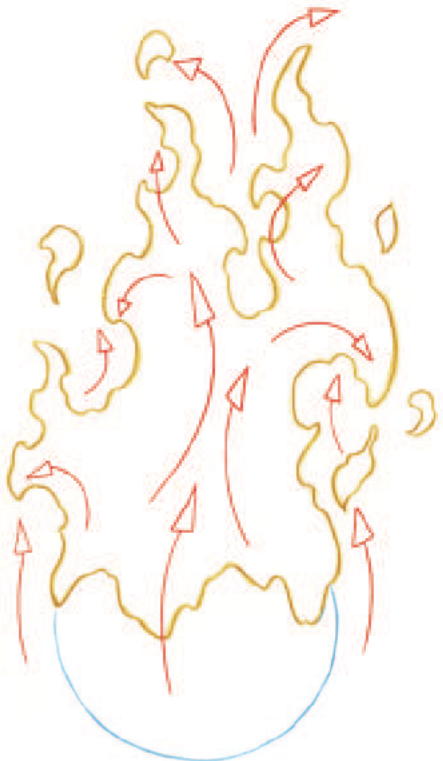
2

Now I have treated each of the individual triangles as if they are flags being blown from below. I have added the curving bottom line shape of the fire, and indicated the directional flow of the air currents causing the flags to flutter and wave. Note that there is cooler air pushing down from above the fire as well, that causes additional turbulence.



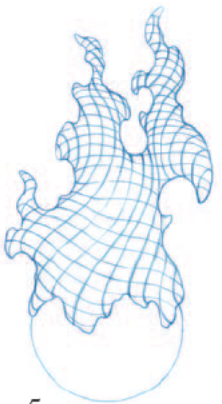
3

Now things start to get a lot more dynamic, as turbulence is added to the picture. This turbulent energy creates small spiraling eddies of air that twist, pull, and squeeze the triangular shapes, causing the familiar arcing shapes that we see so frequently in fire. Note that on the left side of the fire the twisting currents rotate counter-clockwise, while those on the right revolve in a clockwise direction. Opposing arcs become very apparent at this stage, and are an outstanding feature of almost all good, elemental effects design.



4

At this stage in the drawing, the same air currents and turbulence that came into play in Figure 3 are now used to add the more detailed, wavy, wiggly shapes that really start to describe this shape as a fire. Opposing arcs are always a very important aspect of these designs. As always, make sure that the size and positioning of these shapes avoid being overly repetitive.



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At this point, we must think of our two-dimensional diagram as a three-dimensional object. I have illustrated, first with an overly technical wire-frame drawing, one way of approaching this stage in our drawing. We can then add our interior shapes which do much to describe the volume and directional energy of the fire. With more experience, one would generally begin a fire drawing with a very volumetric sketchy approach, skipping the more technical process shown here in the first four examples. But this is an excellent foundation for beginning to understand the principles at play in a fire.

7

Finally, we can add our elegant touches to the fire design, tweaking and pushing our design to make it as dynamic as possible. The final outline, combined with the interior shapes, makes for a pleasing (in this case relatively simplified and stylized) rendition of a typical campfire-size fire.

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In this fire animation, I have added a ghost image of the following drawing, so that you can see clearly how the motion progresses. As the overall shape of the fire undulates with the familiar, whip/wave motion, smaller pieces emerge from the bottom sides of the fire and react to the turbulence being created by the hot and cool air currents colliding as the fire sucks in cooler air from around it, in order to fuel itself with oxygen. This animation also shows clearly how large pieces of the fire are pinched, or squeezed by the cooler air as they rise, causing them to break off as they rise. Note that the small shapes that break off from the main body of the fire do not last more that two or three drawings. Larger pieces may last longer.

Another fire animation trick, clearly illustrated here, is how a curling shape that is breaking off and rising quickly away from the main body of the fire will invert its curved shape from one drawing to the next. This gives these little pieces of flame the appearance of flickering and behaving like a real fire. I animated this fire as if it is a well-fueled "ball of fire" so the shape of the base of the fire is much more stable and static than a camp fire would be. Note too, how the secondary inside shapes enhance the overall shape and volume of the fire, working as a sort of shading tool. This animation is a cycle of thirteen drawings, with the last drawing hooking up seamlessly with the first.

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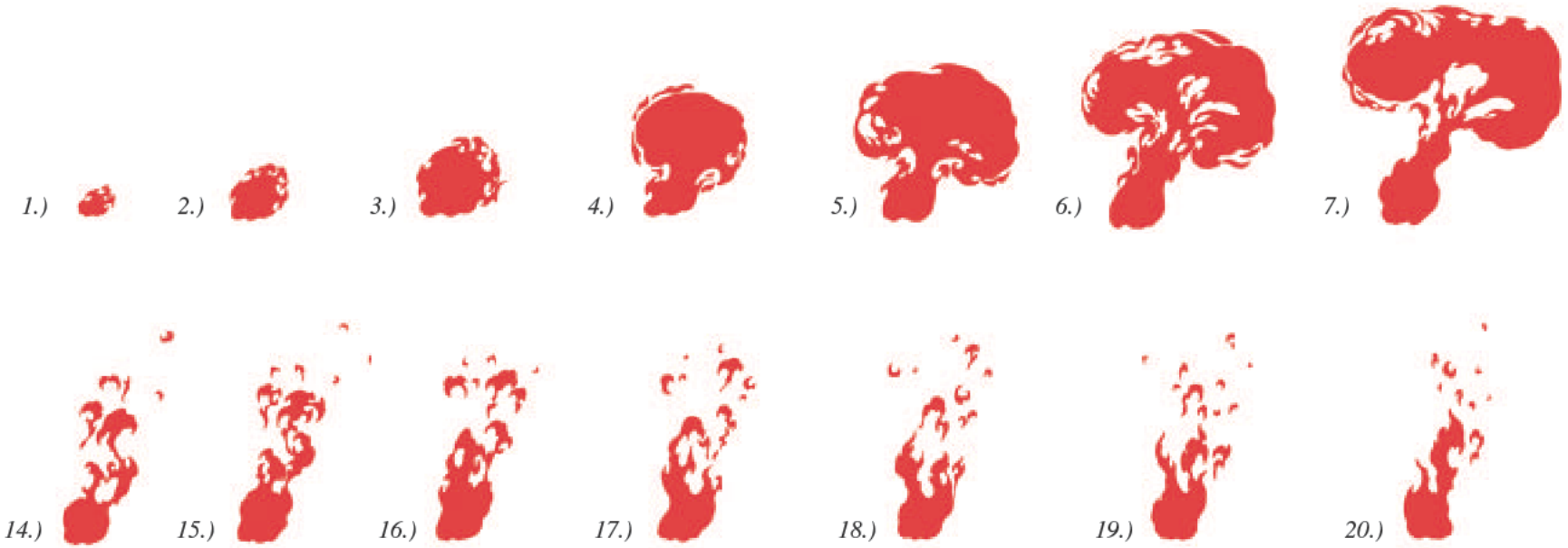
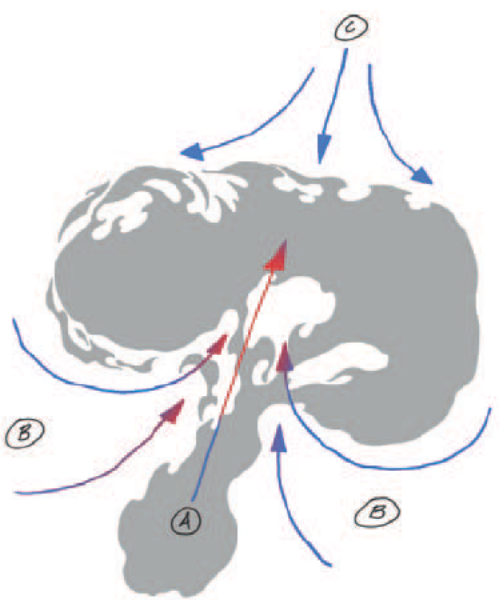
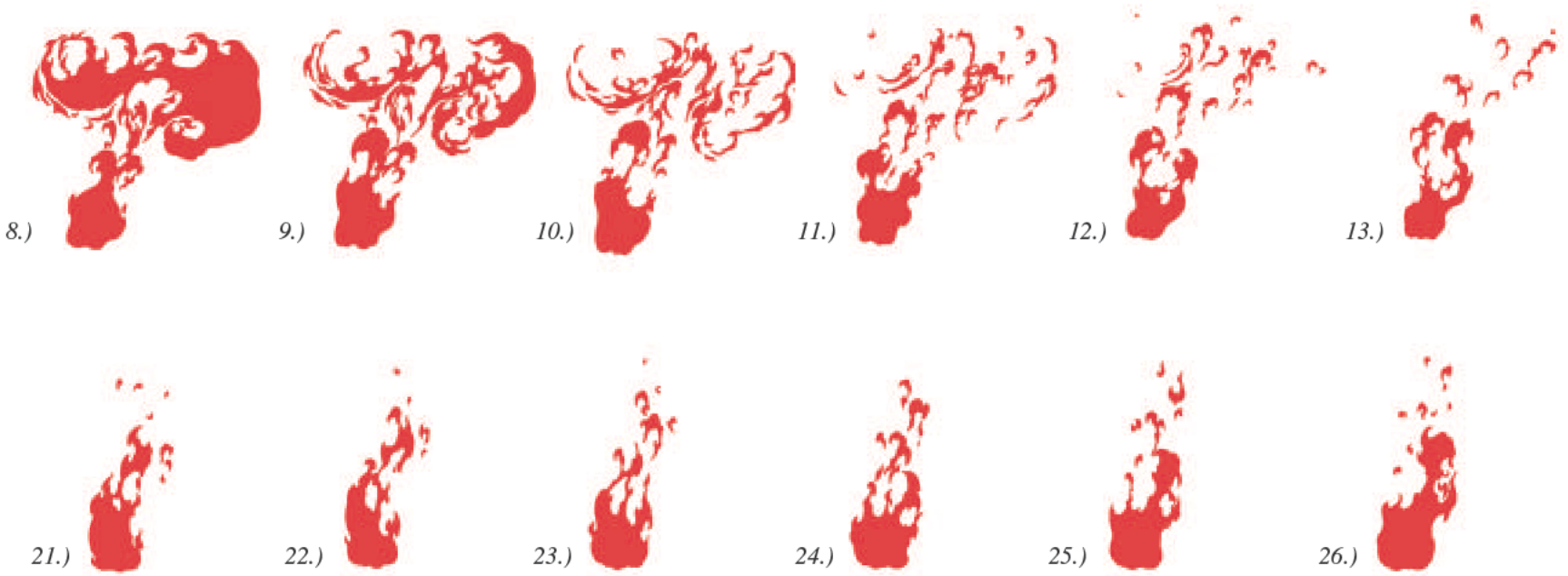


Here I have animated what a fire might look like igniting, as if from a pool of a highly flammable substance like gasoline. Initially, in #1 through 3, the fire simply blossoms quickly upward and outward. It is well fueled with a flammable substance and plenty of oxygen. Outside forces have not yet come into play, but immediately in #4 and onward, we see the familiar mushroom phenomenon beginning to take shape. In this clearly illustrated enlarged view of #7. we see the force of the initial ignition still shooting upward. (A) As the cooler air surrounding the fire pushes it down from above, (C) and the hot fire sucks cool, oxygen-filled air into itself from the air surrounding it, (B) it pulls the cooler air inward and upward, and the collision of these forces creates the mushroom shape.



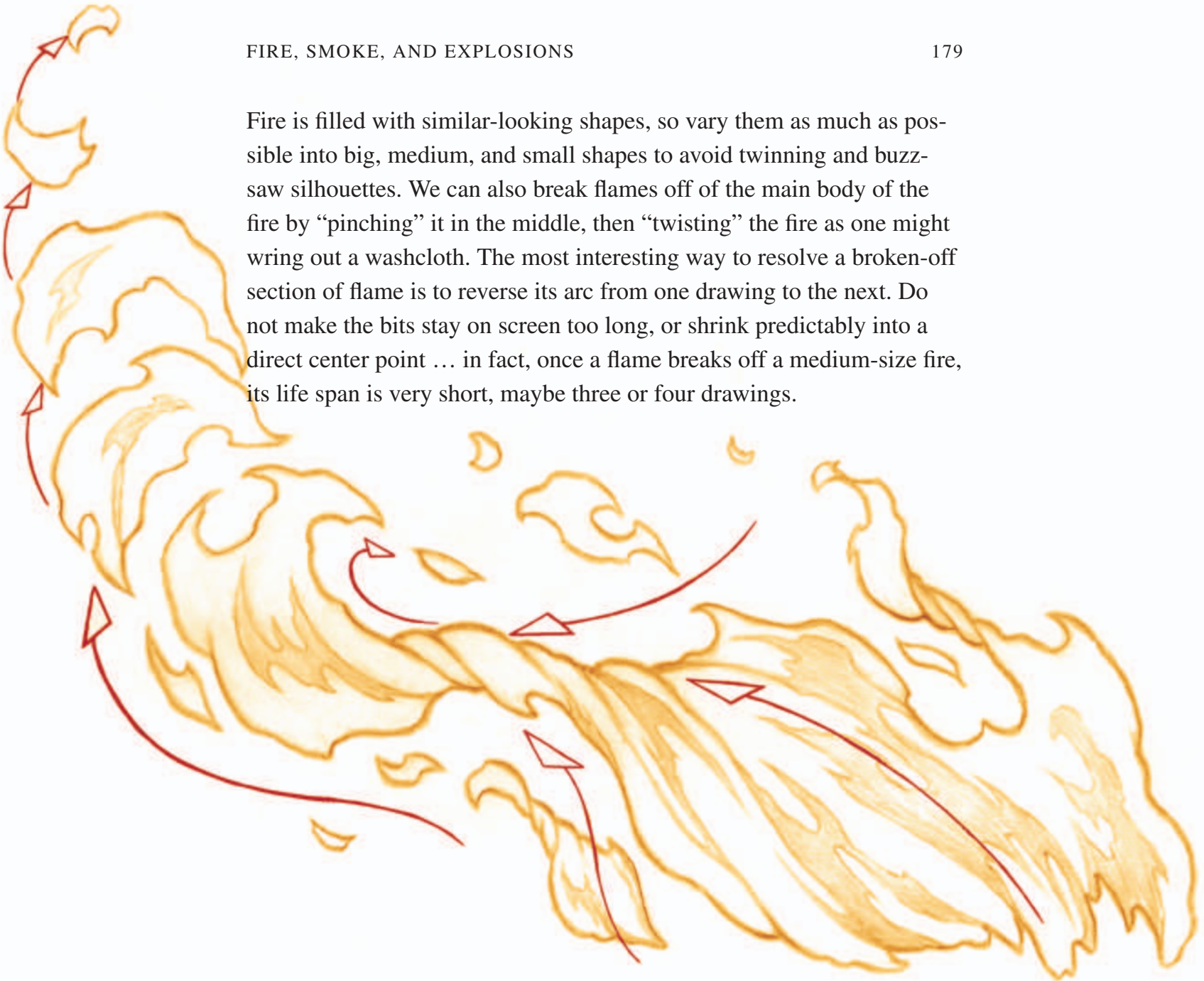


As the fire continues to rise, the intense heat, with no place to go but up, has to push its way through the layers of varying temperatures of cooler air around it. As the hot and cool air interact, spiraling eddies and twisting currents of air are formed, which give the fire its distinct shapes. Every smaller shape of the fire that is created (as outside forces tear it to pieces) takes on a somewhat similar form, as the cooler air above it, and its natural trajectory upward continue to create fascinating and beautifully shaped air currents. Keep in mind always fire is pure energy being expended—and seems to have a life of its own. Its shapes are created by hot and cool air reacting to each other. If the physics of your fire animation don't follow these natural laws of nature, it will look unnatural and strange.

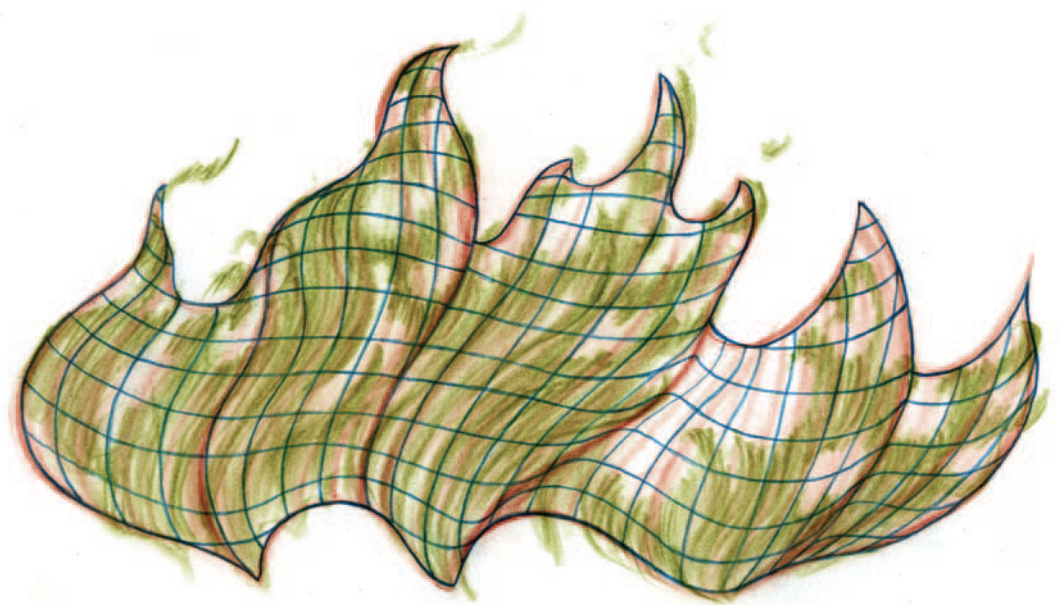
It is important to note that a fire should not increase in size without good fire dynamics being taken into consideration. It's normal for the inexperienced effects artist to unconsciously expand the fire's mass. Without certain tricks under your hat, it's difficult to keep a medium-size fire from expanding as a fire scene progresses. We can achieve consistency in the volume by opening holes from the middle section (as we did previously with water); these will quickly meet together and help to break off large sections of flame. The holes should reflect the direction of the force through their elongated negative spaces. Be careful not to make these negative spaces too water-like! Keep the shapes working with the overall design.

I have indicated X's where there are holes breaking open in this fire. This is a way that we actually mark up our production effects drawings, so that our assistant effects artists can tell the holes from the interior shapes. This is common when animating water and smoke as well.

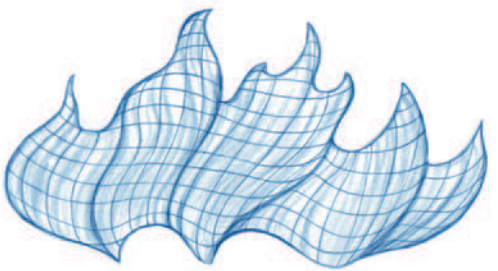
Fire is filled with similar-looking shapes, so vary them as much as possible into big, medium, and small shapes to avoid twinning and buzzsaw silhouettes. We can also break flames off of the main body of the fire by "pinching" it in the middle, then "twisting" the fire as one might wring out a washcloth. The most interesting way to resolve a broken-off section of flame is to reverse its arc from one drawing to the next. Do not make the bits stay on screen too long, or shrink predictably into a direct center point … in fact, once a flame breaks off a medium-size fire, its life span is very short, maybe three or four drawings.



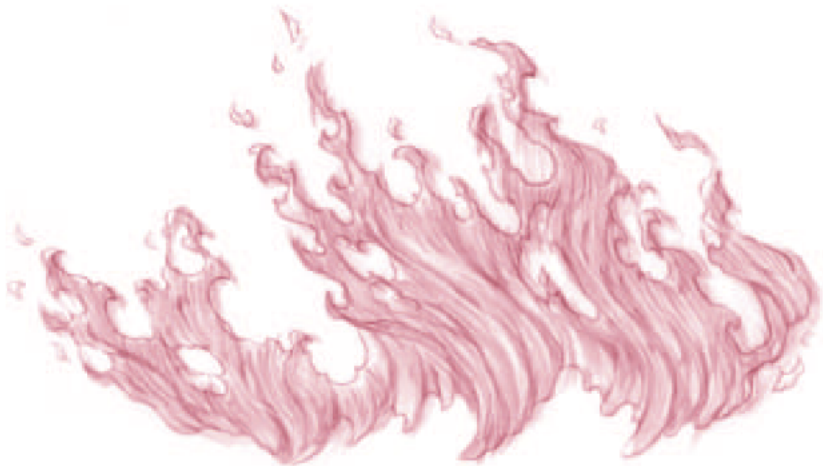
The drawing to the right, shows the underlying geometric structure of our "flags." While I would never go to the trouble to illustrate this when actually animating a fire, it is important to understand the fundamental structure of what you are working with! With time and practice, an effects artist intuitively feels this underlying structure within every drawing he or she creates.



Here is another approach to how to start drawing and animating a fire. As with all special effects, start out with the most basic shapes as in this example on the left. If you turn this drawing on its end, it looks like five flags. Approached with this technique, you can now animate the fire, just as if it was a group of flags, using the basic flag-waving motion, and not having to worry about all the difficult little details of the final fire drawings.

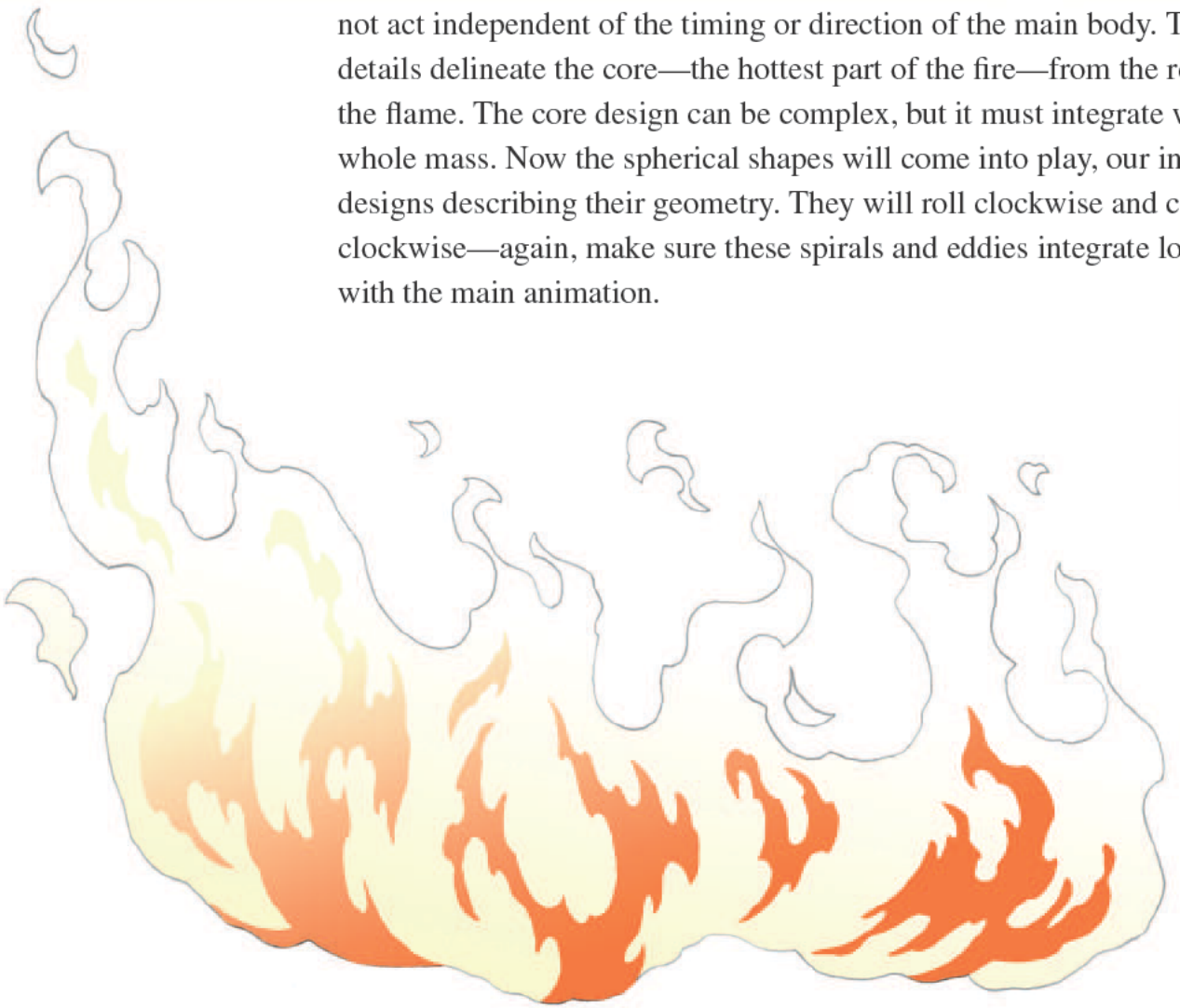


The really "fun" part of the effects drawing experience is the sketching in of the intricate details of your effect. It is important in the case of fire, as it is with water, smoke, etc., to make sure that every detail follows closely the overall flow of the underlying structure. The smaller shapes you see breaking off of a fire will just look like bizarre disconnected shapes if they do not adhere to the natural "flow" of the drawing. When these shapes are drawn with the directional flow of the energy driving the effect, the results are fantastic. This is a very key point to understanding and drawing all effects animation—as I emphasized so emphatically in Chapter 2! Understand the energy behind the effect, and stick to it with every stroke of the pencil.



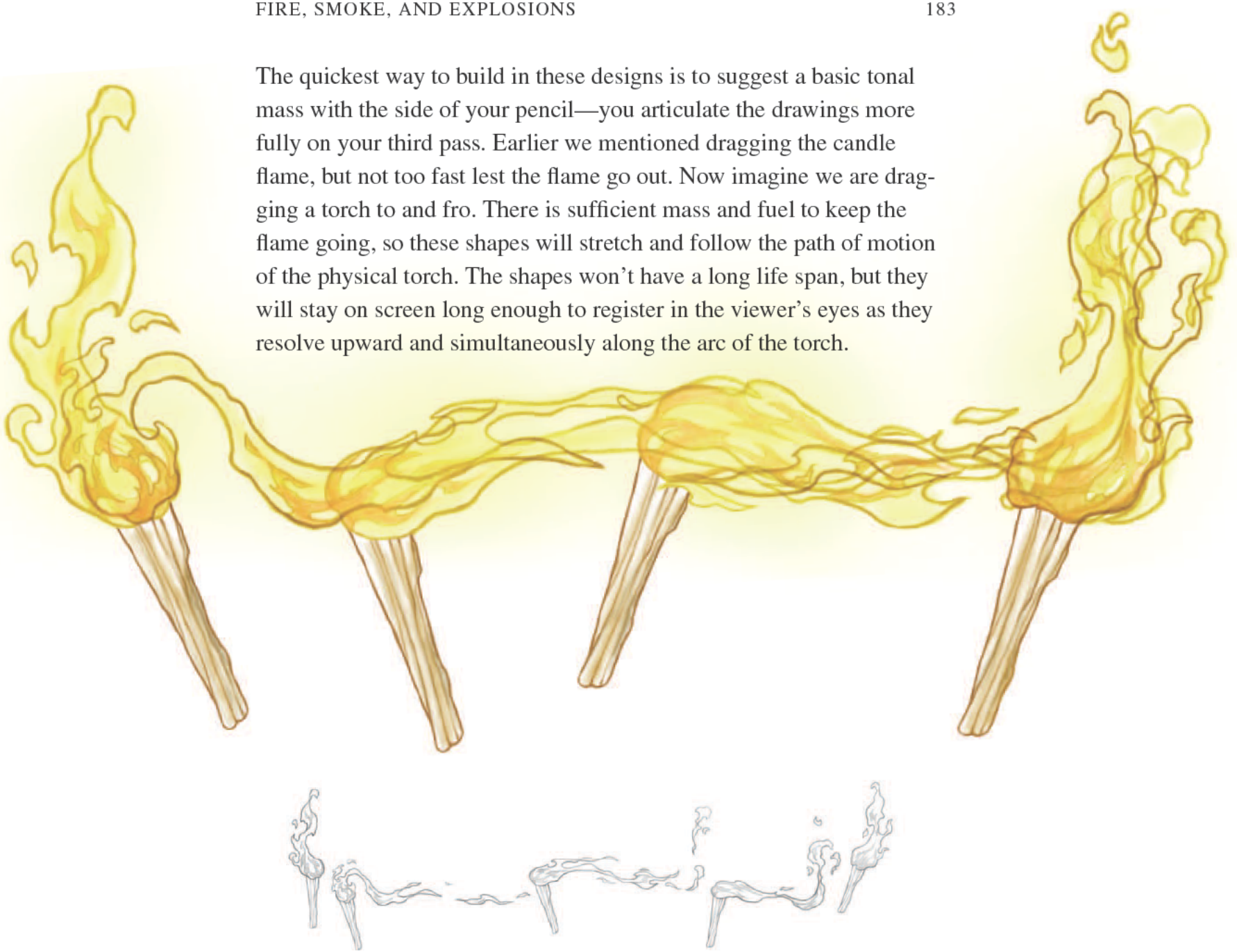
Now with our details roughed in over the basic structure of our fire drawing, we can tidy up the sketchy lines, and define each shape with its final intricate detail. At this stage as well, it is of the utmost importance to conform to the structure and energy that we built on from the very first rough stages of our fire drawing. Every little piece of the fire must feel as if it is being carried along by the energy of the fire's heat, rising up as it sucks in cooler air and oxygen from around it—the hot and cool air's interaction causing the beautiful swirling eddies and air currents which give every shape in the fire its unique and elegant quality. It is physics, and it is complex, but it is also natural and sublime. If we spend some time observing fire and "feeling" it, all of this will make sense, it need not be a technical nightmare! Try to feel *it, and have fun with your drawing!*

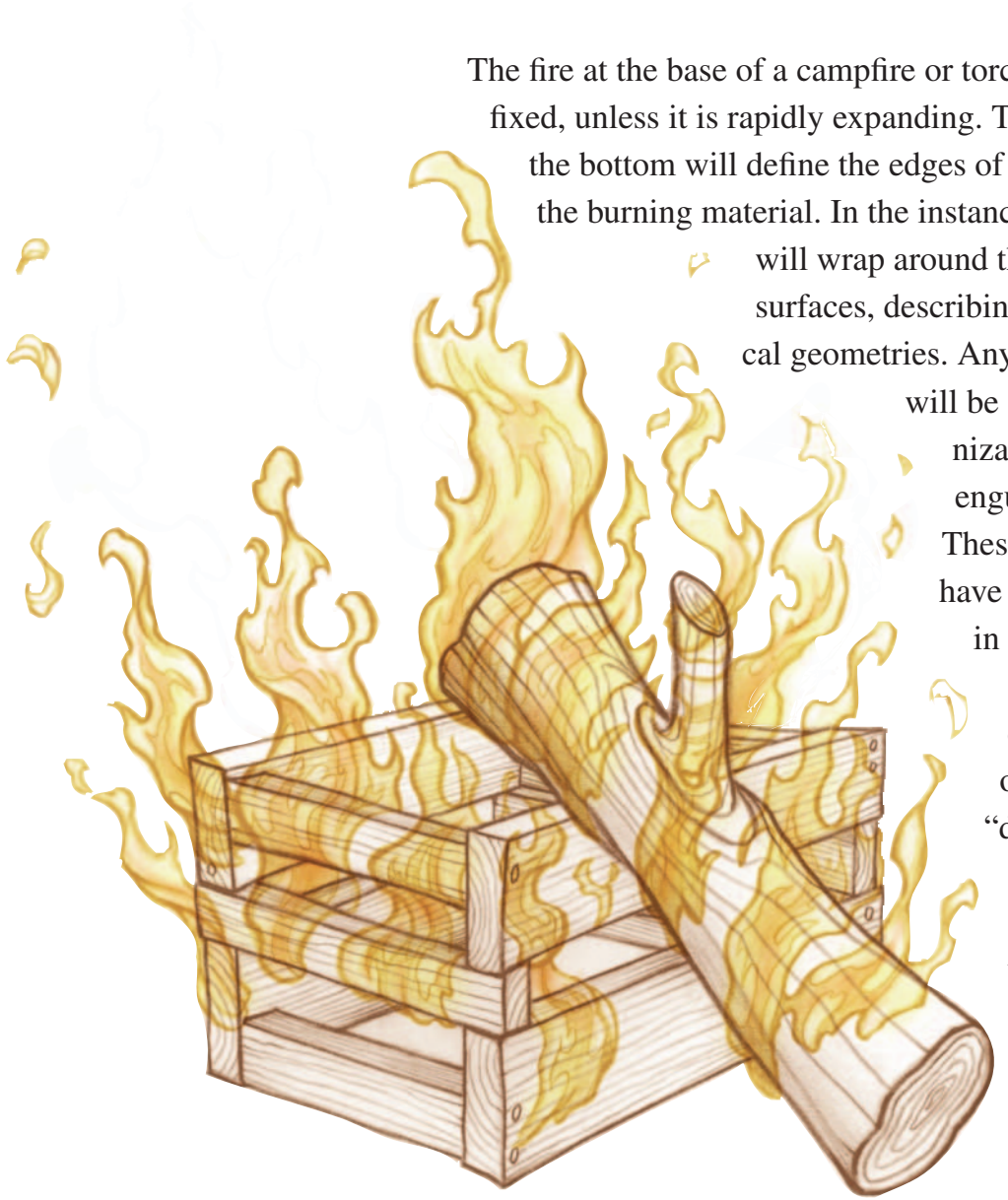
Now that we've worked out the basic mechanics of the main silhouette, let's investigate what happens with the interior details. While there will be overlapping shapes within the fire, never forget that these details cannot act independent of the timing or direction of the main body. These details delineate the core—the hottest part of the fire—from the rest of the flame. The core design can be complex, but it must integrate with the whole mass. Now the spherical shapes will come into play, our interior designs describing their geometry. They will roll clockwise and counterclockwise—again, make sure these spirals and eddies integrate logically with the main animation.



In this example I've illustrated the importance of the interior shapes of a fire drawing. As we see in the basic line drawing to the left, the outer line of the fire alone does little to describe its volume and directional energy. The interior shapes and the way they move, do much to describe the overall shape and energy of the flame.

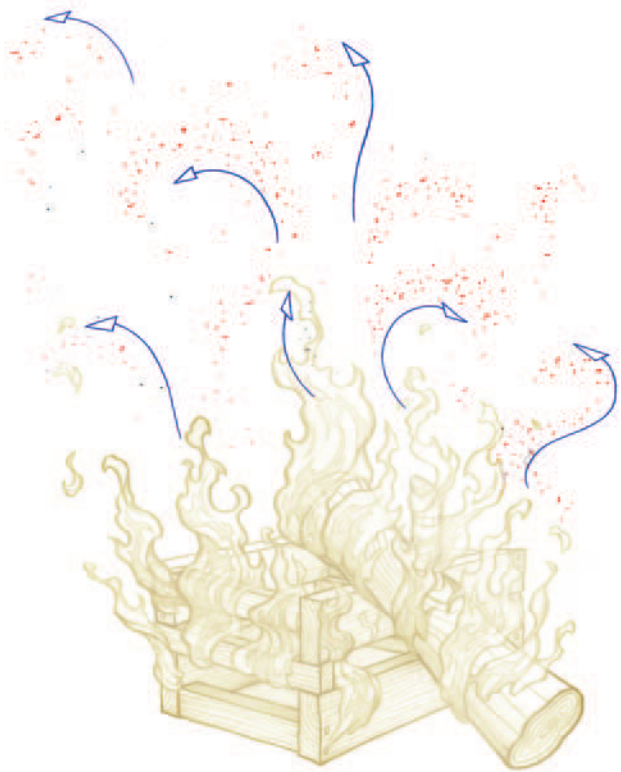
The quickest way to build in these designs is to suggest a basic tonal mass with the side of your pencil—you articulate the drawings more fully on your third pass. Earlier we mentioned dragging the candle flame, but not too fast lest the flame go out. Now imagine we are dragging a torch to and fro. There is sufficient mass and fuel to keep the flame going, so these shapes will stretch and follow the path of motion of the physical torch. The shapes won't have a long life span, but they will stay on screen long enough to register in the viewer's eyes as they resolve upward and simultaneously along the arc of the torch.





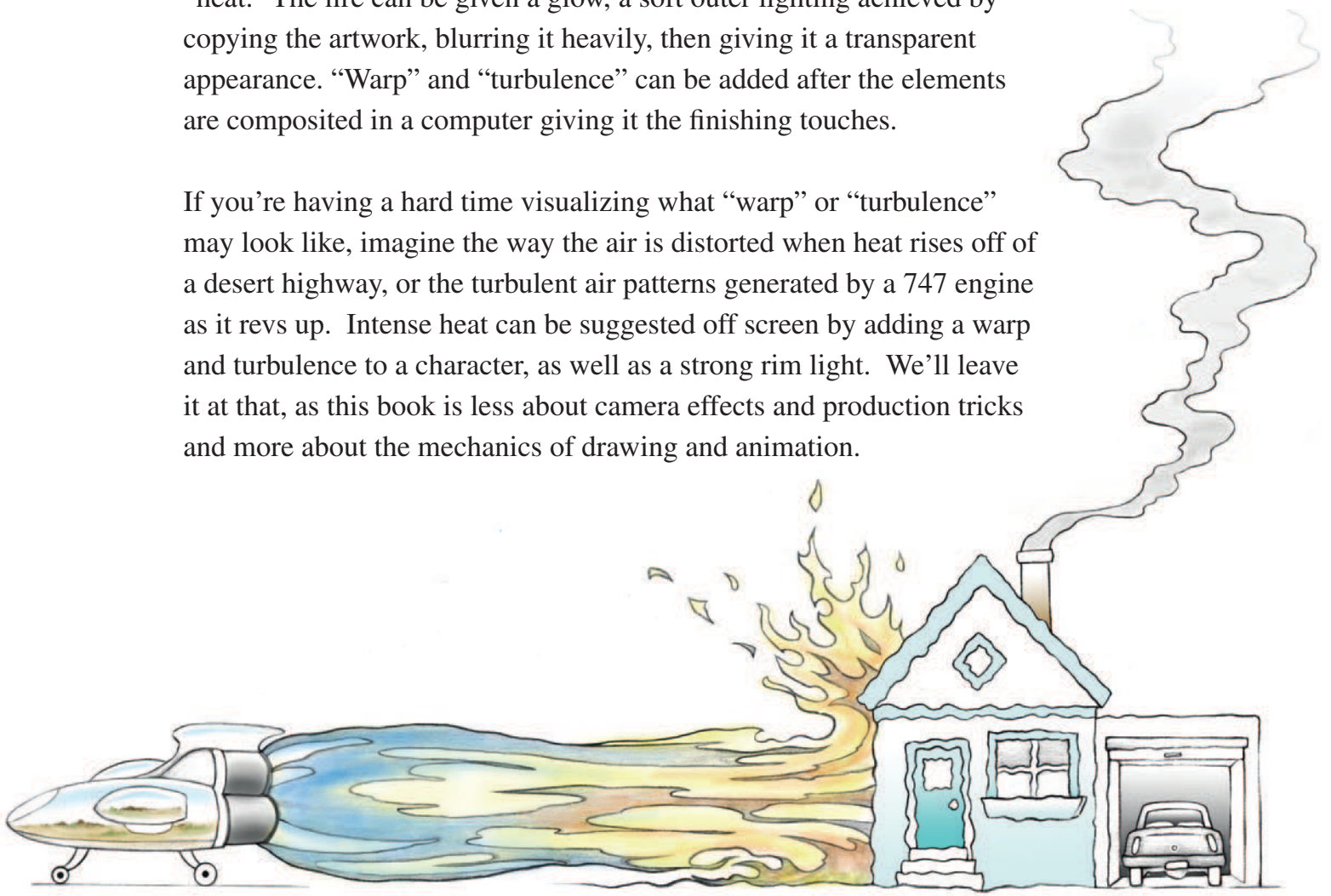
The fire at the base of a campfire or torch is relatively fixed, unless it is rapidly expanding. These shapes on the bottom will define the edges of the surface of the burning material. In the instance of logs, the fire will wrap around their respective surfaces, describing their cylindrical geometries. Any object on fire will be somewhat recognizable, even when engulfed in flames. These shapes will also have a liquid feel, both in terms of design and timing—the contact point of the fire will "dance" in a more deliberate, slower manner than the upper portion.

Before we move onto large-scale fire, we'll touch ever so briefly on a couple of other important additional fire elements. Once we get to any fire mass beyond a candle flame, we will begin to see cinders rising with the fire. These particles will have some randomness, but their direction will be dictated by the structural arcs and forces that move within the main fire. We should also be aware of the spacing of these elements, and course, vary their sizes.



When fire is eventually colored, there are ways to reinforce the idea of "heat." The fire can be given a glow, a soft outer lighting achieved by copying the artwork, blurring it heavily, then giving it a transparent appearance. "Warp" and "turbulence" can be added after the elements are composited in a computer giving it the finishing touches.

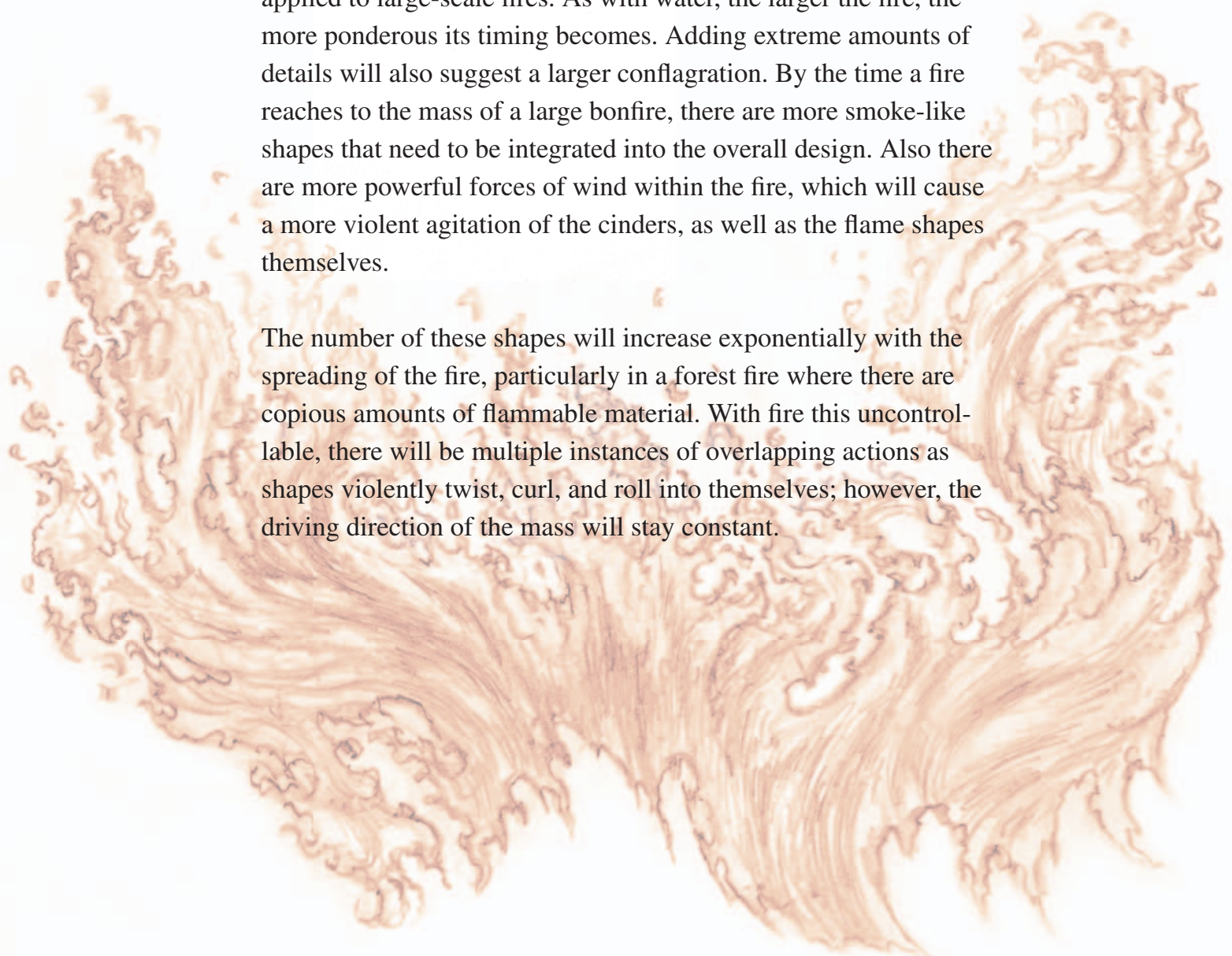
If you're having a hard time visualizing what "warp" or "turbulence" may look like, imagine the way the air is distorted when heat rises off of a desert highway, or the turbulent air patterns generated by a 747 engine as it revs up. Intense heat can be suggested off screen by adding a warp and turbulence to a character, as well as a strong rim light. We'll leave it at that, as this book is less about camera effects and production tricks and more about the mechanics of drawing and animation.



Here is an example of a jet engine causing some heat turbulence. Of course the house would also be reduced to ashes in pretty short order!

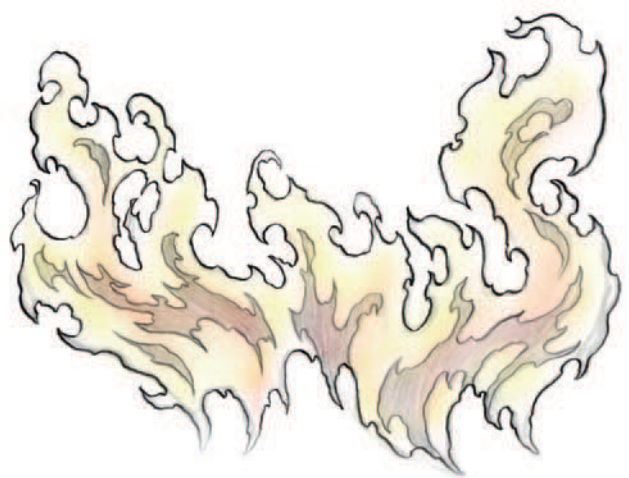
Almost everything we mentioned about medium-size fires—can be applied to large-scale fires. As with water, the larger the fire, the more ponderous its timing becomes. Adding extreme amounts of details will also suggest a larger conflagration. By the time a fire reaches to the mass of a large bonfire, there are more smoke-like shapes that need to be integrated into the overall design. Also there are more powerful forces of wind within the fire, which will cause a more violent agitation of the cinders, as well as the flame shapes themselves.

The number of these shapes will increase exponentially with the spreading of the fire, particularly in a forest fire where there are copious amounts of flammable material. With fire this uncontrollable, there will be multiple instances of overlapping actions as shapes violently twist, curl, and roll into themselves; however, the driving direction of the mass will stay constant.

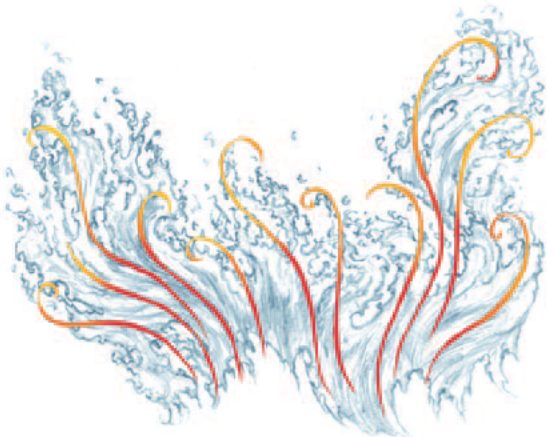


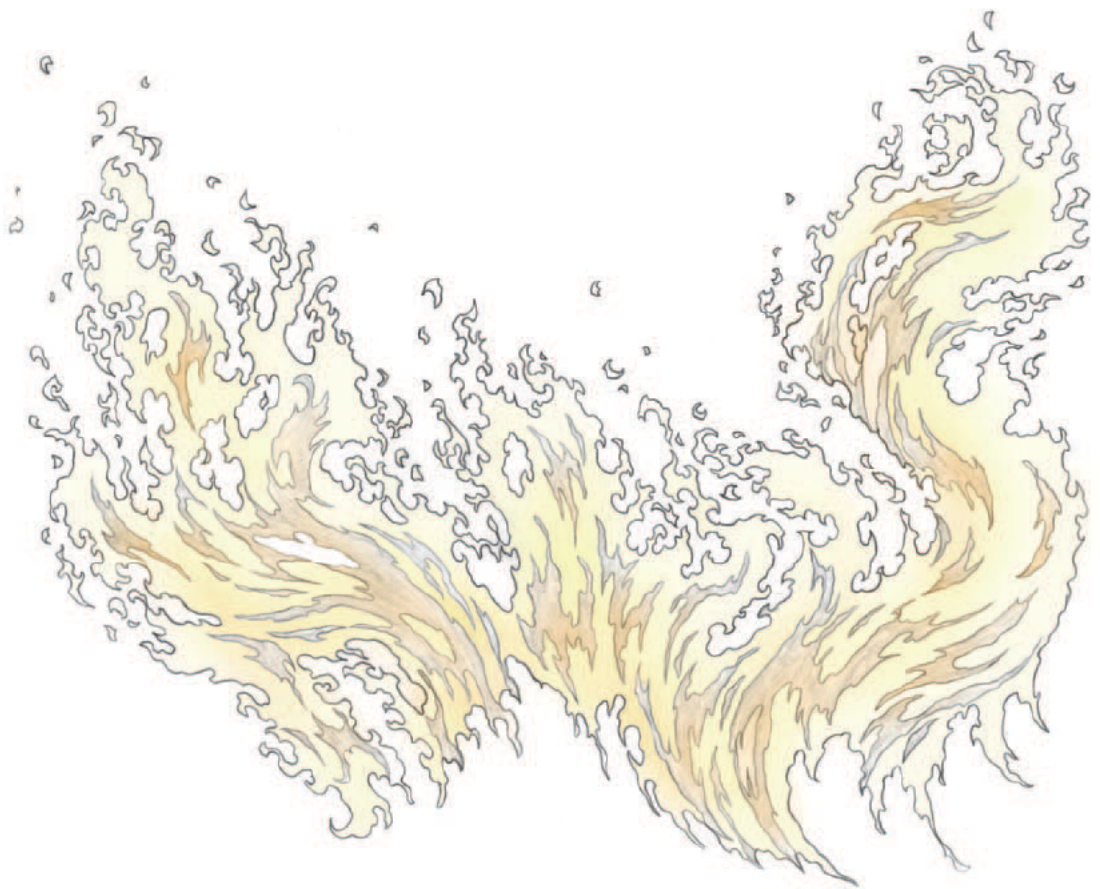
This rough drawing of an enormous fire shows how freely and energetically one must draw in order to describe this firey mayhem. You can see clearly the directional energy of my pencil strokes that helped construct the shapes in a way that clearly conveys the violent turbulence of the fire.

This smaller drawing to the right, has exactly the same underlying structure as the huge fire above, but by simply leaving some of the details out, it ends up looking like an average-size campfire. This illustrates perfectly the fractal nature of all pure elemental energy. Whether at the macro- or microcosmic level, the same patterns are apparent.



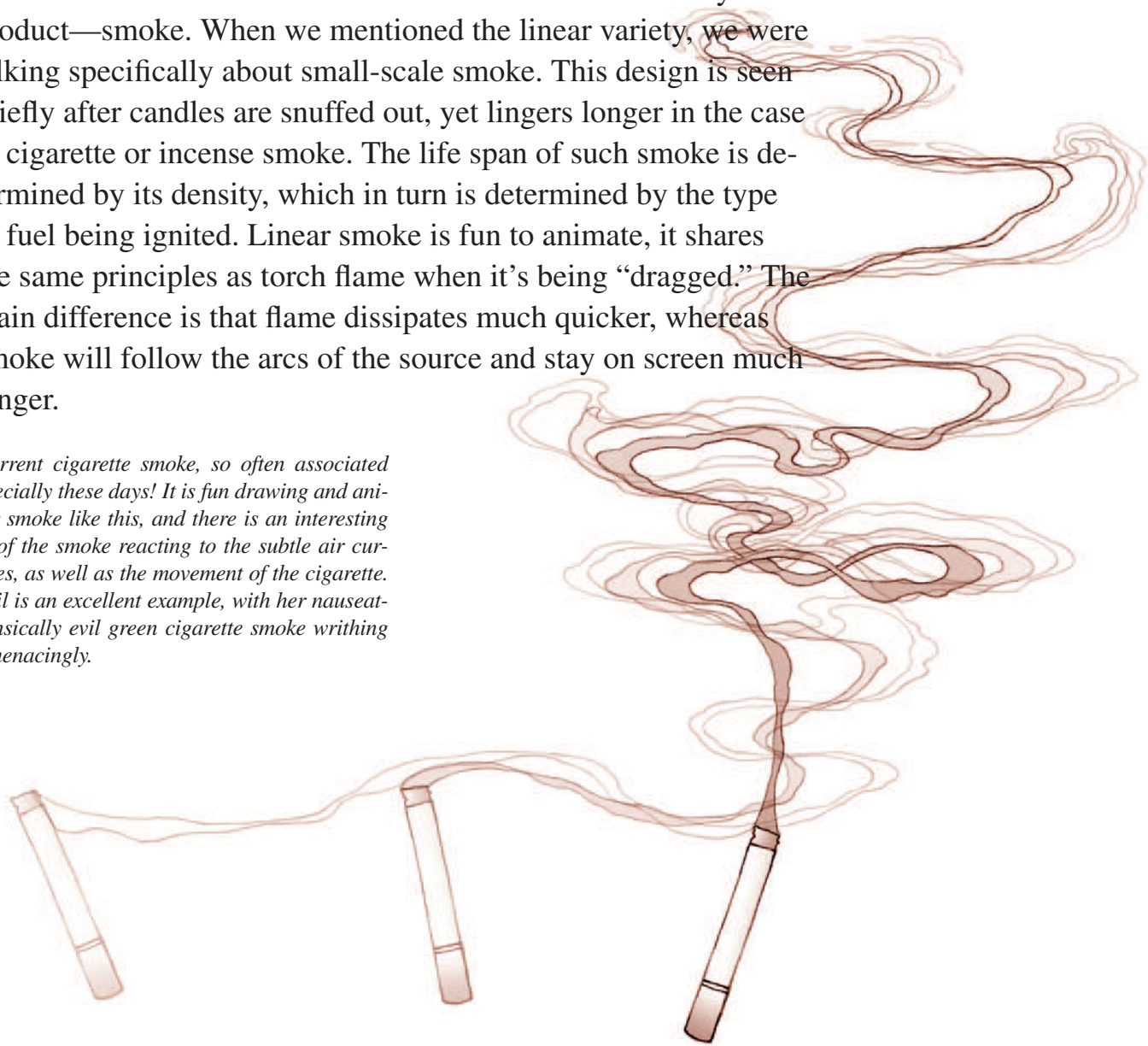
In this drawing I have illustrated some of the directional energy of this fire with a series of elegant plant-like strokes, rather than textbook type arrows, to better emphasize the natural aspect of this energy. Again pointing out the relativity of all nature's design, whether it is a fern unfolding deep in the forest, a raging forest fire, or an elegant lily in a garden.



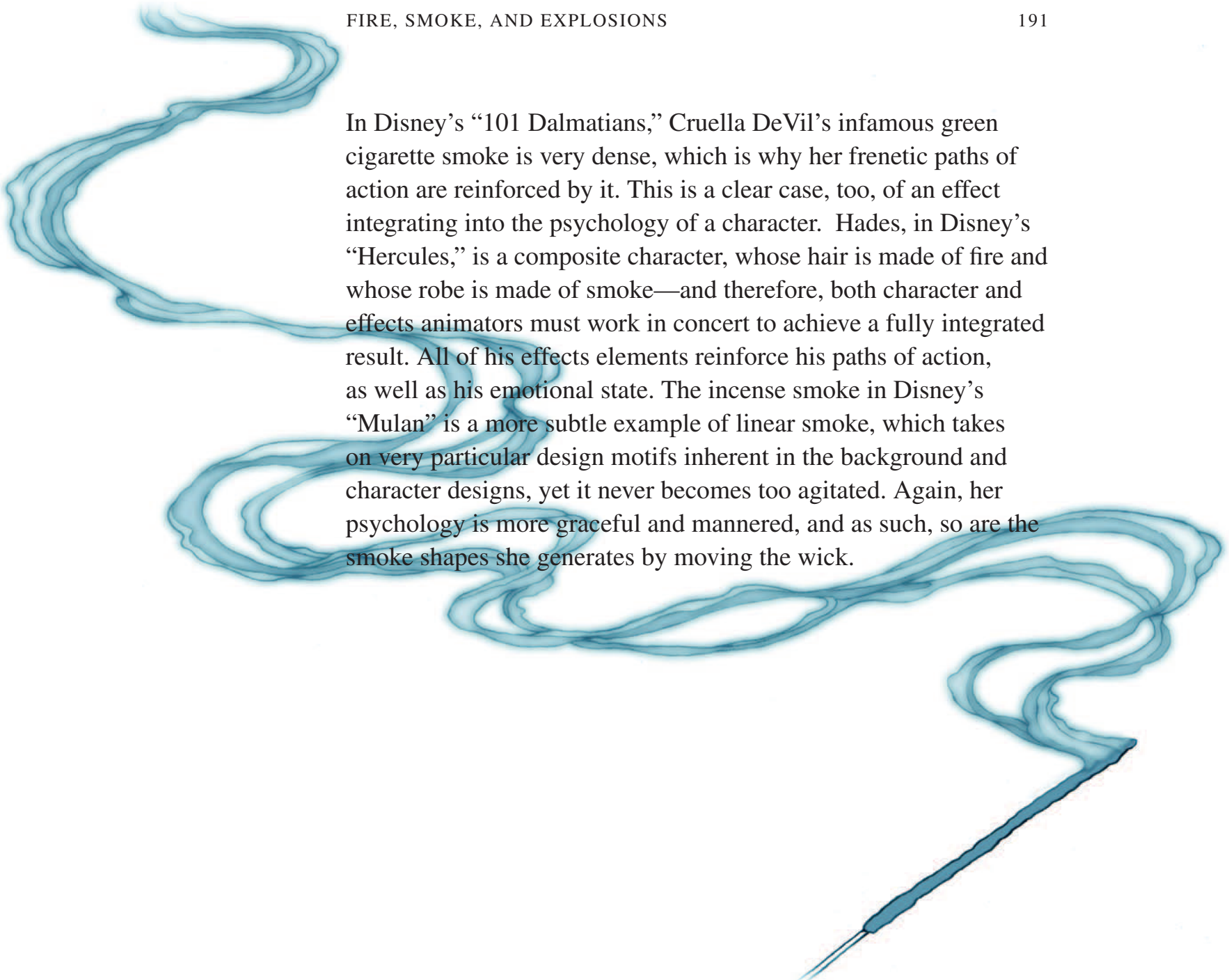


In this clean version of the rough sketch on the preceding pages, I paid more attention to the inside shapes, which help describe the volume of the fire, the actual three-dimensional shape of it, while at the same time accentuating the directional energy. As in the earlier examples in this chapter of a medium-size fire, to animate something of this level of detail we must start out with the simplest of shapes, and apply the simple rolling wave and whipping flag animation principles before adding the more intricate and tiny details, each of which must animate with that same whipping, curling motion.

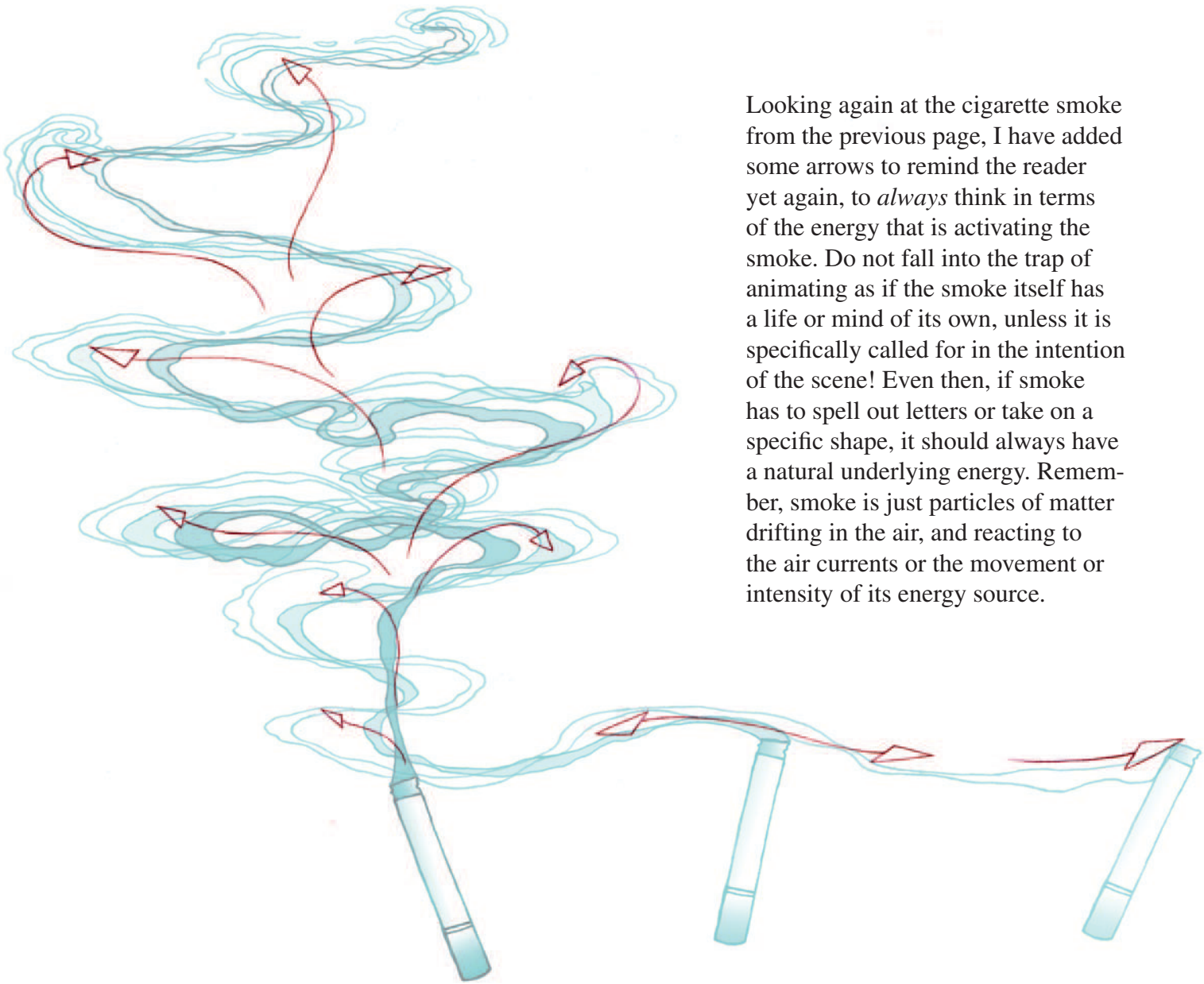
We can't finish this discussion until we address the fire's by-product—smoke. When we mentioned the linear variety, we were talking specifically about small-scale smoke. This design is seen briefly after candles are snuffed out, yet lingers longer in the case of cigarette or incense smoke. The life span of such smoke is determined by its density, which in turn is determined by the type of fuel being ignited. Linear smoke is fun to animate, it shares the same principles as torch flame when it's being "dragged." The main difference is that flame dissipates much quicker, whereas smoke will follow the arcs of the source and stay on screen much longer.



Ah, the abhorrent cigarette smoke, so often associated with evil, especially these days! It is fun drawing and animating linear smoke like this, and there is an interesting combination of the smoke reacting to the subtle air currents as it rises, as well as the movement of the cigarette. Cruella De Vil is an excellent example, with her nauseating and intrinsically evil green cigarette smoke writhing and curling menacingly.

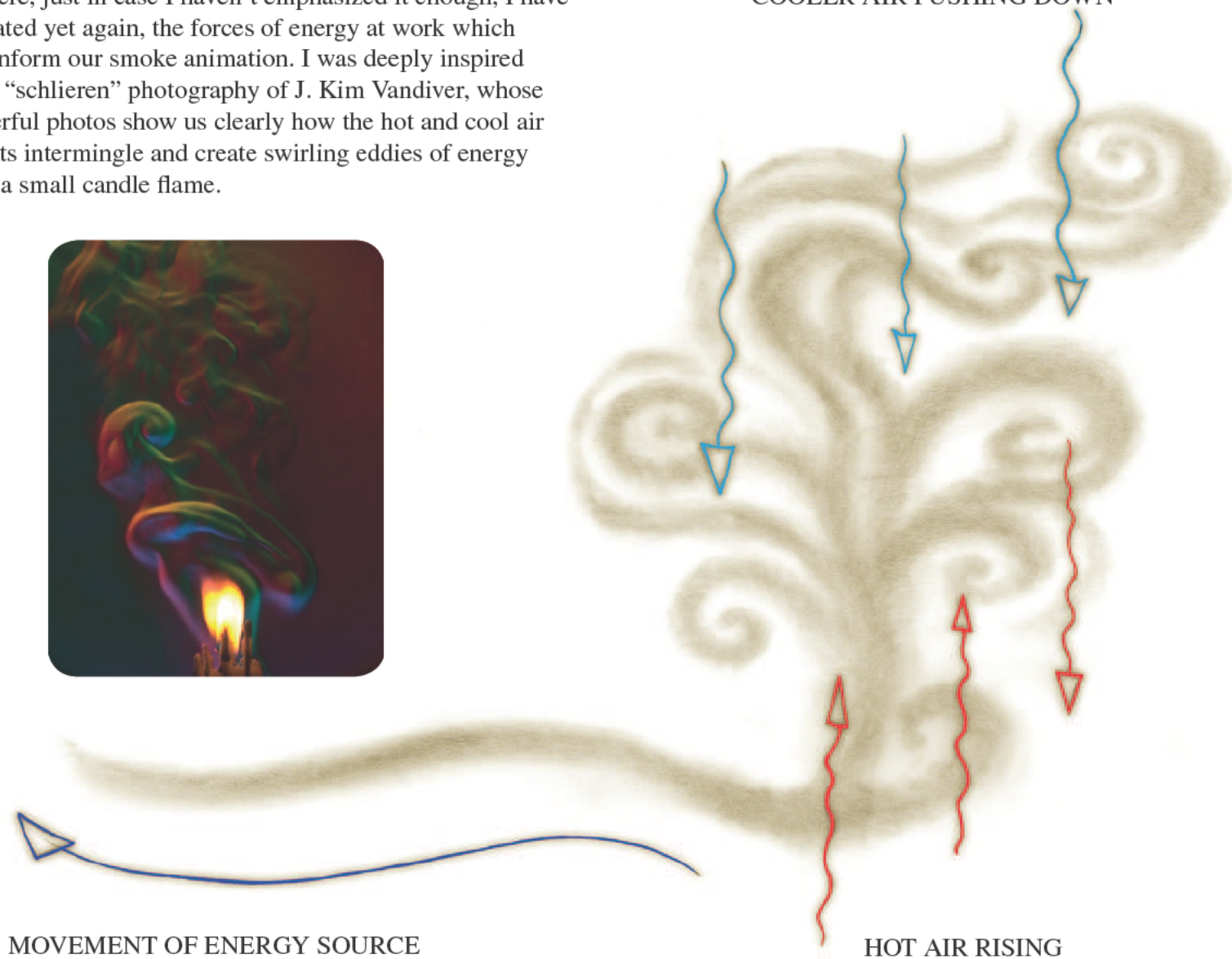


In Disney's "101 Dalmatians," Cruella DeVil's infamous green cigarette smoke is very dense, which is why her frenetic paths of action are reinforced by it. This is a clear case, too, of an effect integrating into the psychology of a character. Hades, in Disney's "Hercules," is a composite character, whose hair is made of fire and whose robe is made of smoke—and therefore, both character and effects animators must work in concert to achieve a fully integrated result. All of his effects elements reinforce his paths of action, as well as his emotional state. The incense smoke in Disney's "Mulan" is a more subtle example of linear smoke, which takes on very particular design motifs inherent in the background and character designs, yet it never becomes too agitated. Again, her psychology is more graceful and mannered, and as such, so are the smoke shapes she generates by moving the wick.



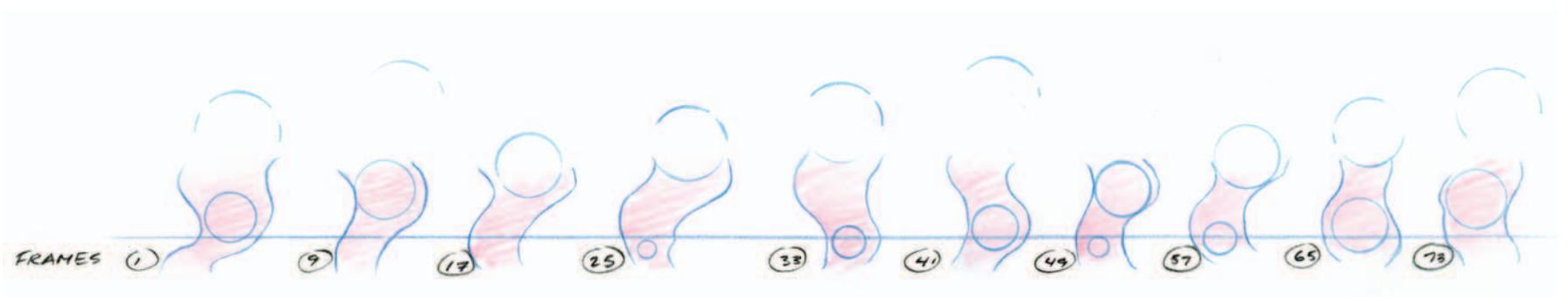
Looking again at the cigarette smoke from the previous page, I have added some arrows to remind the reader yet again, to *always* think in terms of the energy that is activating the smoke. Do not fall into the trap of animating as if the smoke itself has a life or mind of its own, unless it is specifically called for in the intention of the scene! Even then, if smoke has to spell out letters or take on a specific shape, it should always have a natural underlying energy. Remember, smoke is just particles of matter drifting in the air, and reacting to the air currents or the movement or intensity of its energy source.

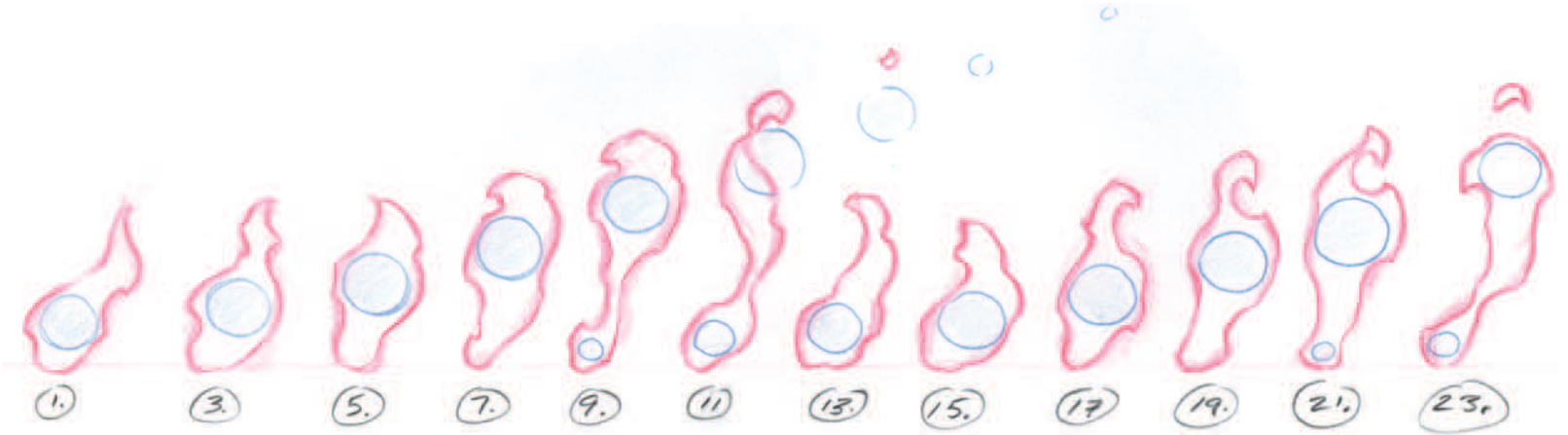
And here, just in case I haven't emphasized it enough, I have illustrated yet again, the forces of energy at work which must inform our smoke animation. I was deeply inspired by the "schlieren" photography of J. Kim Vandiver, whose wonderful photos show us clearly how the hot and cool air currents intermingle and create swirling eddies of energy above a small candle flame.



The drawing sequence below illustrates one of my simplest formula for animating smoke. I have numbered each drawing with the actual frame numbers, and this sequence would represent 74 frames, or just over two seconds of animation. This simple formula can be applied to fire animation as well, as I will illustrate on the following page.

I have combined the classic upward flag-like wave motion, along with an emerging ball shape, that kind of "rolls" upward through the volume of the smoke in time with the wave action. This ball continually expands as it rises as smoke will in relatively calm air with a hot source pushing it upward, and cooler air above slowing it down and pushing it outward. Variations of this simple approach can be combined with a far more complex smoke design as well, but this is a great little exercise to get a good feeling for how a simple column of smoke can be simply but well animated.



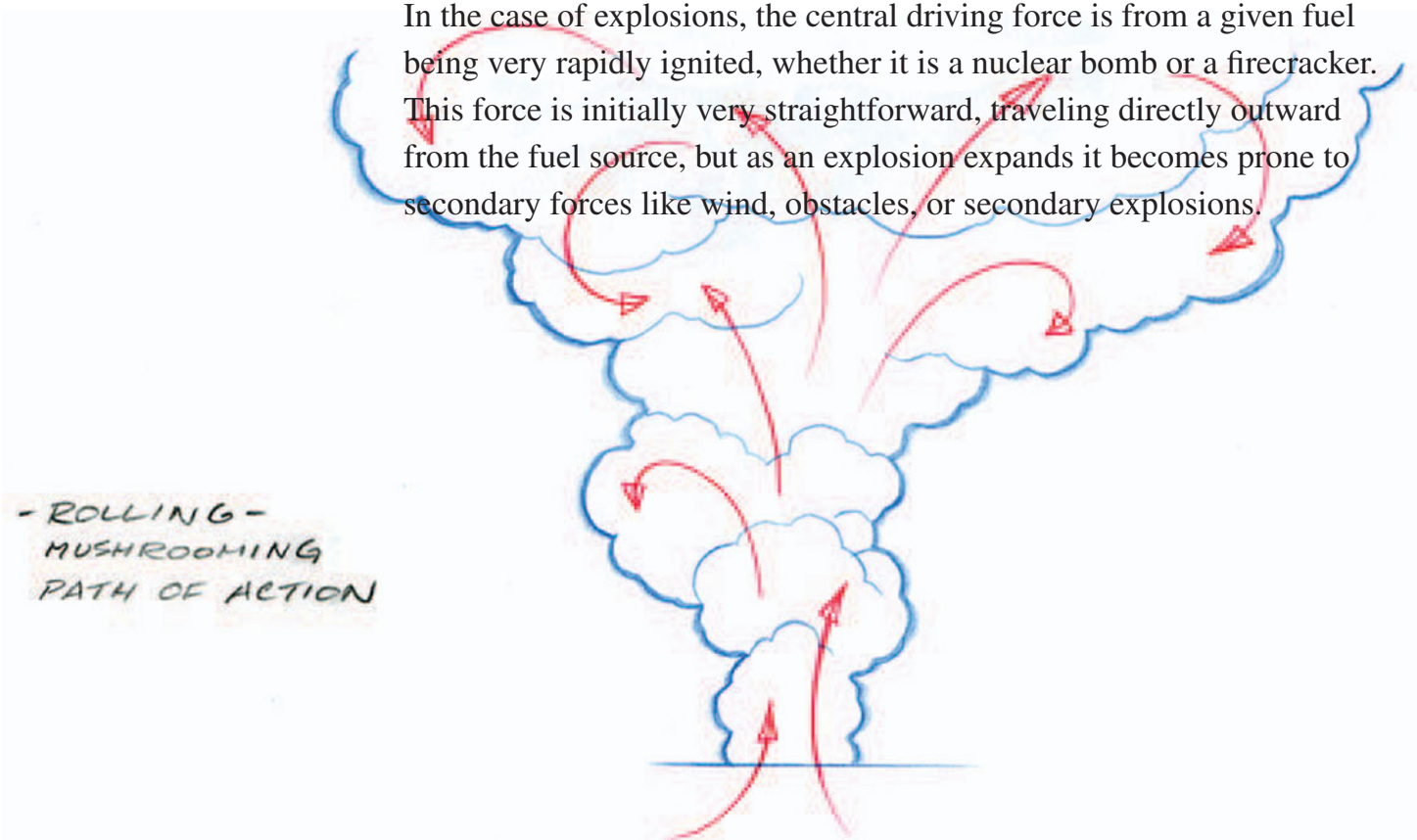


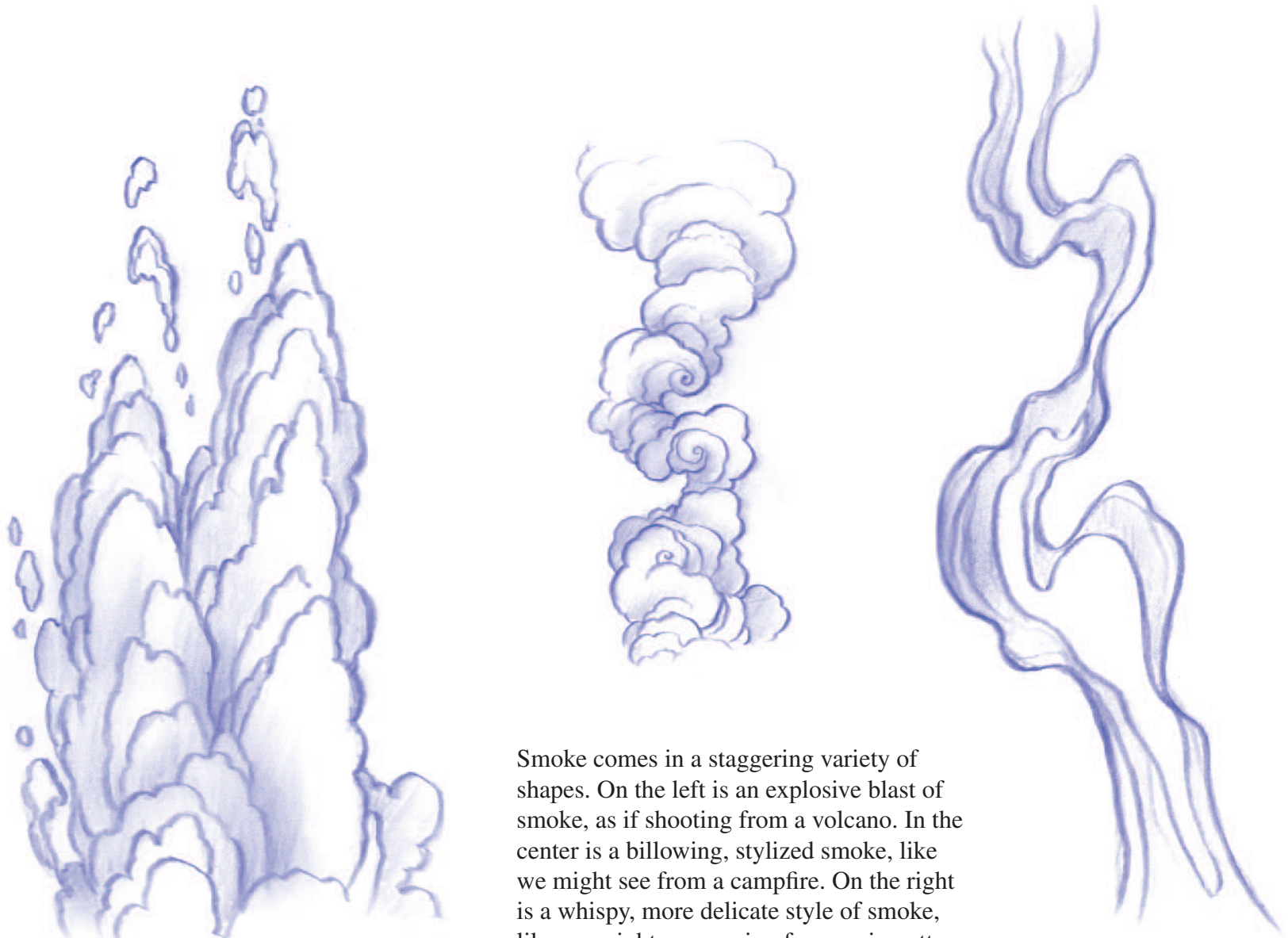
This sequence is very similar to the smoke sequence on the preceding page. This time I have combined the upward flag-waving motion with the rolling ball, and worked it all into a fire design. This sequence would represent approximately 24 frames, or only one second of animation, so it is moving much faster than the smoke on the last example—as fire does.

Note that, as the large piece breaks off the main body of the fire, the shapes almost wrap themselves around the ball shape as they break off and then dissipate. In the more detailed smoke animation examples that follow, we will look at how the hot and cold air currents that collide around smoke and fire's perimeter shapes create additional vortexes and swirling eddies of energy that influence the velocity, shape, and the paths of action of the smoke. Starting out with simple, stylized, medium-size smoke and fire like this is a great way to understand the fundamental principles of animating energetic and convincing-looking smoke effects.

What we are seeing when we look at a plume of smoke above a flame is actually the shapes of moving air currents which are filled with smoke particles—the waste by-product of a material reaching its combustion point. So the temperature changes, air currents, availability of fuel in the air, and the nature and amount of the material being burned all conspire to create the particular fire and smoke that we're observing.

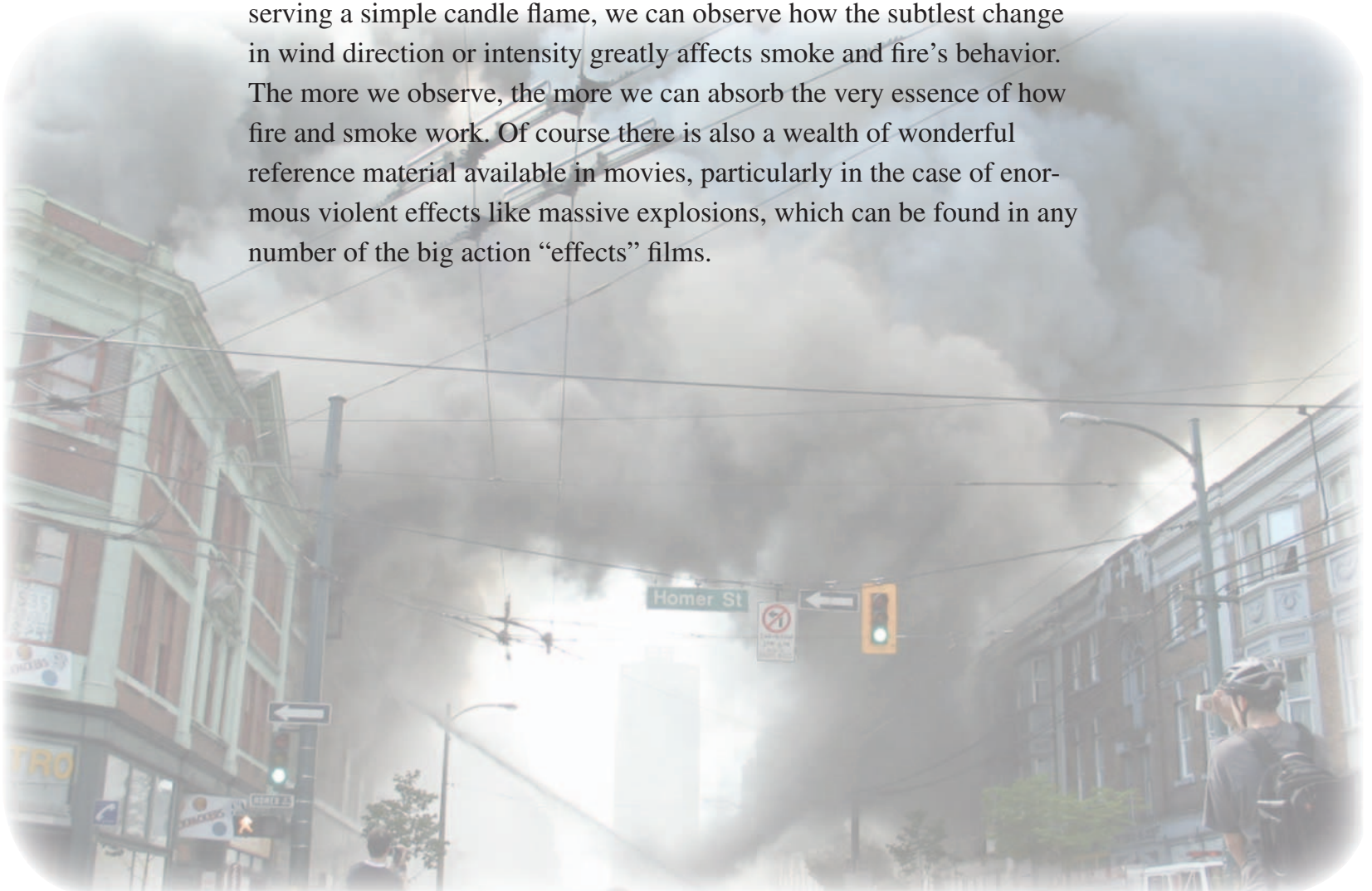
In the case of explosions, the central driving force is from a given fuel being very rapidly ignited, whether it is a nuclear bomb or a firecracker. This force is initially very straightforward, traveling directly outward from the fuel source, but as an explosion expands it becomes prone to secondary forces like wind, obstacles, or secondary explosions.

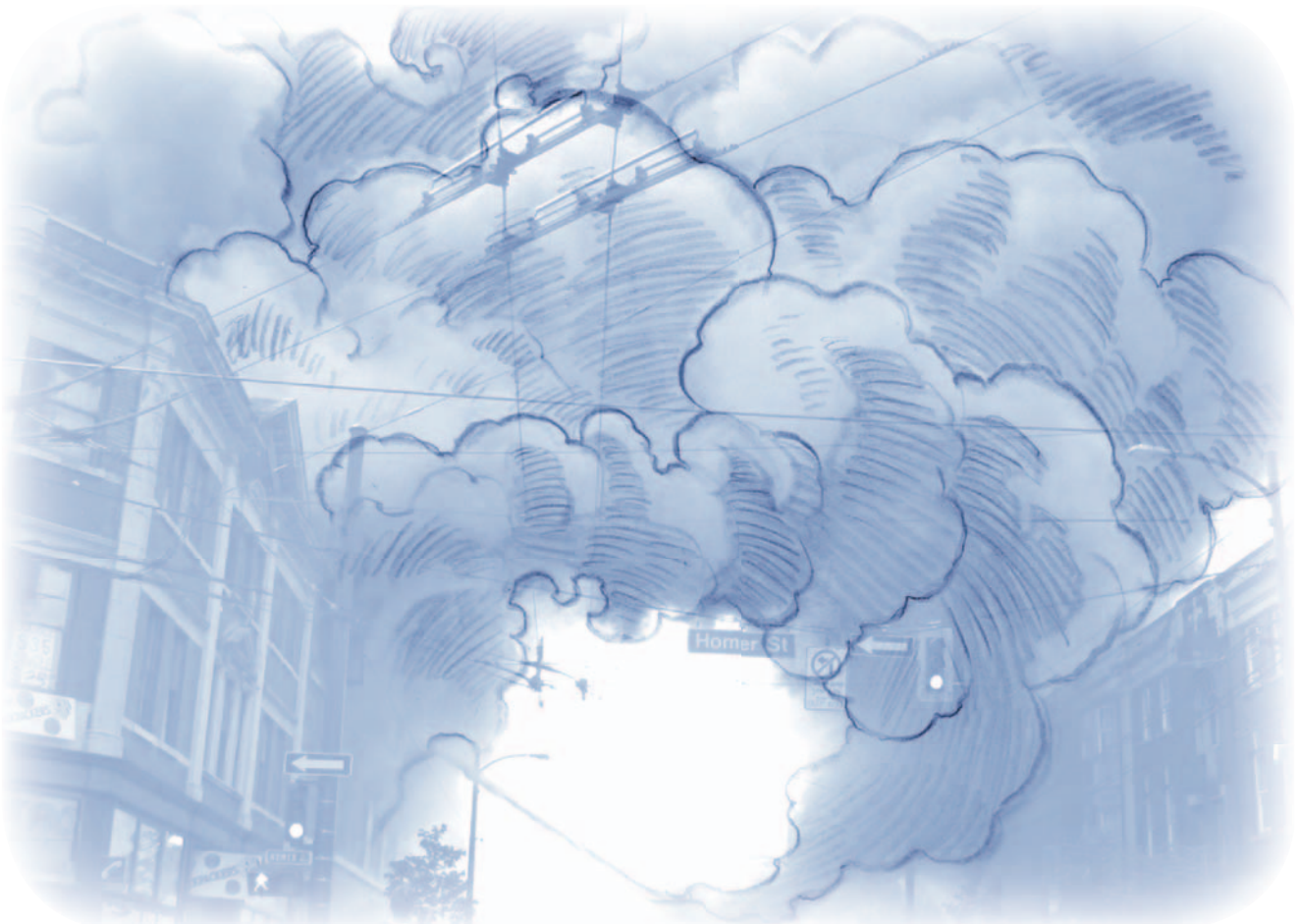




Smoke comes in a staggering variety of shapes. On the left is an explosive blast of smoke, as if shooting from a volcano. In the center is a billowing, stylized smoke, like we might see from a campfire. On the right is a whispy, more delicate style of smoke, like we might see coming from a cigarette or stick of incense.

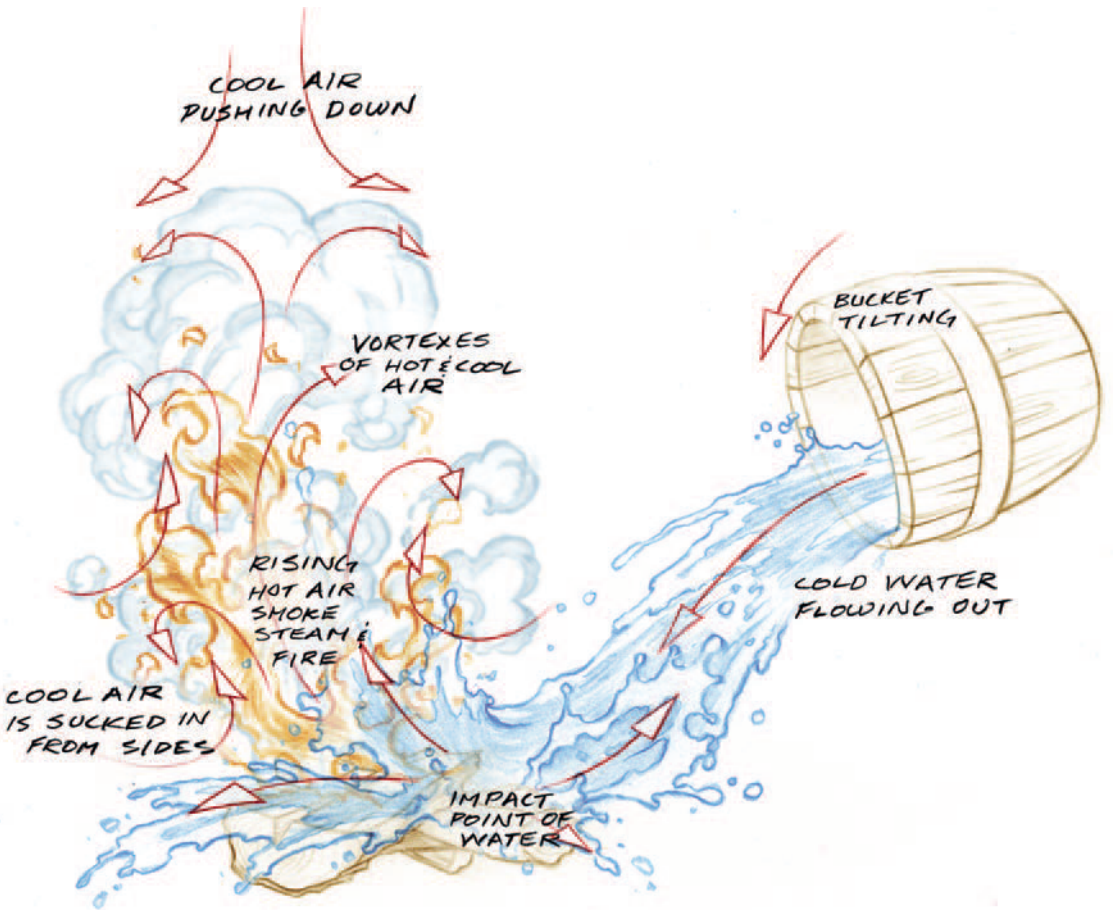
The best way to really understand how these forces move in time and space is, as with all natural phenomenon, through observation. Staring fascinated into a pile of burning leaves, watching a building burn, or observing a simple candle flame, we can observe how the subtlest change in wind direction or intensity greatly affects smoke and fire's behavior. The more we observe, the more we can absorb the very essence of how fire and smoke work. Of course there is also a wealth of wonderful reference material available in movies, particularly in the case of enormous violent effects like massive explosions, which can be found in any number of the big action "effects" films.



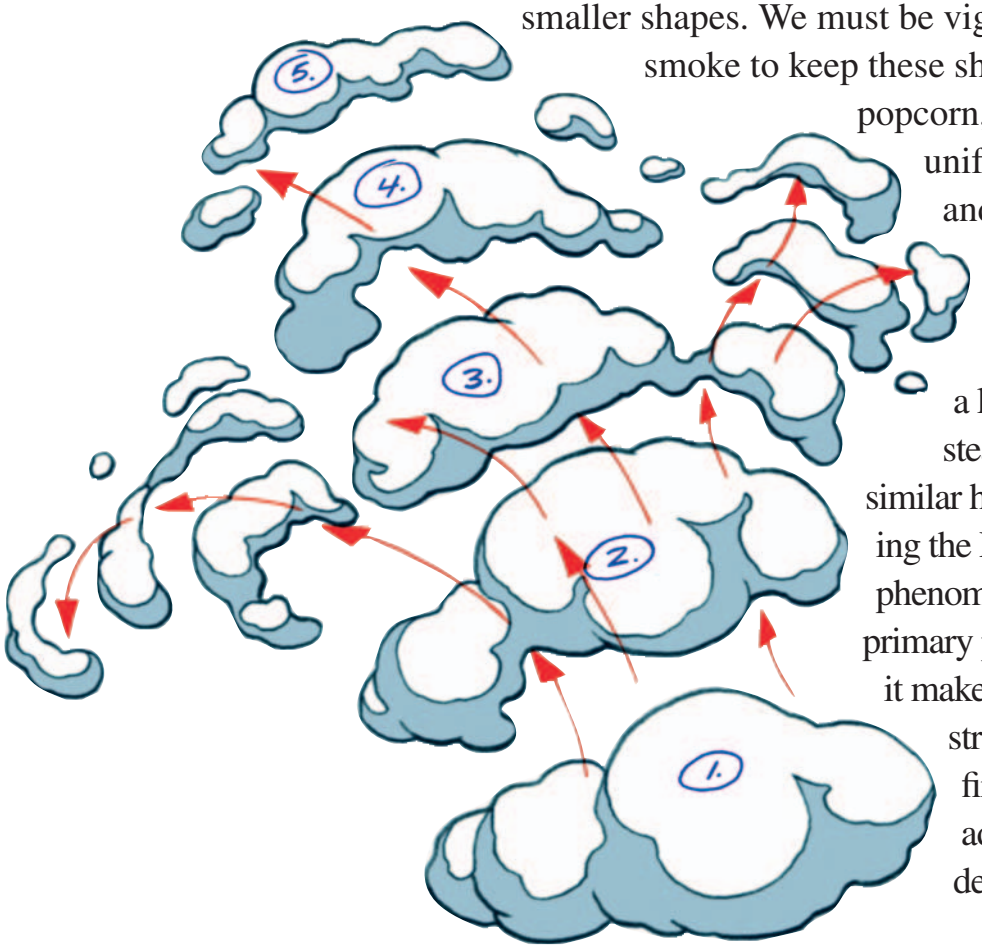


Anyone keeping his or her eyes open for great reference material will be rewarded with a wealth of fantastic stuff. With persistence, hard work, and lots of drawing, we can translate this reference material and our understanding of it, into beautiful special effects animation.

Steam technically falls into the category of water, but it behaves like smoke in many ways. As with the notion of turbulence and warp to indicate distortion, steam initially behaves the same way, eventually morphing into more discernible cloud-like shapes. If steam is being generated by a teakettle, there will be very little tonal variety within the silhouettes. However, once we get to the steam created by a nuclear power plant, it is virtually indiscernible from large-scale smoke (except the color is white). The steam generated from our mouths is a small-scale event, therefore the silhouettes will seem somewhat "flat." Unlike smoke, steam is water evaporating, which means that it can have the quality of "disappearing" as it resolves.

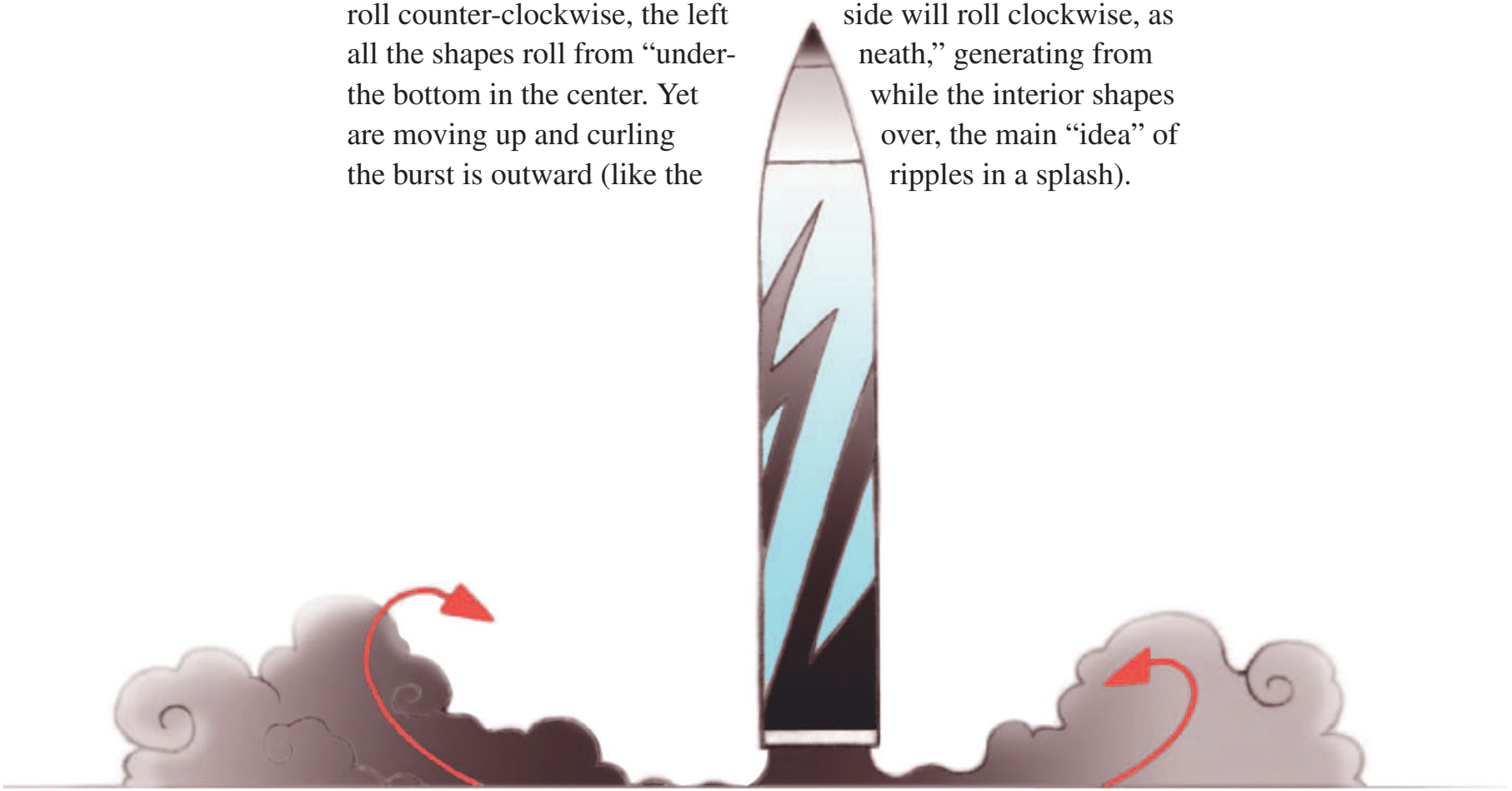


This illustration that I created in Chapter 3, could be smoke, could be steam, or both. In an actual scenario like this, there would probably be a great deal of smoke and steam at the same time. The steam would dissipate and disappear much faster than the smoke. The smoke would continue to rise and spread out into the air above. Often when driving in a rural area, you can see smoke from relatively small campfires or brush burn-offs, from miles away!

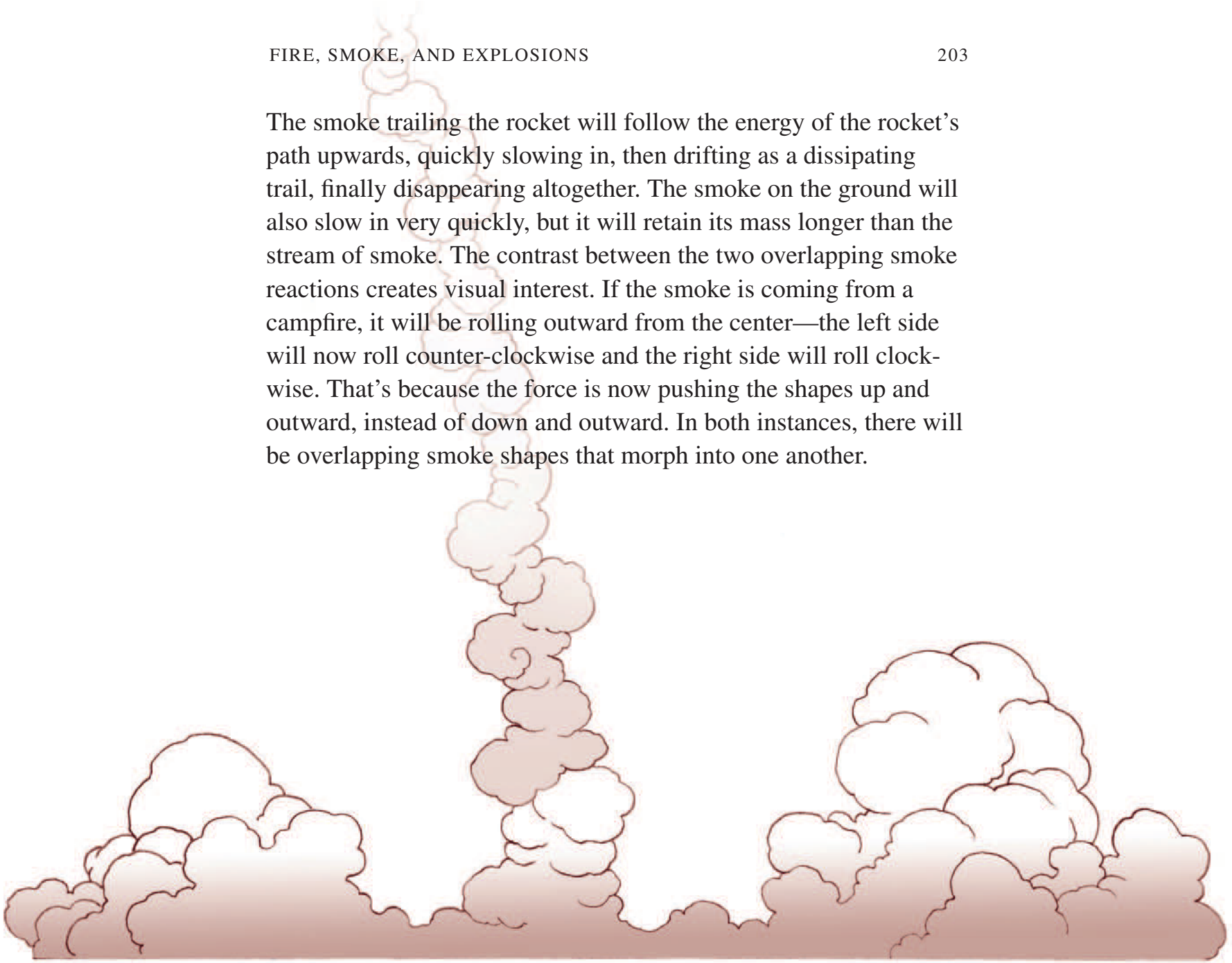


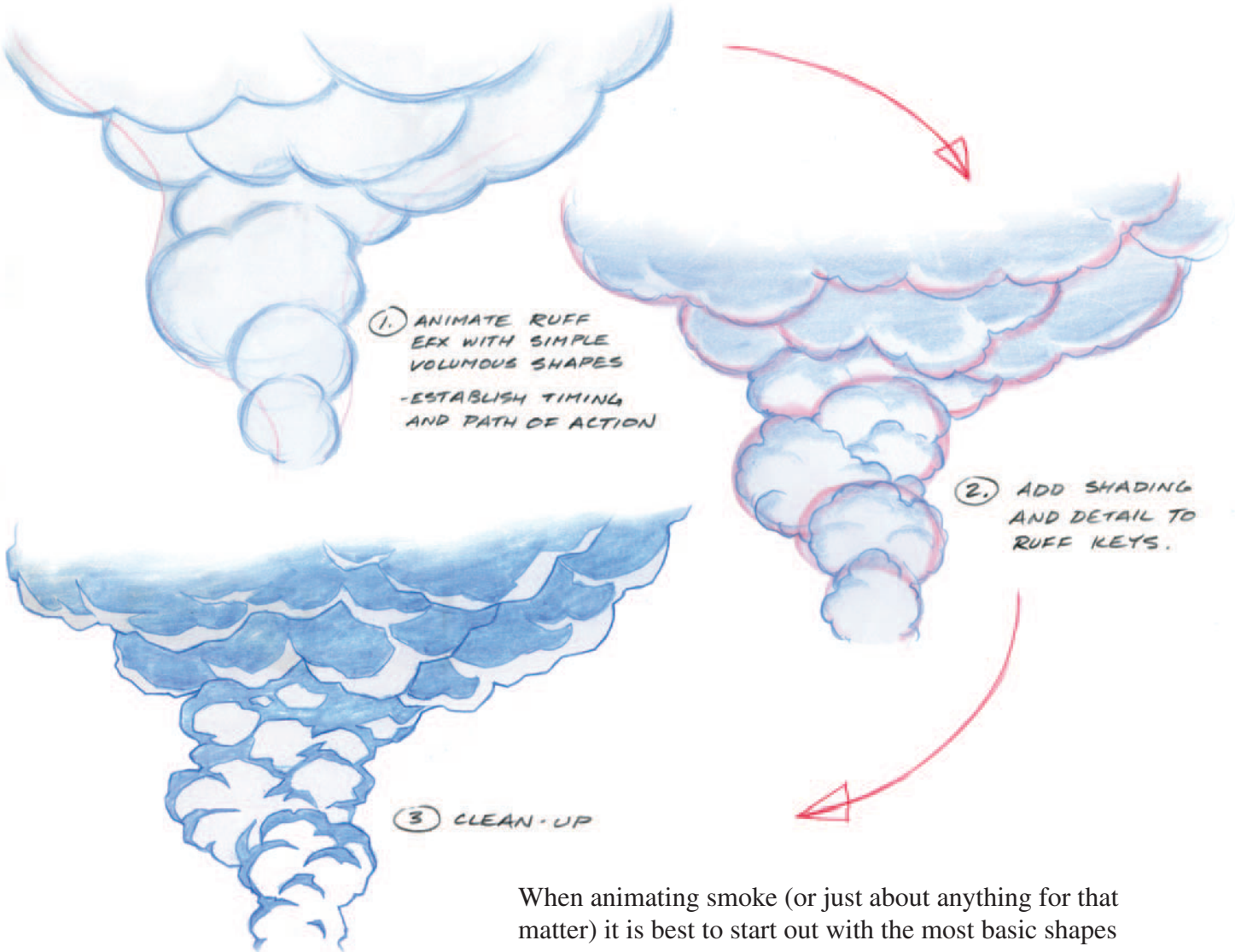
Smoke can dissipate also, but it is prone to breaking down into smaller shapes. We must be vigilant in small-scale smoke to keep these shapes from looking like popcorn, where the shapes are uniform in size, design, and spacing. Bilateral symmetry is the kiss of death in organic effects. If you can draw a line bisecting a path of steam or smoke into two similar halves, you are approaching the Rorschach inkblot test phenomenon. Try varying the primary path of action so that it makes the direction less straight-up-and-down. Like fire, pinching shapes can add more interest in the design of the silhouette.

In medium-size smoke, we should always be aware of the rolling direction, which is determined by the energy of the forces at play. As with fire, smoke rolls in clockwise and counter-clockwise directions, but it does so very specifically. For instance, if a model rocket is set off in your backyard, the energy is pushing the smoke down and out on both sides; this means the right side will roll counter-clockwise, the left side will roll clockwise, as all the shapes roll from "underneath," generating from the bottom in the center. Yet while the interior shapes are moving up and curling over, the main "idea" of the burst is outward (like the ripples in a splash).



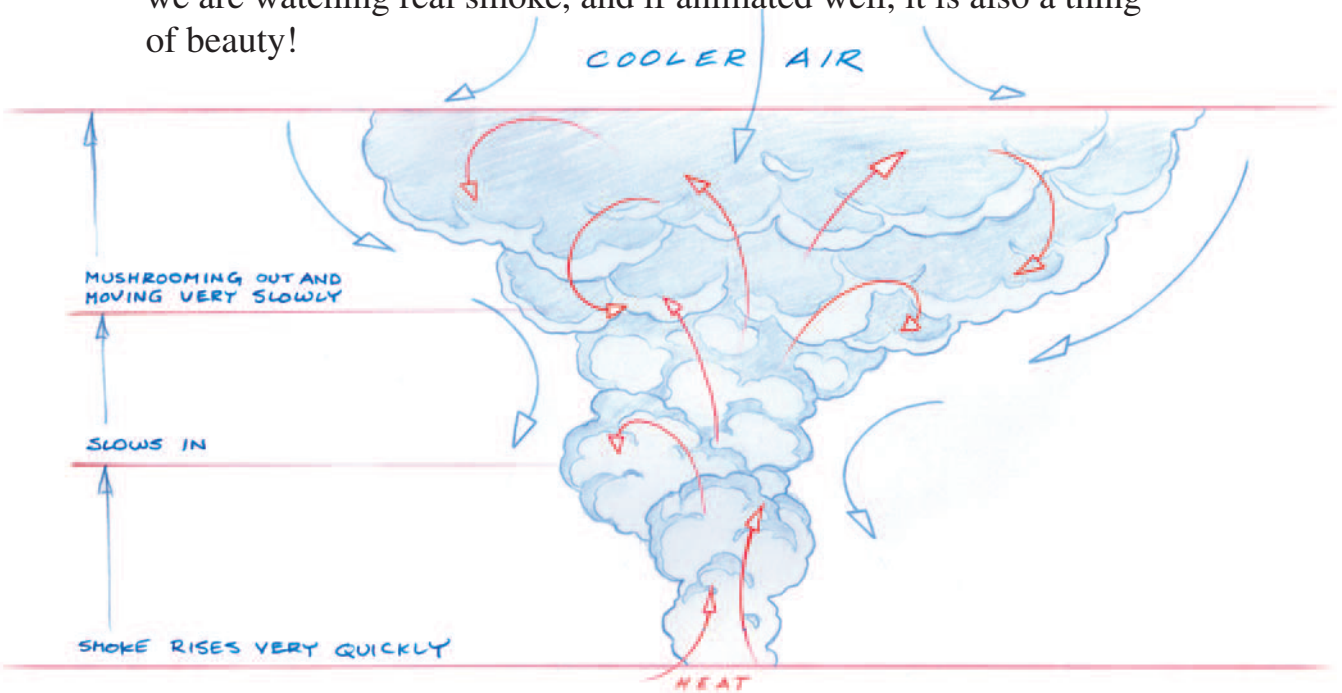
The smoke trailing the rocket will follow the energy of the rocket's path upwards, quickly slowing in, then drifting as a dissipating trail, finally disappearing altogether. The smoke on the ground will also slow in very quickly, but it will retain its mass longer than the stream of smoke. The contrast between the two overlapping smoke reactions creates visual interest. If the smoke is coming from a campfire, it will be rolling outward from the center—the left side will now roll counter-clockwise and the right side will roll clockwise. That's because the force is now pushing the shapes up and outward, instead of down and outward. In both instances, there will be overlapping smoke shapes that morph into one another.





When animating smoke (or just about anything for that matter) it is best to start out with the most basic shapes and get the action and timing working first, before adding detailed shapes and lighting. In the upper left-hand corner, you can see the subtle wave action lines in the drawing.

To animate smoke that is rising up from a fire, one needs to be very aware of the physics involved in order to give the animation a nice, believable energy. In this diagram, we see that initially the smoke rises up very fast, driven upward by the intense heat of its source fire. As the heat begins to subside, the smoke slows down, reacting to the cooler air it is running into. The warm and cool air colliding will cause the smoke to meander and twist slightly at this point. As it rises further, it spreads out, moving slower and slower. The cool air above forces the smoke back on itself, creating a slow rolling effect as the smoke being pushed down gets sucked back into the rising warm air once again. It is a beautiful movement to see when we are watching real smoke, and if animated well, it is also a thing of beauty!



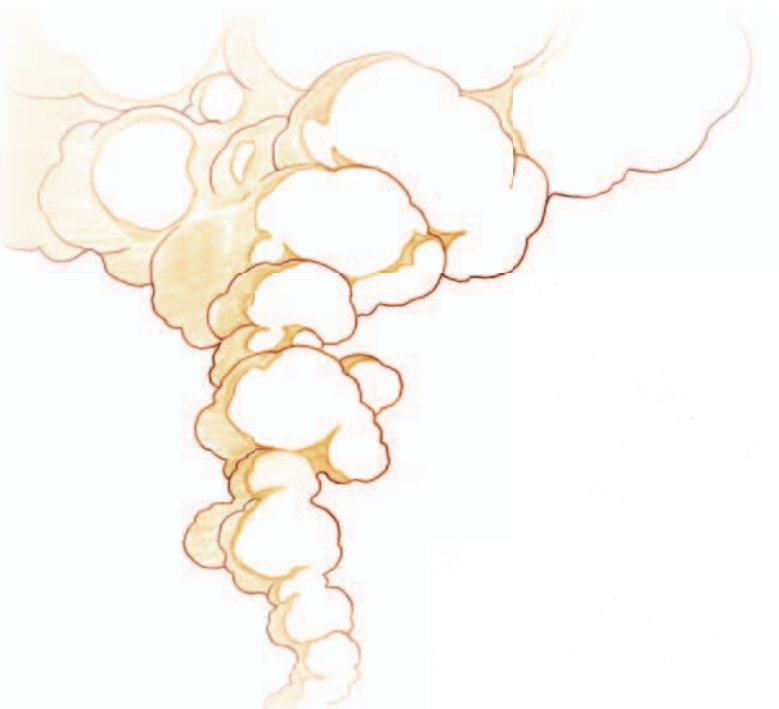
In Chapter 2, we spoke of "implied geometries," how a tone describes the internal structure of a lit object. Smoke is generally one-third shadow and two-thirds light. This tone will accurately delineate the interlocking spherical shapes and suggest how they fit together, even where there is no drawn line to connect them. Be careful not to make your smoke look like a bag of marbles—vary the shapes! Often, the bigger shapes will be the ones containing the most energy, and therefore, they will consume some of the smaller shapes as they roll upward and around.

The initial sketchy smoke drawing below shows the loose drawing style that is best to "sculpt" and construct smoke designs, well describing the internal masses.

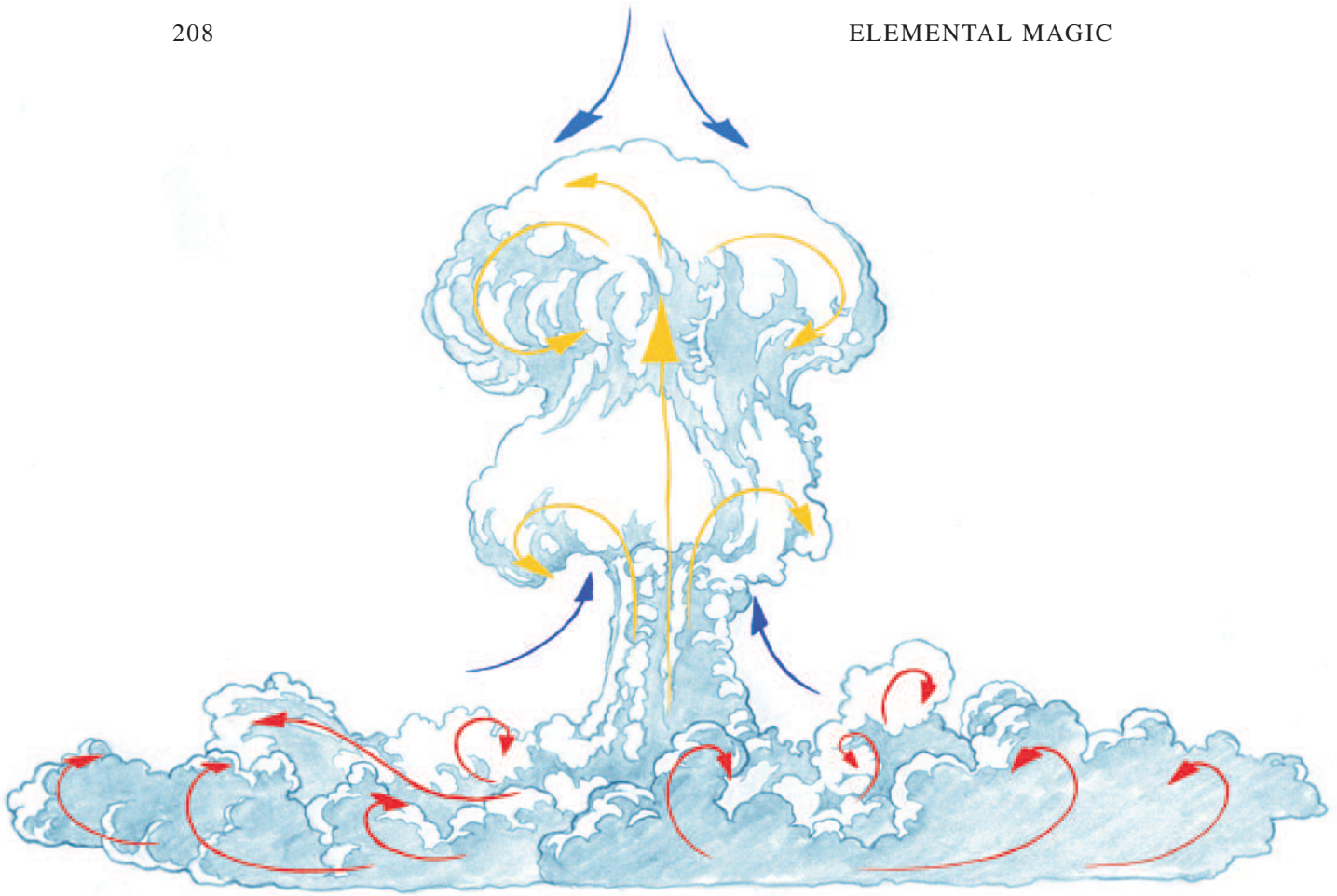
In this drawing, we see how I have applied a clean line over the initial rough drawing, keeping it simple, yet describing the most important shapes clearly defined.

In the drawing below, I have added shadows to the smoke as if it was lit from above. The shadow areas, or "tones" as we call them, do much to describe the underlying geometry of the smoke. 'Implied geometry' through lighting the form.

Be careful also, not to literally think of smoke as rising circles; there is an element of "fluid randomness" which must play out in the animation. Drawing with the sides of our pencils, we can define these internal masses quickly, without getting mired in details. The line quality of smoke is a series of "shaky" lines falling about a rounded edge, whereas fire has fluid lines accentuated by sharp points. Smoke is "married" to the forces generating it, whether its from a long-burning source or an explosive event. This means it will be affected by the same forces of hot and cool air, wind and gravity.



Here the smoke is lit from below, and the "tones" still do much to describe the inner implied geometry of the smoke. The final example on the right, is lit from the right, again, clearly describing the underlying shapes elegantly.

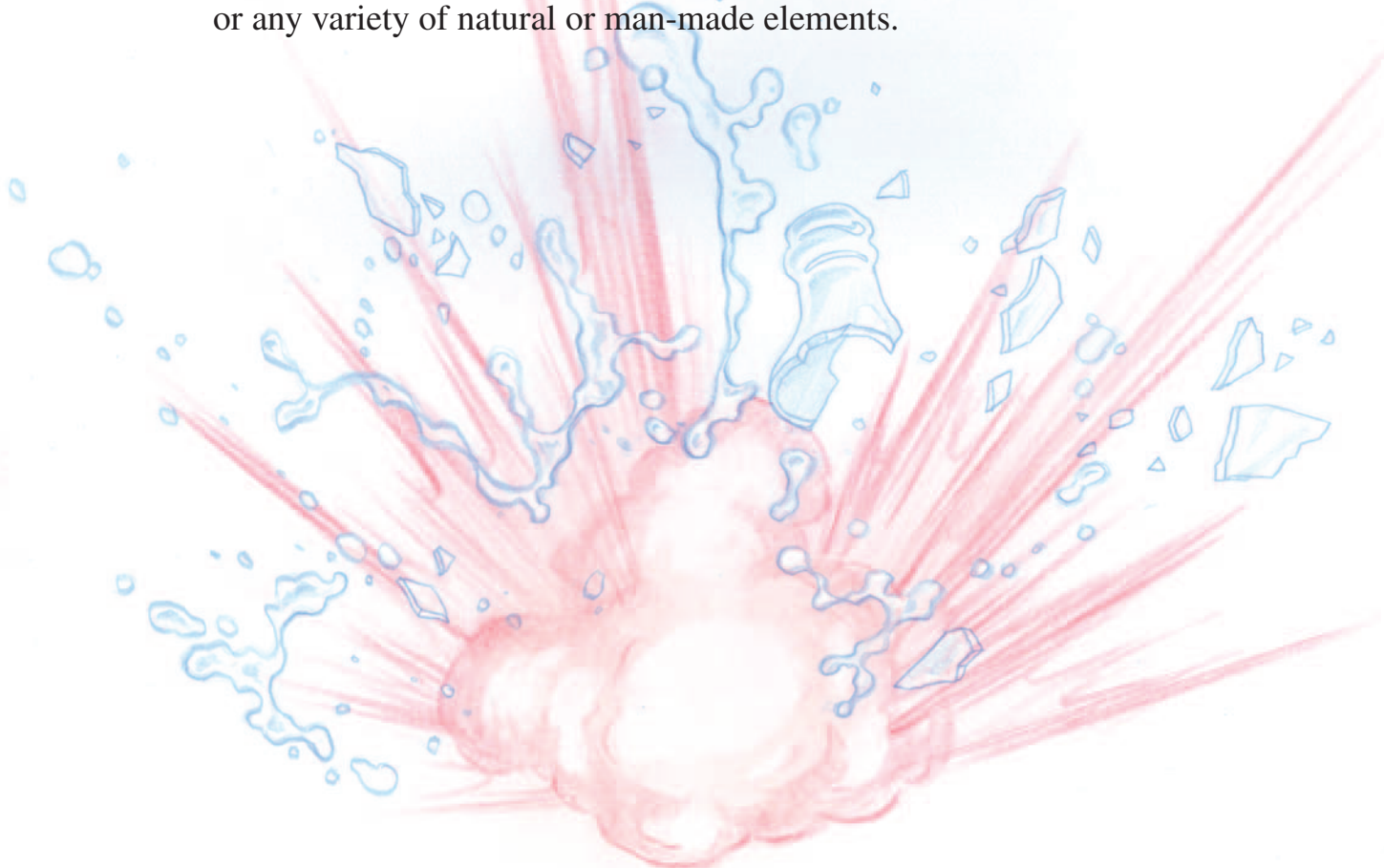


By the time we attempt a large-scale plume of smoke, we've already learned that slower timing plus extra detail equals a greater scale (and lots more drawings!). We can achieve this by adding more internal, churning and rolling shapes (with a finer articulation of tone as it describes them), as well as a more particular refining of the smaller "shaky" shapes as they wind around the rounded edges.

Let’s take our model rocket example and find a similar large-scale version. A nuclear blast comes to mind. Think about it—there is a strong directional force moving upward, which causes a reversed series of explosive plumes on the ground. The only difference is that a nuclear explosion then hits a ceiling of cooler air as it rises, which makes it roll out into that iconic mushroom shape. The internal rolling of such a massive blast is exceedingly violent, yet the movement is still extremely slow. A volcanic eruption is very similar, with enormous amounts of internal overlapping shapes rolling into themselves as the overall mass gains volume.

This is a series of drawings from an animation sequence of a fire-based explosion, superimposed over each other. This reveals the shapes of the underlying energy of the explosion, an almost flower-like, blossoming shape.
Elegant forms of pure energy.

And so we arrive at explosions, which, while being a necessary topic of discussion in this chapter, can fall into several different categories, as not all explosions have an igniting fire at their core. Explosions can be caused by any sort of pressure building up within a space that can no longer contain that pressure, such as steam pressure building up in a pressure cooker, or a river swollen with rainfall pressuring a dam. Explosions can be full of liquid, gases, or any variety of natural or man-made elements.

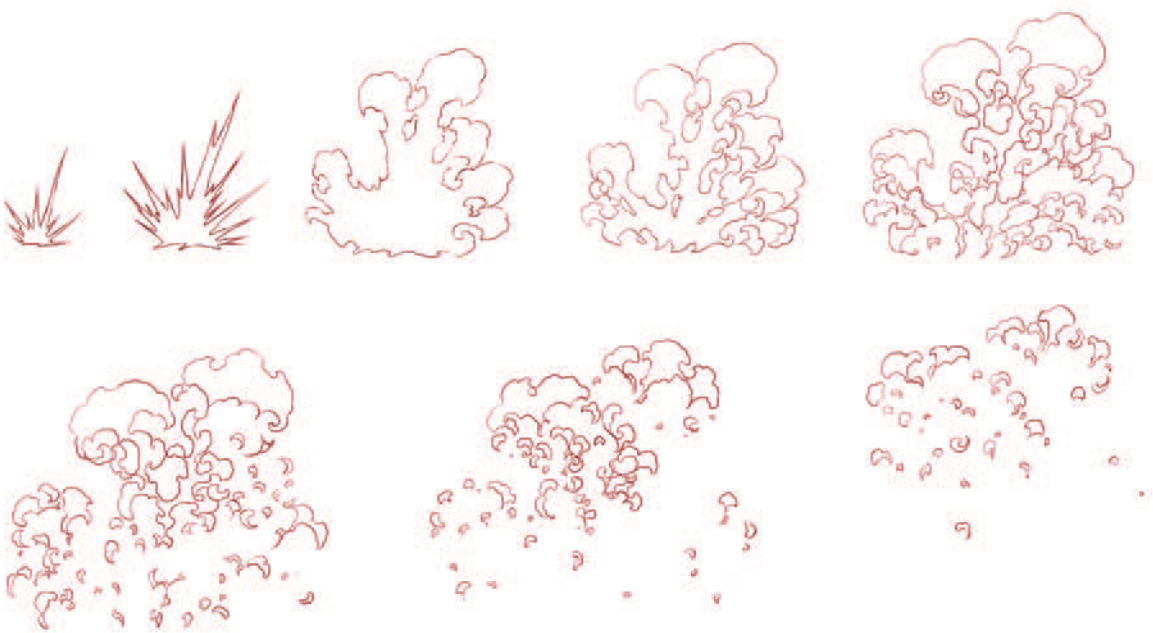


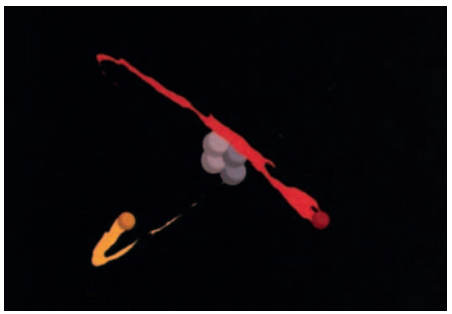
Or an explosion can be caused by an extremely abrupt and violent impact of two or more objects, like a comet colliding with the earth's surface, a car crashing into a telephone pole, or even something as simple as a glass bottle or a teacup, falling and exploding on impact with a tile floor. And of course there are all kinds of fantastic explosions caused by the imaginative devices of our science fiction and fantasy worlds that drive the animated special effects we see in films today.

Explosions come in an infinite variety of sizes, shapes and materials!

However the vast majority of explosions we see, are caused by heat and/or incendiary devices, and at their core contain vast amounts of energy capable of creating incredibly high temperatures, and subsequently, fires and a great deal of smoke as well.

Regardless of the cause behind them, the anatomy of all explosions is very similar. The initial action is an extremely abrupt outward burst of energy from the source of the explosion, which then collides with the resistance of its surroundings—be it air, water, or objects with which it collides. This slows the initial outward force of the explosion substantially, and then as the explosion continues to expand outwards it eventually slows down completely. Its shapes are completely affected only by the surrounding atmosphere, air or water currents, and to some degree the material that initially caused the explosion.





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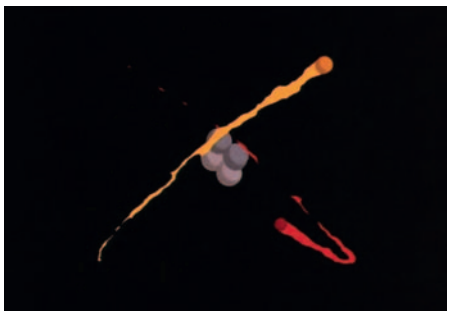

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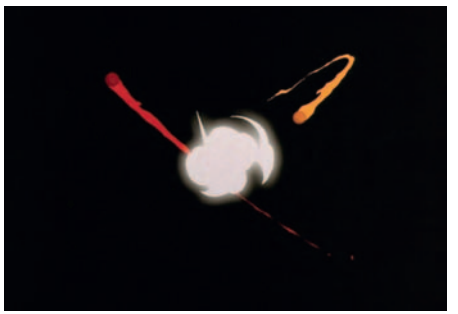

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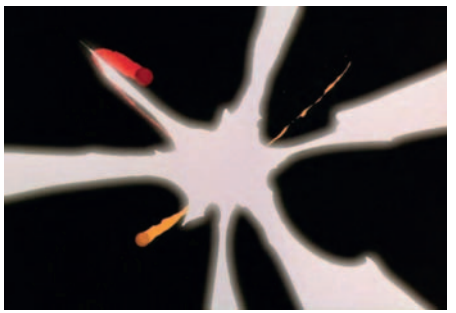

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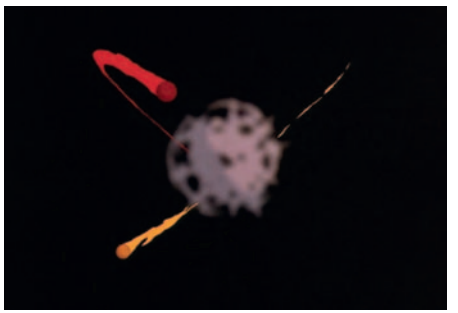

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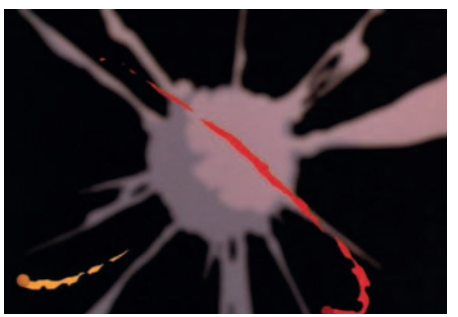
This absolutely fabulous explosion from Michel's Gagne's classic "Prelude to Eden" is a perfect example of Michel's unique design and approach to special effects animation. Starting with the spinning nucleus, this explosive birth of a small planet begins with a bright white central design, almost abstract, growing quickly as it shoots out random beams of light. There are only 6 frames of this initial animation, a quarter of a second, bringing us to frame 8 in this image sequence. Frame 8 is an anticipation frame, with no central light at all. Frames 9 to 16 of this sequence represent a full 150 frames of animation—over 6 seconds! This is a fantastic formula for dynamic explosions of all kinds. The initial explosion is extremely sudden, only 3 to 6 frames, and then the explosive shape forms in 3 to 6 frames. From that point on it settles and expands extremely slowly. This particular scene is a pretty subtle explosion, not really a large-scale cataclysmic event, but the timing dynamics could apply to any size of explosion.



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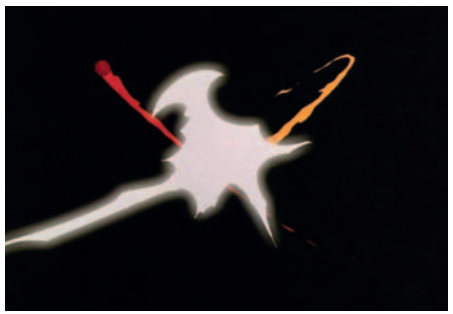

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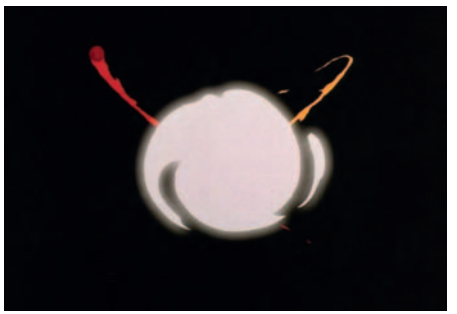

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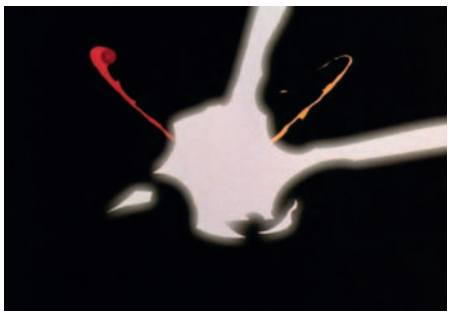

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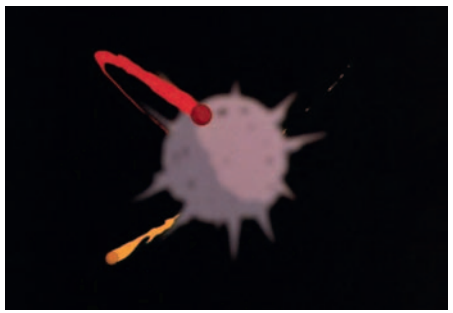

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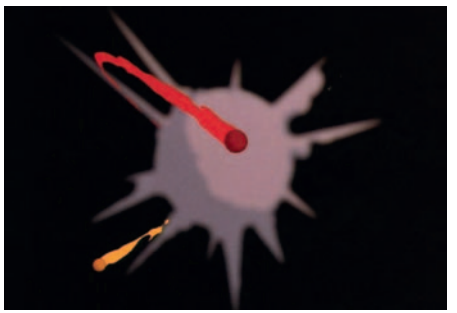

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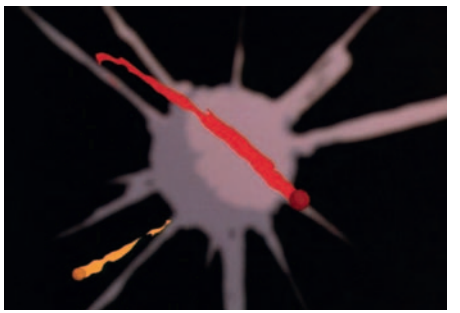

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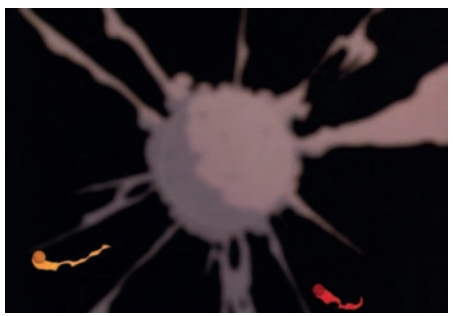

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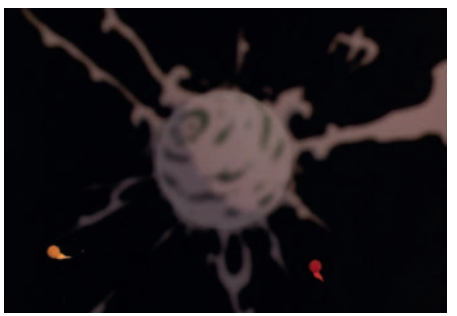

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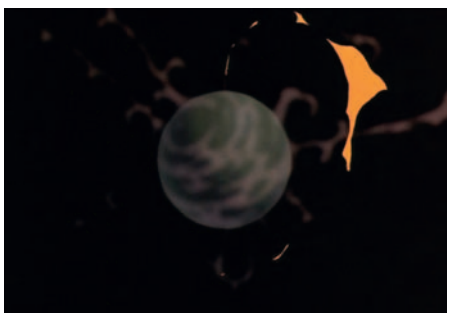

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Another explosion from Michel's Gagne's "Prelude to Eden"—this time a much more violent and dynamic style of explosion—with fantastic use of "flash" frames to shock and jar the viewer's eye. In frames 1 and 2 we see a bolt of energy enter from frame right. Frame 3 is the initial explosion, which expands extremely quickly through to frame 6. Michel has used alternating frames of off-white and bright red to create a strobing intensity in the scene. Then there are five frames of alternating red and white frames, which create a powerful sense of the explosion having filled the frame entirely, blasting our eyes with impact! From frames 12 through 16 we begin to see the edges of the explosion flickering and lessening in intensity. This carries on for about 8 frames past the final frame of this sequence. These frames are the real-time actual frames of this scene representing two-thirds of a second of animation screen time. A simple, yet extremely effective, animated explosion!

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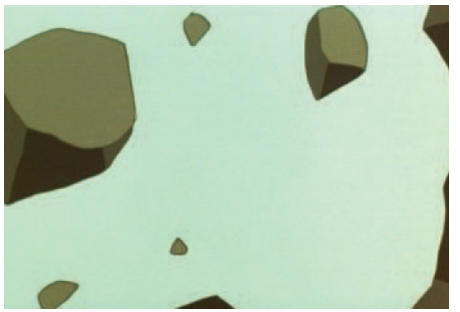

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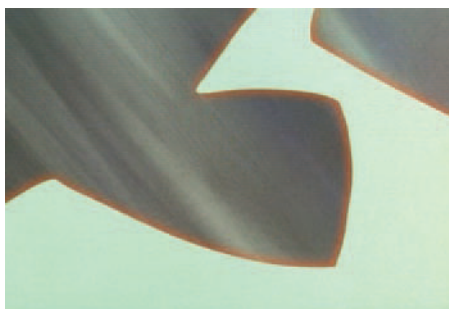

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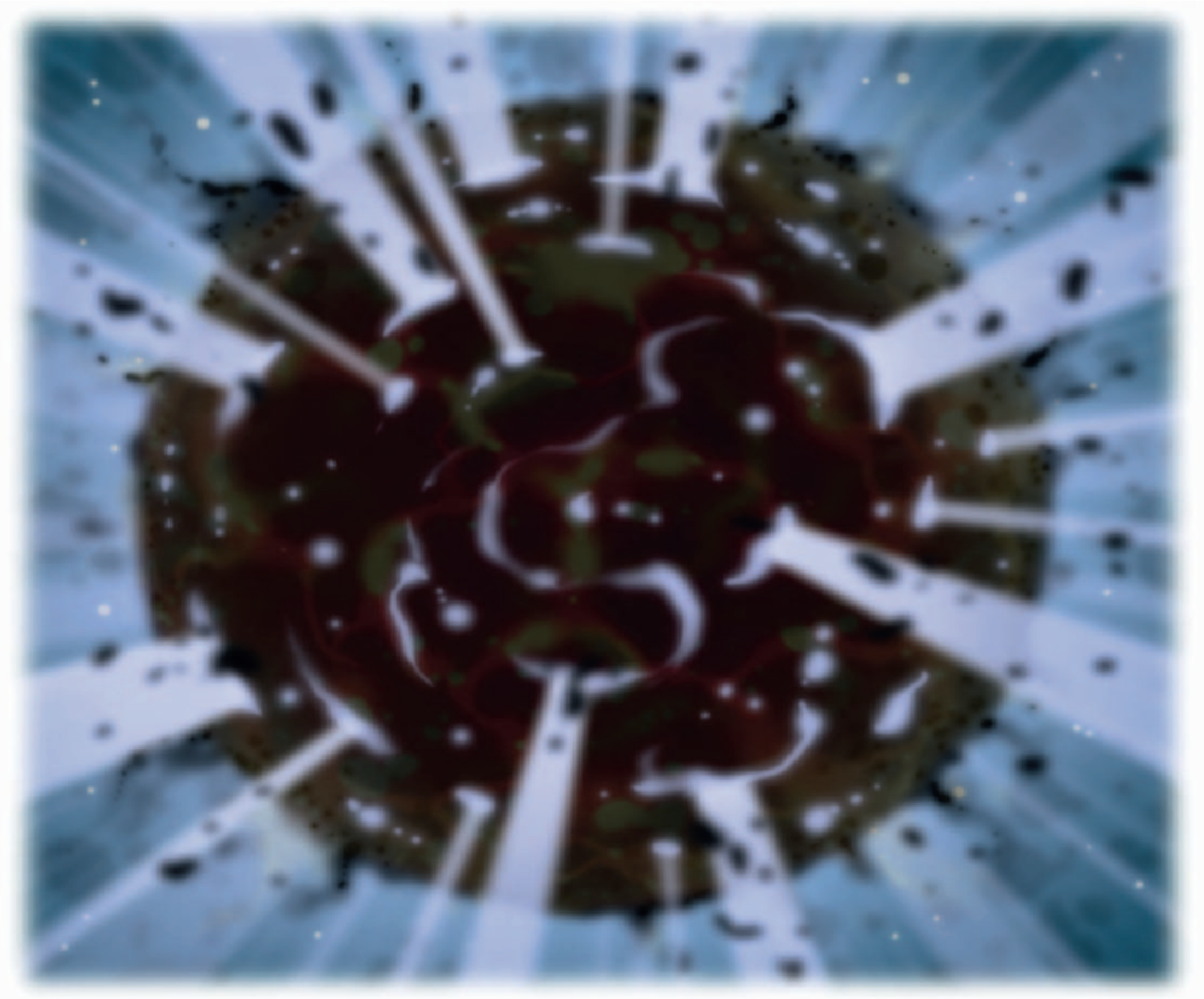


Turning Up the Digital Heat

Having looked at fire, smoke and explosions in a great deal of detail from hand-drawn point of view, what have we learned that can breathe new life into the fire, smoke and explosions that we animate digitally?

As I have explained in previous chapters, the most important things we need to consider when animating elemental effects using any technique, is what I call the *classical* animation principles of strong design and exaggerated dynamics. And I feel that nowhere are these principles more important than with fire and smoke.

When we approach fire digitally, our starting point is very often a preset or "canned" fire effect that has been created already by the software manufacturer, using particles or points in our virtual 3D space that are emitted or shot out of a particle generator. The generator, is usually a simple geometric volume that serves as the birth place for our particles. This particle-emitting software will have various forces of energy that we can bring into play to make the particles move. Sometimes they are called "fields," which is a physics term, but I like to think of them as forces of energy. There are things like wind, gravity, vortex, turbulence, and then there is usually a baffling array of newly invented terms—many that are unique to whatever software package you happen to be

using. There are no real standards of terminology from one software package to another, so we very often need to learn a whole new set of descriptions to understand the forces that we are working with. I would love to have an opportunity to plea with the brilliant folks who create these software interfaces, would it be too much to ask of you just to use simple English terminology, like, just speak to us in the same language that we all speak to each other in? That would make some of these software packages learning curves a lot less steep!

But let's move on. The computer-generated fire that we usually end up seeing is usually moving quite simply straight upward, with some added back and forth movement, or turbulence, to give the fire that twisting and curling look created by hot and cool air that I described earlier in this chapter. A lot of computer-generated fire looks pretty darned good, and there is no mistaking what we are supposed to be looking at. This is helped of course by its color. When we see yellow and orange flame-shaped objects shooting upward, it tells our brain "fire" pretty easily.

Some of the computer-generated fire I have seen is downright amazing to look at. With things called "expressions," a kind of code that can introduce a variety of variable mathematical equations to our animations—computer programmers are able to add wildly variable dynamics to any given effects simulation, adding chaotic and unpredictable turbulent forces that whip and curl the fire effects in a pleasantly random and realistic way. In

may cases, the fire animation that we get right out of the box, using software packages like Maya, Particle Illusion, XSI, or RealFlow, looks good enough to simply add to a scene and leave it at that.

However, if we bring to the table our knowledge of how classical animation actually improves on nature—by exaggerating and pushing the dynamics both in the design and the timing of our work, we can take our fire animation to a much higher level and make it stand apart from the standard, out-of-the-box computer-generated fire that we generally see in CGI cartoons today.

Once again, dare I emphasize that if our environments and our characters are designed imaginatively, why not our special effects elements? Of all the elements, I find that fire truly lends itself to being treated like a character. It is an observation that all of us may have had as children while watching a campfire that fire does indeed seem to have a life of its own. It crawls, creeps and devours, leaping and shooting, curling and flipping around, seeming to taste things and caress them as it advances and grows. Fire drives our imagination powerfully, as an elemental force, wild, dangerous, unfathomable in its mysterious nature. It consumes things and spits them out as an empty, spent, and lifeless shell. So if we are animating fire, we have an opportunity to tap into that mesmerizing living energy. We can breathe even more life into it, by going back to the classical animation principles we have learned, and injecting that energy into the mathematical equations of our digital fire.

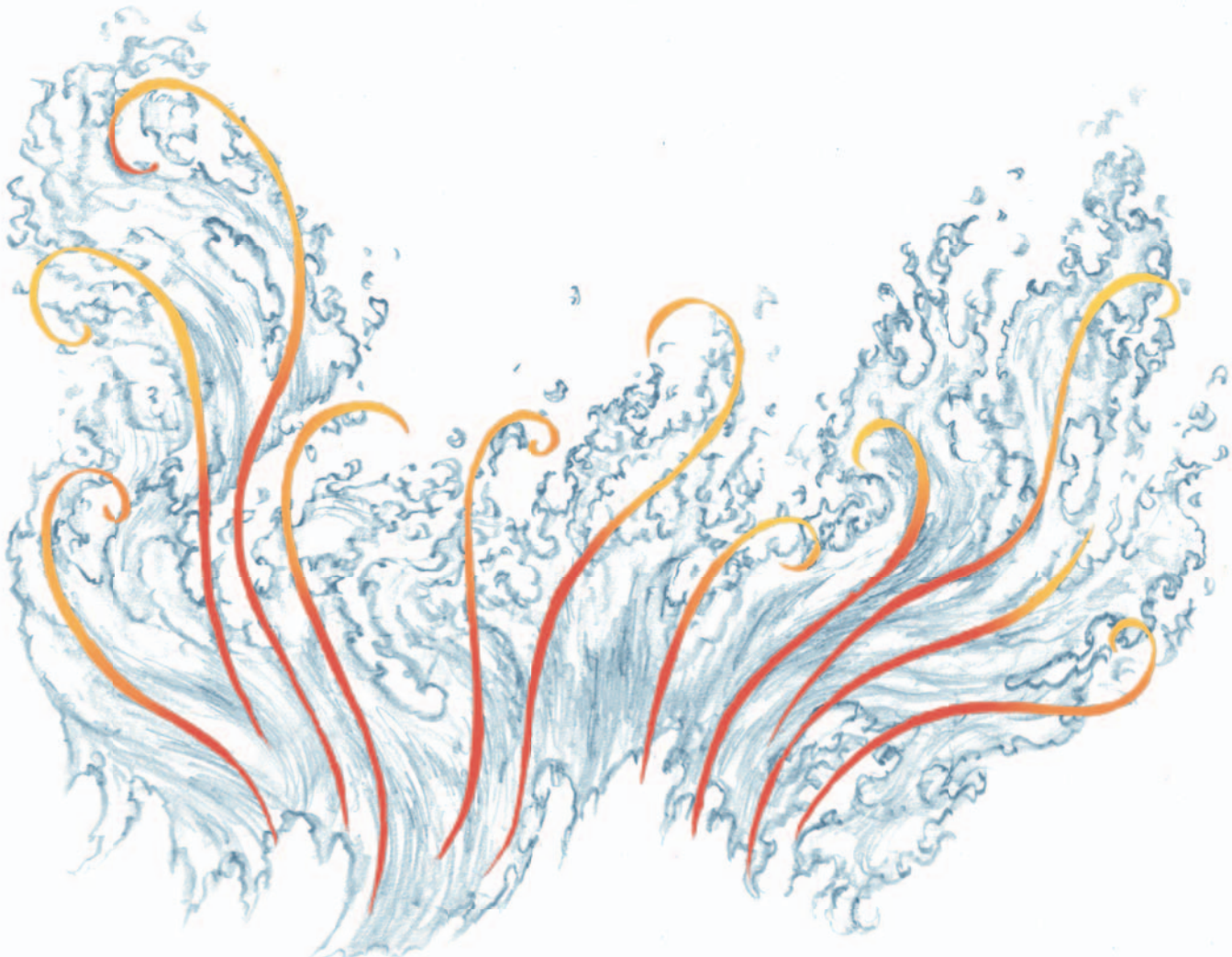
From a technical standpoint, that can be easier said than done, most particle-generating computer programs are wickedly complex affairs, and learning how to tweak the inner workings of a particle simulation can be daunting. But there are easier tools to use as well. Simply by adding more dynamic key frames and a more dynamic scale to our fire animation, begin to add to its dynamic appeal. Exaggeration is a key principle to remember and turn to every time we animate. Always exaggerate your animation, unless the scene specifically calls for a very subtle and understated effect. It helps to understand that exaggerated images or poses in animation may be only a frame long and are "felt" more than "seen." They serve to help the viewer feel the energy in the effect. The exaggerated image is visible for too short a time for the viewer's mind to isolate and concentrate on it, but as part of the stream of images that make up the full event, they add a great deal of dynamic energy to the overall animated effect.

When a fire ignites, push the size of its shapes as it mushrooms outward, scale it way up and then back down again. In the case of a character holding a burning torch, vary the volume and size of the flame as it is moved around. A torch being moved sometimes will create a burst of energy to the fire as it gets more oxygen, and then it will simmer down as the action slows. There are countless ways to improve on any fire effects animation, simply by adding to its dynamic interface.

Smoke also can benefit enormously from our understanding of dynamic design and animation. All too often I have seen 3D computer-generated smoke animation that is extremely uniform and uninteresting looking even in very big budget live-action films. It seems that as long as it sort of *looks* like smoke, nobody seems to care if it doesn't really *act* like smoke. Long columns of smoke rising in the distance after a battle has been waged frequently look like perfectly symmetrical columns rising upwards, as if there is no hot and cool air to create more interesting shapes as would invariably happen in reality. And as animators, we need to exaggerate even reality and push the design and movement of our smoke animation even farther.

I have been warned repeatedly as an author to avoid repetition as much as possible. Good advice for animators too! But repeating my message of exaggerated dynamics, is a cornerstone of this book. I don't think it can be over-emphasized when it comes to animation. Do not let the computer do your thinking or animating for you! If you do, the results will be mediocre, lifeless and uninspired. Bring your imagination, and your knowledge of what makes great animation great, to every element that you create.

Fire and its smoke is wildly exciting, incredibly dynamic and unpredictable stuff. And if if we inject the power of our imaginations into it, it can be *even more* exciting for our audiences to watch!



Chapter 6

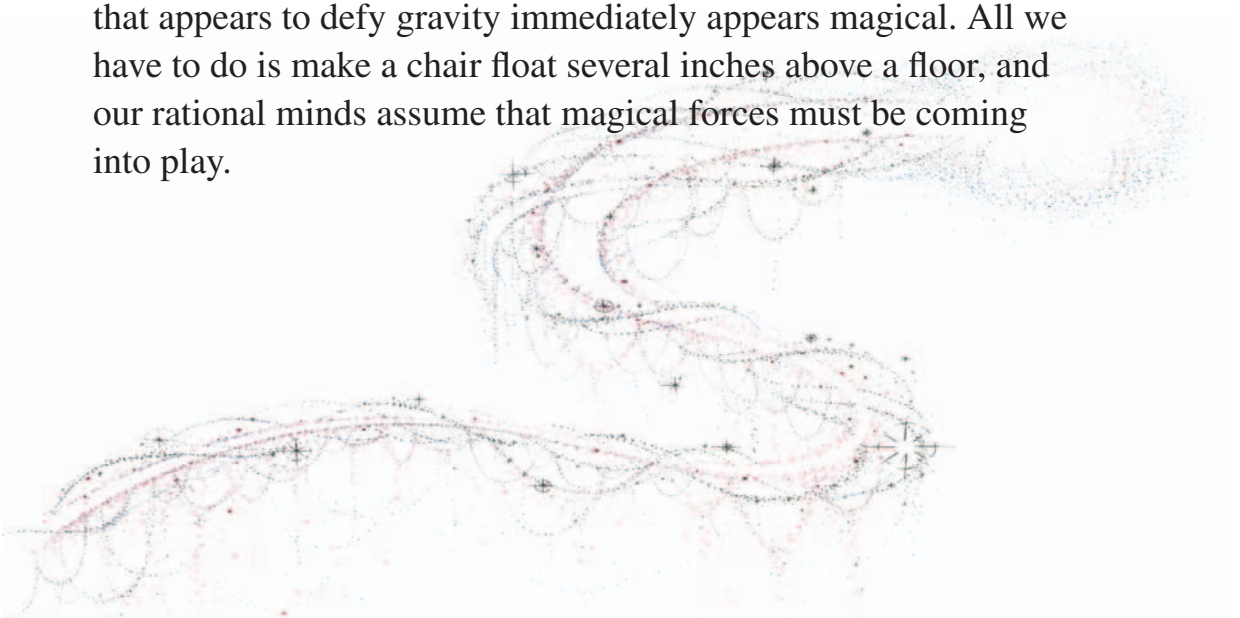
Magic

Often when I try to describe special effects animation to my friends who are not in the animation business, I will mention “pixie dust” as one of the common effects elements that we deal with. This inevitably brings oohs and aahs of familiarity, enchantment and curiosity. Humankind has always had a love affair with magic and it seems to transcend the humdrum routines that we’ve mostly come to accept as reality. In the world of special effects animation, it is no different. I have yet to know an animator who doesn’t feel a wonderful creative release when working on “magic” effects.

With magic we are able to shed the rules of physics that govern most effects animation, and invent our own set of physics or lack thereof. Magic sets us free on the page. Free to defy gravity and transform objects at will.

We are free to conjure up visions from our subconscious, free to break out from the boundaries of what is possible in our mundane day-to-day life! Magic is by far the most open-ended category of special effects animation. An entire book could easily be written about this fascinating artform.

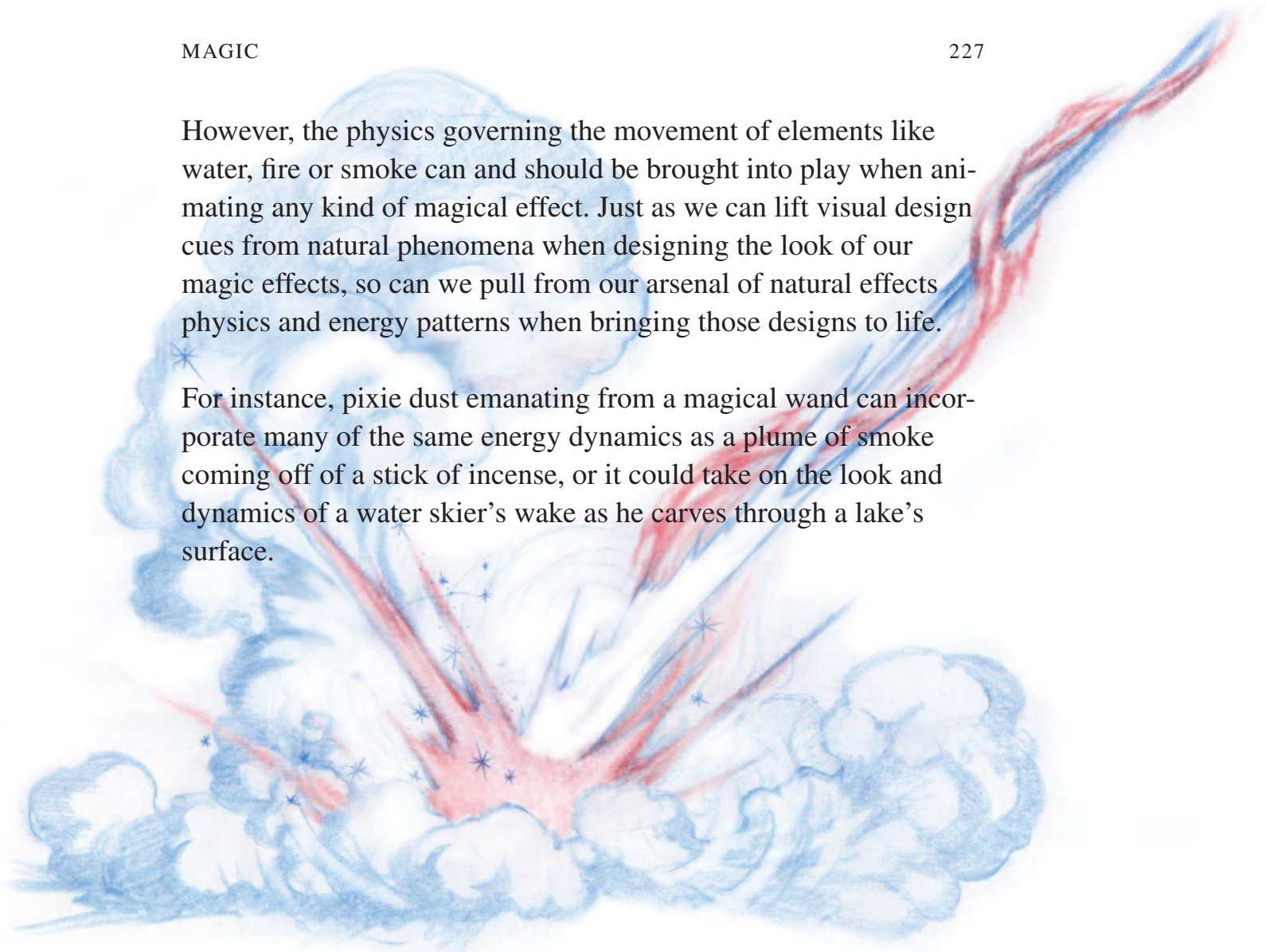
Sometimes what helps something to appear magical is the simple fact that it does not follow the same set of natural principles that govern the physics of our material world. For instance, any object that appears to defy gravity immediately appears magical. All we have to do is make a chair float several inches above a floor, and our rational minds assume that magical forces must be coming into play.



This is the same drawing from the preceding page. Here it is right side up, as it was originally drawn. I flipped it upside down on the previous page to illustrate just how well it can work, to break the rules, and defy all natural physics, when we are creating magical special effects.

However, the physics governing the movement of elements like water, fire or smoke can and should be brought into play when animating any kind of magical effect. Just as we can lift visual design cues from natural phenomena when designing the look of our magic effects, so can we pull from our arsenal of natural effects physics and energy patterns when bringing those designs to life.

For instance, pixie dust emanating from a magical wand can incorporate many of the same energy dynamics as a plume of smoke coming off of a stick of incense, or it could take on the look and dynamics of a water skier's wake as he carves through a lake's surface.

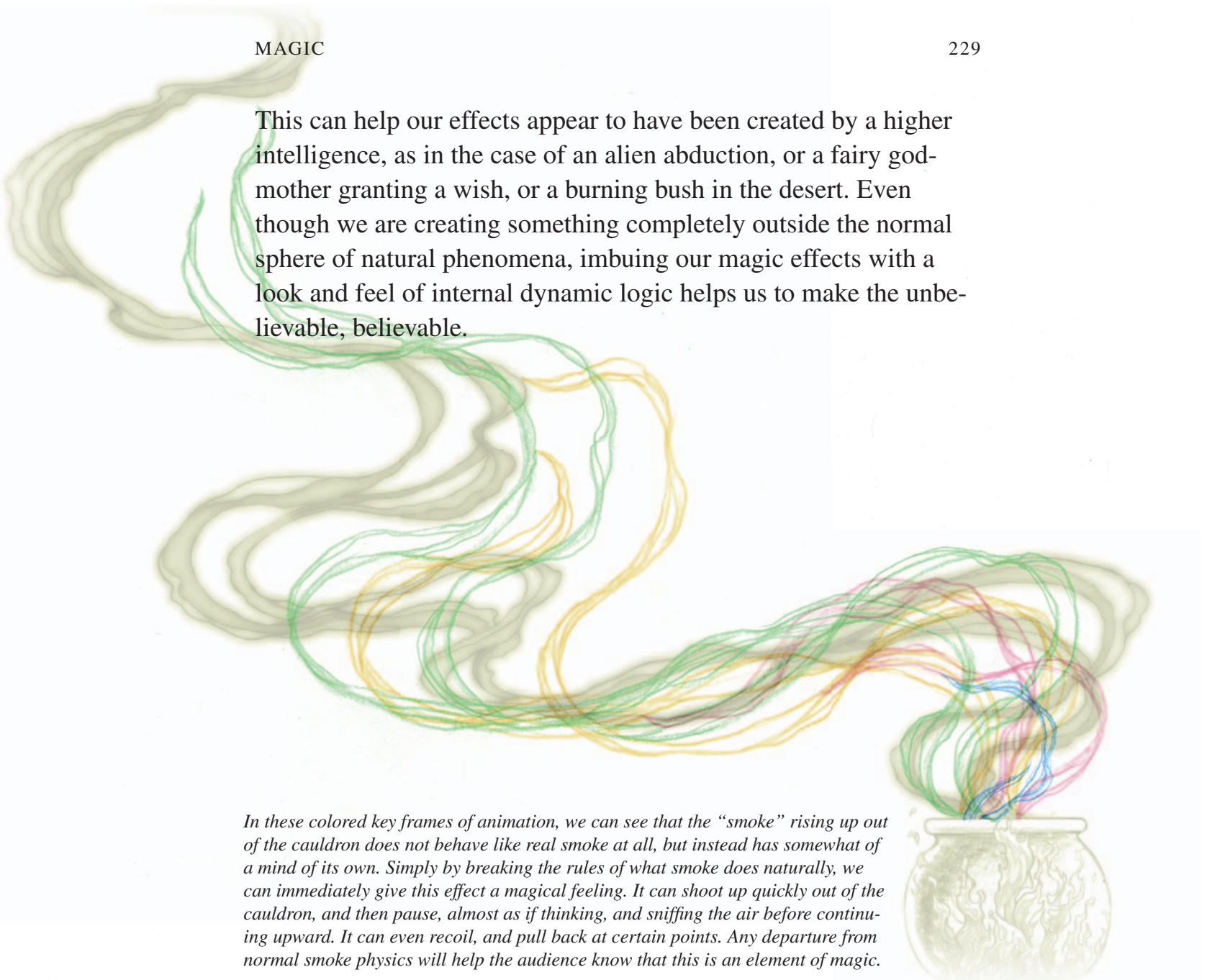


Where magic is involved, a director frequently won't really know what he or she is looking for until they see it! That is when a special effects artist with a good knowledge of elemental energy, and a loose drawing hand, can come up with something fresh to spark the director's imagination.

The difference in the dynamics is that when we are animating pixie dust, we can depart from the worldly physics of elements like smoke or water at any moment in our animation and create a magical dynamic of our own. This is of key importance when animating magic effects. To invent and describe our own set of dynamic rules, so that whatever the magical effect we create, no matter how bizarre or unusual, it will appear to have an internal logic of some kind governing its design and movement.

One of the best ways to proceed when animating something that you want to feel magical, is to start out with a solid design based on what you know about creating good organic special effects elements. Then, animate it in a completely different way, giving it a life of its own. I used this smoke elsewhere in the book to illustrate linear smoke design. But if we animate this smoke as if it was rising up like a snake out of a basket, pausing, hesitating, shooting forward and such, it would not feel like smoke at all, but more like a self-willed spirit rising up out of the cauldron.

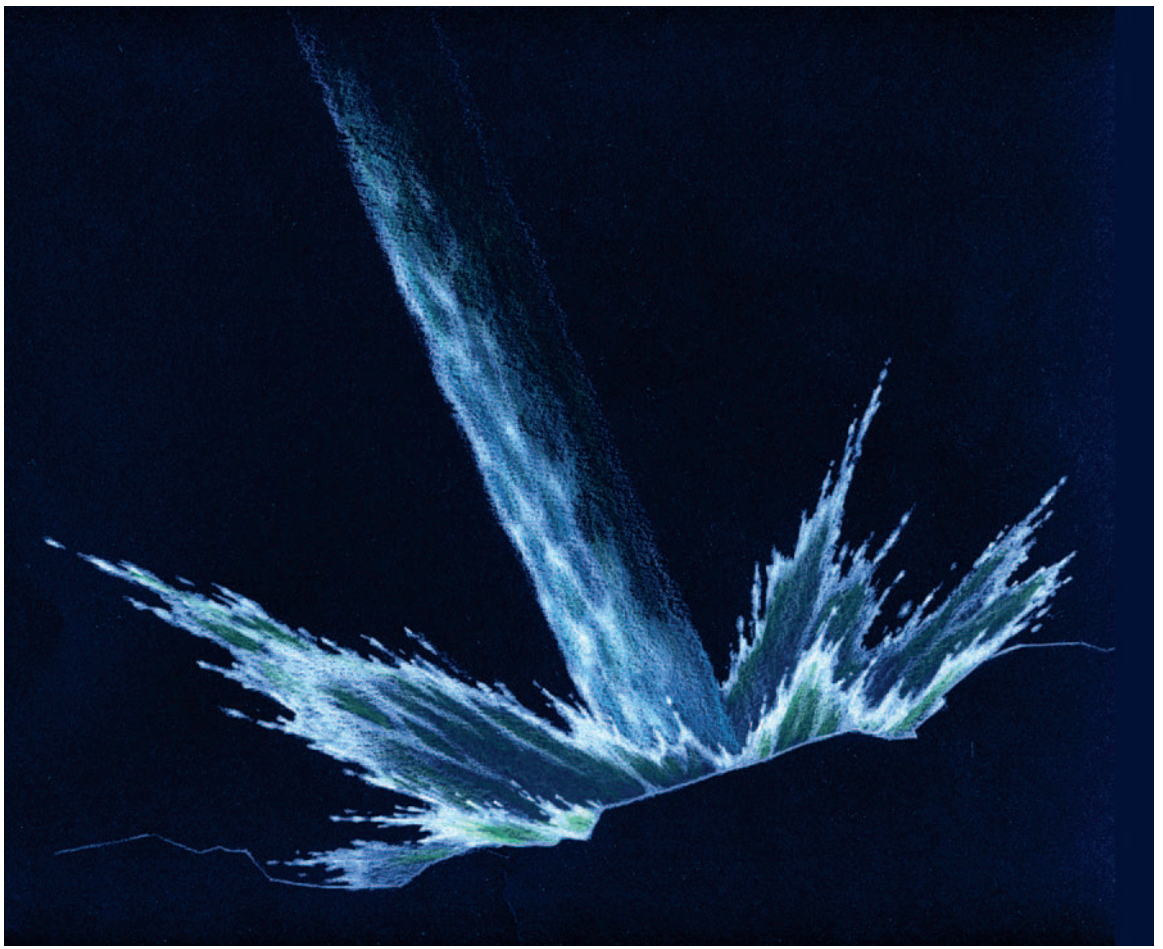
This can help our effects appear to have been created by a higher intelligence, as in the case of an alien abduction, or a fairy godmother granting a wish, or a burning bush in the desert. Even though we are creating something completely outside the normal sphere of natural phenomena, imbuing our magic effects with a look and feel of internal dynamic logic helps us to make the unbelievable, believable.



In these colored key frames of animation, we can see that the "smoke" rising up out of the cauldron does not behave like real smoke at all, but instead has somewhat of a mind of its own. Simply by breaking the rules of what smoke does naturally, we can immediately give this effect a magical feeling. It can shoot up quickly out of the cauldron, and then pause, almost as if thinking, and sniffing the air before continuing upward. It can even recoil, and pull back at certain points. Any departure from normal smoke physics will help the audience know that this is an element of magic.

Magic effects can be represented in an infinite variety of ways. Defining and categorizing them is impossible in a way, and its infinite nature discourages me every time I try. So I'll touch on the few definable principles that make magical effects animation look magical, and from there, perhaps this most ethereal and mysterious form of special effects animation elements will reveal itself to us.

Let's begin with the design aspects of magic effects. While the parameters may be less restricted than those of a naturally occurring physical phenomenon, we really need to keep in mind all of the design principles which apply to other more natural types of effects. Good design is good design after all. Watch out for the tendency to create repetitive or twinning shapes. Always avoid perfect symmetry (unless of course it is specifically called for by the director). Keep your silhouettes dynamic and interesting to look at, play inventively with positive and negative spaces, and always think of the core energy underlying your designs.



Here is a concept drawing that I did for directors Aaron Blaise and Bob Walker for Walt Disney Pictures feature film, "Brother Bear". The idea was that a beam of liquid light would shoot down from the sky and create a pool of magical liquid with a beam of light in the center, that the film's hero, Kenai, would then wade into. This was the beginning of the film's "transformation" sequence in which Kenai in transformed into a bear.

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Here we see frames from the transformation sequence that ended up in the film, very similar in look to the original concept drawing. The liquid tendrils in the first frame were created entirely in 3D CGI, and then as the beam became more intense and focused, as we see in the second frame, additional hand-drawn elements were added around the CGI element. Many tests and months of development went into making the CGI and hand-drawn effects integrate together to achieve the look of the original concept sketches.

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As the beam impacts the ground, the splashes that explode outward were entirely hand-drawn effects animation. This helped to keep the overall look of this magical effect very well integrated with the art direction of the film. It was felt that an entirely computer-generated magical element would have lacked the organic feel that we were able to achieve with hand-drawn effects. But no matter what technique is used for the final effect, great effects animation still depends on a fertile, well-informed imagination and a skilled drawing hand at the stage of the initial concept.

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Even when we are creating shapes that are intended to look repetitive, an effort should always be made to inject some variety into our shapes. We find that the shapes inherent in the natural phenomenon we study can adapt well to the world of magical elements.

A wizard conjuring up some sort of a spiritual entity from a bubbling cauldron magical brew may very well produce a shape that is very much liquid-like in its appearance, and might resemble a waterfall falling upwards! A common dynamic of many magical effects is a spiraling vortex of energy, much as we might see in a small whirlwind or a mighty tornado.

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In "The Sorcerer's Apprentice" segment of Walt Disney's classic Fantasia, the wizard conjures up a fantastic bat-like demon creature that then morphs into a beautiful butterfly. This is magical animation at its best, starting out as simple rising smoke, but then taking on a very magical life of its own.

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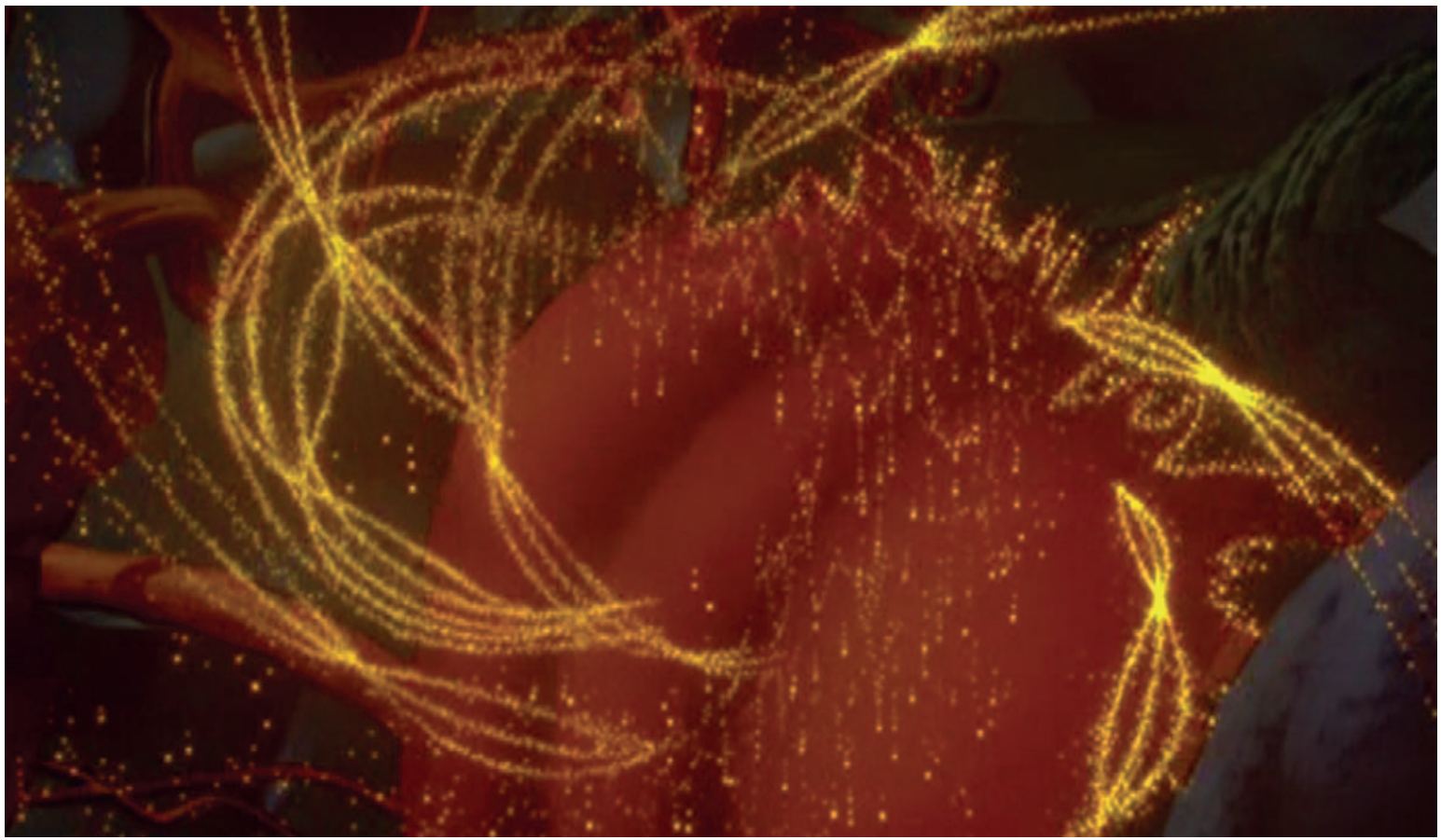
This energy vortex could very well be added to our magical wizard's apparition. This spinning vortex of liquid magical energy might appear to be filled with smoke or steam, or pixie dust behaving much like snowflakes. The possibilities are endless, fantastic, challenging and fun! We can imbue our effects designs with visual cues lifted from all the other natural effects elements, and add our own twists and turns wherever we please.

Attempting to imply scale and perspective in our magic effects can be extremely difficult because we don't have any of the normal reference points that we do when we are creating effects elements which commonly occur in real life. So we need to exaggerate our scale and perspective even more than we do with special effects in general, when we are animating magic.

We can approach this in a number of ways. In order for our magic effects to appear massive, we can add a great deal more detail, which can be an effective way of suggesting scale. We can give our animation even more scale by slowing it down a great deal, almost making it slow-motion.

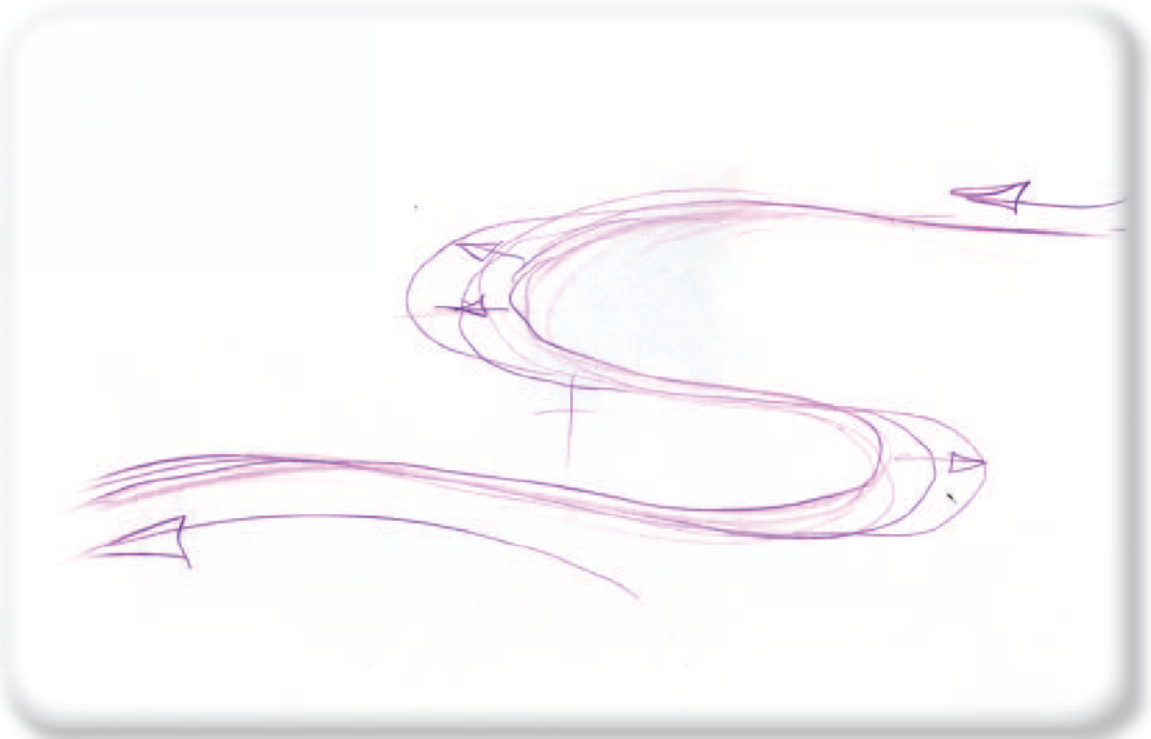
In cases where magic effects are moving in perspective, it is important to really exaggerate our drawings as much as possible, pulling closer effects out toward the viewer, and pushing and pinching into forced perspective the effects which are farther away. Don't forget, we can always push our drawings a little farther!

This pixie-dust sequence is from the Don Bluth film, "Thumbelina" that was released in 1994. As a directing special effects animator, I was given the task of animating a great deal of pixie-dust. At the time, CGI animation was still very much in its infancy, and creating pixie-dust like this using 3D particle emitters, as is frequently done today, was not yet an option. This scene in particular was extremely difficult and painstaking, as it was over 30 seconds long and all of it had to be drawn on one's, that is to say, 24 drawing per second, because the art-work was panning and would strobe disturbingly if not done this way. In the last couple of decades I have had the opportunity to work more and more often with computer-generated special effects, and in particular, on pixie-dust. When I was special effects director at the Walt Disney Animation Services department, one of my tasks there was to digitally create pixie-dust that looked exactly like classically animated

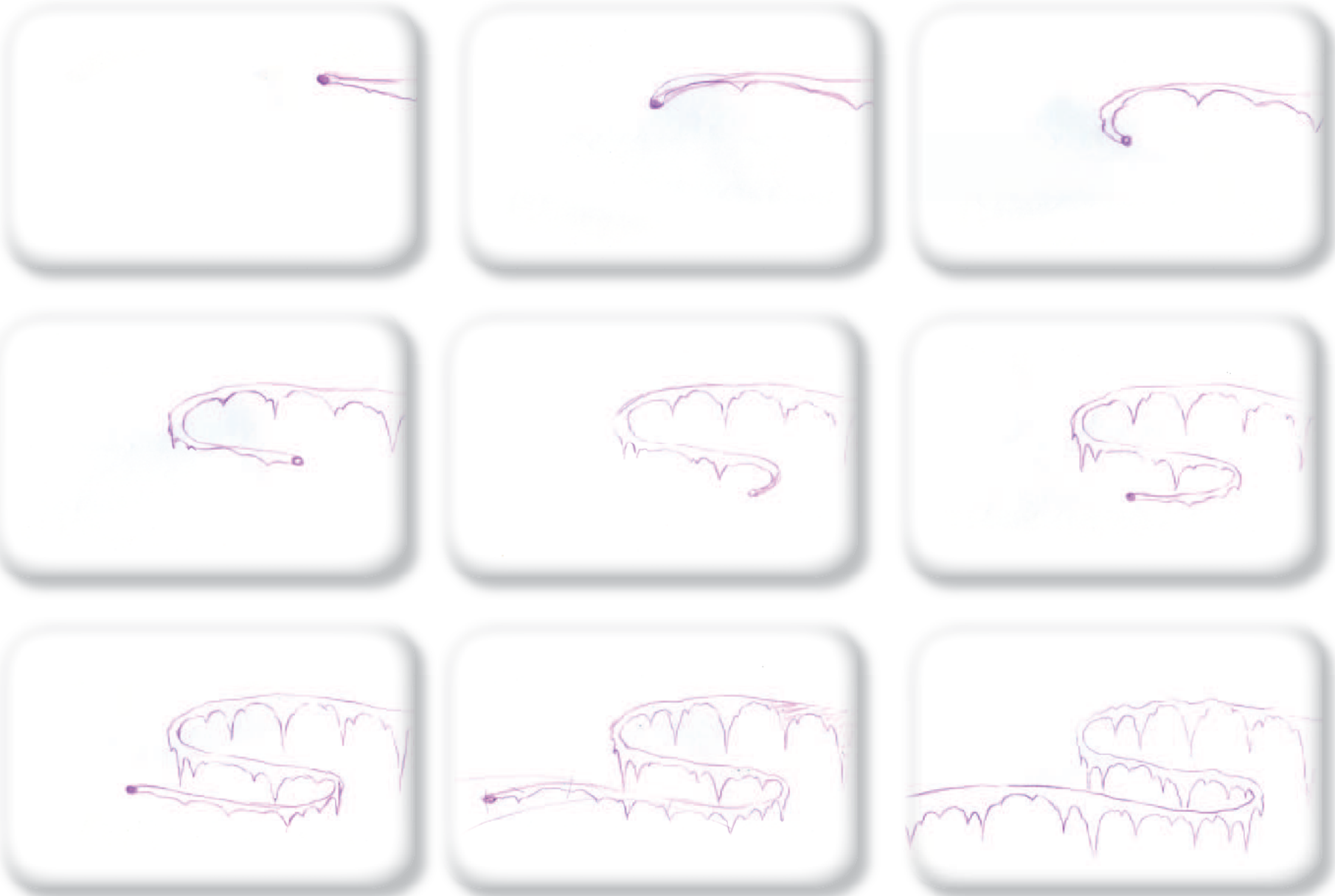


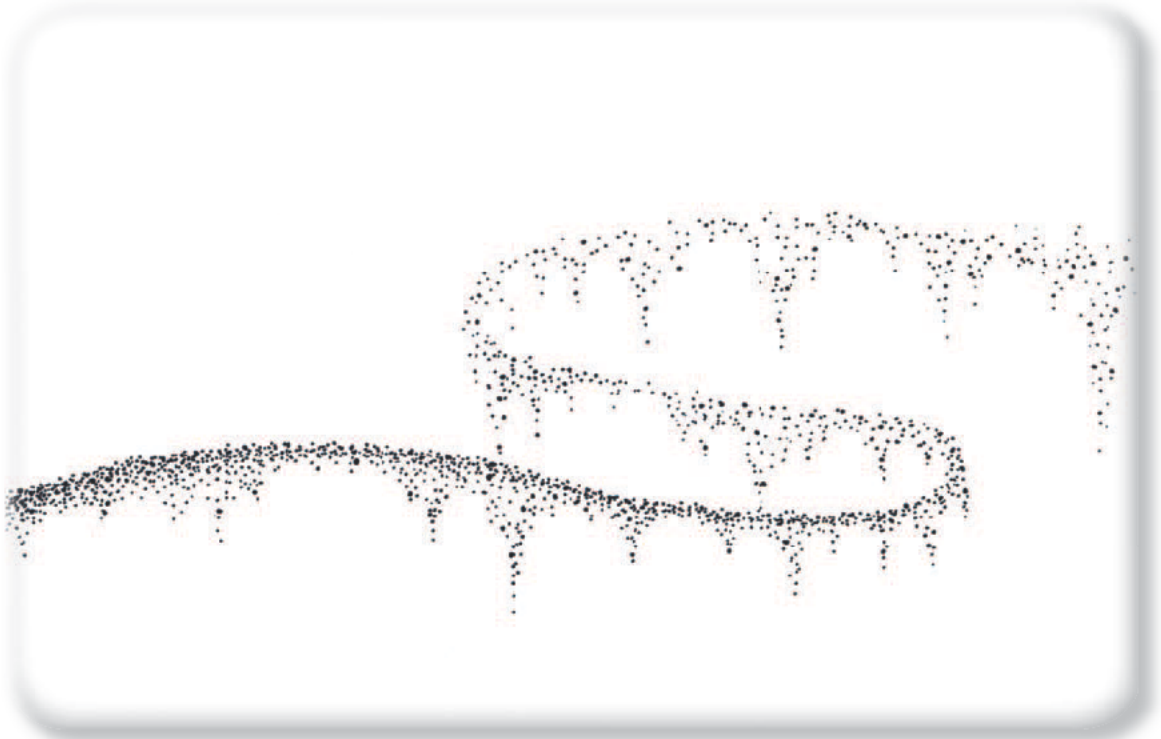
pixie-dust from the Disney classic, "Peter Pan". What I found out was that to create pixie-dust as complex, variable, and organic—as what is pictured here, was extremely difficult using CGI particles. The difference was that as I was drawing the pixie-dust by hand, I was able to improvise. I split the pixie-dust into several streams, spiraled them around each other, and varied the speeds and behaviors of all the particles intuitively, spontaneously, and at will.

To do this with computer animation, the behavior of every individual particle needs to be described to the computer. This is not impossible, but when we are doing it by hand, it is a split second creative decision, rather than a technical exercise of figuring out precise mathematical coordinates. Fantastic organic-looking effects can be achieved digitally, but tweaking individual particles intuitively, and unpredictably on the fly, is far more easily achieved by hand.

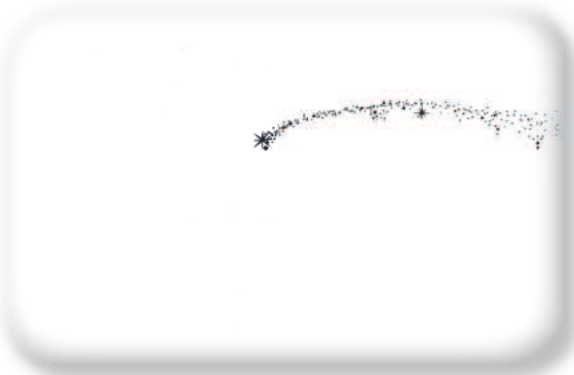


When animating classical pixie-dust, it is best to begin with a very rough path of action—loose and fluid. This drawing will provide the underlying structure of your effect. When you are ready to animate, don't start out by drawing lots of little dots of pixie-dust! As with any and all special effects animation, the looser and more energetic your initial drawings are, the more fluid and dynamic your animation will be. So forget about the details at first, and just animate with as much fluidity as possible.





Once you are satisfied with your rough animation, the difficult task of adding thousands of pixie-dust points begins. However, this need not be as tedious as one might imagine. Keep in mind that a certain amount of randomness will add to the magical, sparkling effect that makes pixie-dust look so fantastic. Only about 10% of the dots you draw actually need to follow through precisely from one drawing to the next, the majority can be very random, as long as they support and follow the basic structure of your animation.



Digital Magic

Magic, of all the elements discussed in this book, is perhaps the element most well suited to digital techniques. One of the greatest challenges that face computer animators is to coax something that looks natural out of a machine that by its very nature creates perfectly logical (by mathematical standards) and frequently un-natural looking movement. Character animation, and effects animation created on the computer by less experienced artists, who lack the foundational knowledge of classical animation techniques, looks lifeless and overly smooth.

Left to its own devices, and without the dynamic use of strong, pose-driven animation, a computer will invariably give us a mathematically sound, yet artistically flat performance, no matter what it is that we are attempting to animate. In the world of magic, however, computer animation can be a truly fantastic tool, because here what we are attempting to create is something that feels un-natural. It is for this reason, that so much of the computer-generated magic that we see in feature films in the 21st century really does work extremely well.

The case for computer-animated special effects magic is not hard to make. Experiencing the simplicity with which computers can create extremely complex magical effects, that would never have been possible in the days of classical hand-drawn animation, is an exhilarating experience for an *old-school* animator. However, not all is so wonderful when attempting to create magical special effects on a computer.

As long as the effect we are creating has no basis for comparison—as long as it is purely a flight of fancy—computers are incredibly well suited to the task. But if we are trying to re-create a magical effect that has already been done extremely well in the classical hand-drawn technique, then recreating it on a computer can be a very frustrating and complex affair. Making minute adjustments intuitively, moment-to-moment organic tucks and nudges to our animation, does not fly well on a computer. Computers need to be given all the information ahead of time. Improvisation is not the computer's comfort zone! My own first-hand experience at attempting to recreate convincing *"Peter Pan Style"* pixie-dust at Walt Disney Studios in 1994, using particle-generating CGI technology, showed me just how difficult computers are to manipulate frame-to-frame.

It is not extremely difficult to come up with very interesting, elegant and visually stunning pixie-dust type effects on a computer, but to create something that looks hand-drawn, is almost impossible. So, why even bother? The way I see it, if we want old school Peter Pan style pixie-dust, we should just create it the same way they did way back in 1953. Computers should be used for what they are best at, not for trying to recreate classical animation.

Where computers really shine is when we can input vast amounts of complex data governing countless attributes. Give the computer a path of action, a starting point and an end point, and just let the digital magic unfold.

The basis for particle-generated special effects, is what is frequently called a particle emitter. Imagine a point in space, or a shape like a sphere, that creates points of light or energy, and then spits them out into 3D space. The effects animator has control over an incredible array of forces that act to move these particles in an infinite number of ways. They can be affected by gravity or reverse gravity. They can expand outward at any variation in rate of speed, or they can be caught up in a spinning vortex of turbulent energy. They can change size, shape, and color, as they are caught up in these forces. The variations are as infinite as are the incredible numbers of particles that we can manipulate in this manner, given adequate computing power.

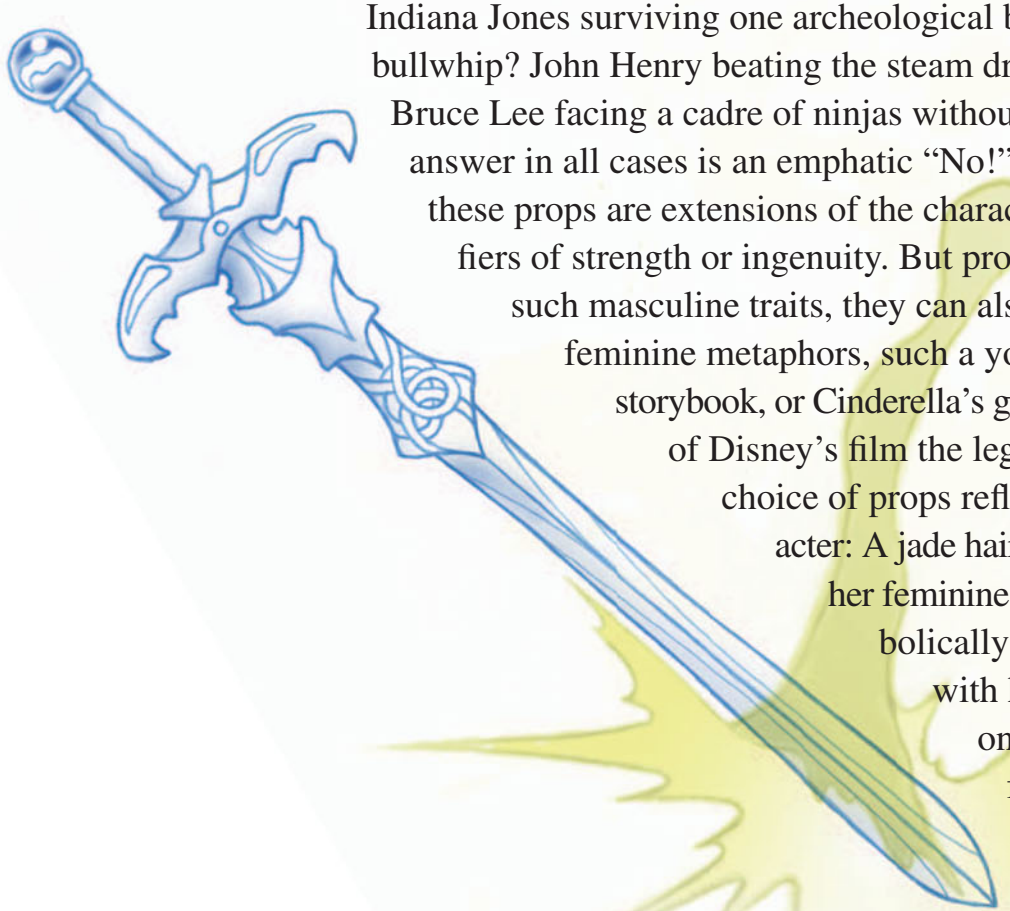
With this incredible technology, special effect artists have been given an entire new world of possibilities. We can now go far beyond the constraints of pencil and paper, into the magical realms of unfolding fractal geometry and vision unlimited by numbers or visual complexity.

However, all of the classical sensibilities that I have touched on in this book still hold the key to creating lively, dynamic and captivating visual effects. If we allow our computers to do all the work, the results will be lacking in vibrancy and vitality. Not enough can be said about the importance of the human, or natural touch, when we are creating animation of any kind, and special effects magic is no exception. In fact, there are no exceptions.

Every day, whether we draw on paper or fire up our computers, the magic we are looking for is in our hearts and in our imaginations. Keep this in mind always. Bring it to the forefront of all you do as a special effects animator, and the magic will appear before your very eyes.

Chapter 7

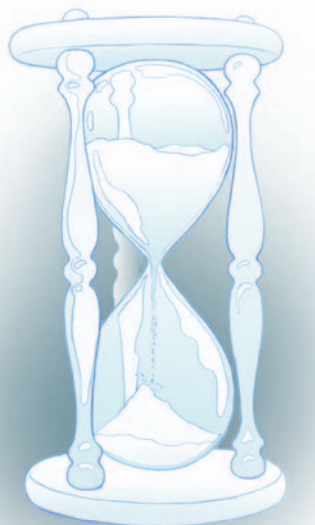
Props and Everything



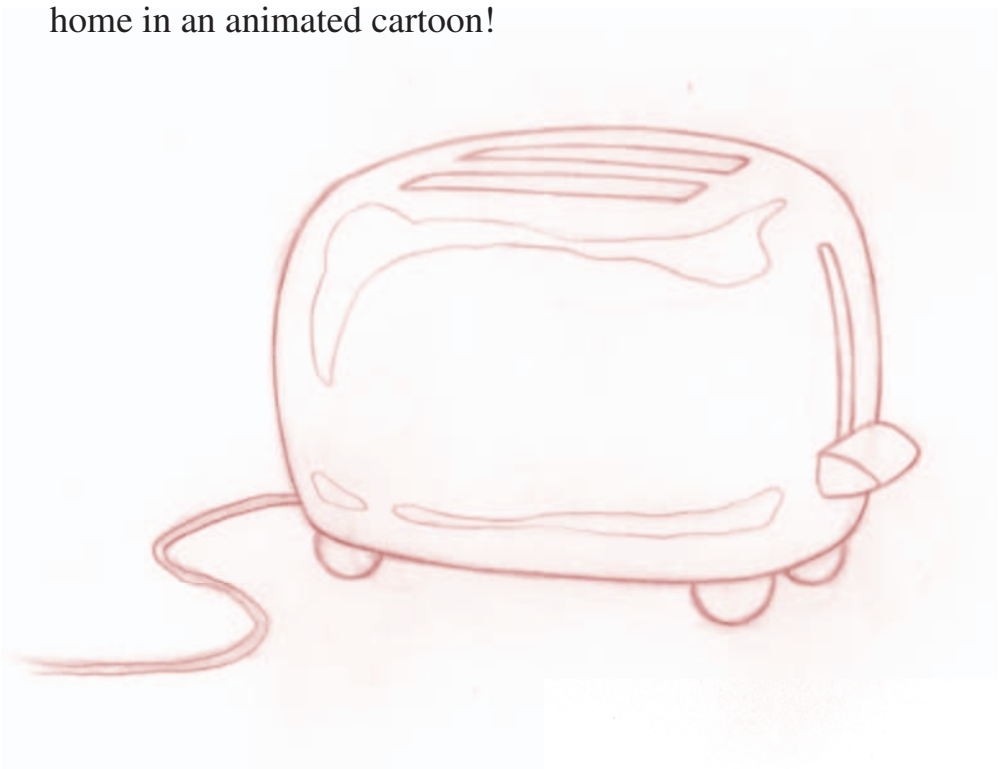
Can you imagine: Arthur ruling Camelot without his trusty sword, Excalibur? Luke defeating the Empire without his light saber? Indiana Jones surviving one archeological boobytrap without his bullwhip? John Henry beating the steam drill without his hammer? Bruce Lee facing a cadre of ninjas without his nunchucks? The answer in all cases is an emphatic "No!" In all five examples, these props are extensions of the character's personality, signifiers of strength or ingenuity. But props don't have to possess such masculine traits, they can also act as more benign, feminine metaphors, such a young girl's doll, a child's storybook, or Cinderella's glass slipper. In the case of Disney's film the legend of Mulan, her choice of props reflect her shifting character: A jade hair comb remains part of her feminine costume until she symbolically casts it away (along with her role as a naïve girl), only to replace it with her father's helmet, armor, and sword (accepting the role of a male warrior).

Props can also be signifiers of magic, wisdom, or judgment—such as Moses with his stone tablets, Gandalf with his staff, Father Time with his hour glass, the Grim Reaper with his scythe, Harry Potter with his wand and broom, or Thor with his hammer, Mjolnir. Props can even represent a character's weakness, imminent danger, or futility: Jacob Marley beset with his heavy chains, the White Rabbit antagonized by his pocket watch, Sisyphus forever pushing his boulder up the mountain, Snow White eating her poisoned apple, or the Little Match Girl shivering with her last warmth-giving matches.

In the world of traditional animation, "props" are defined by their function and utility. For instance, Cinderella's glass slipper is a costume element, but once it is separated from her body it becomes a prop. A boulder, on the other hand, is not a man-made object, it has been shaped over time by natural forces, and as such, it is a background element … but once Sisyphus starts pushing it up the mountain, it becomes a prop. As a rule of thumb, if an organic element, such as a tree branch, a rock, or a pine cone, becomes utilized by another life form, or if it moves as the result of gravity, wind, or pressure, it can be construed as a prop. An inorganic element, such as a toaster, an incense burner, or a computer, can automatically be viewed as a prop because of its inherent function, but it must also be "active" (that is, either a character or a force must utilize it … otherwise, in the world of animation, it would merely remain part of the layout).



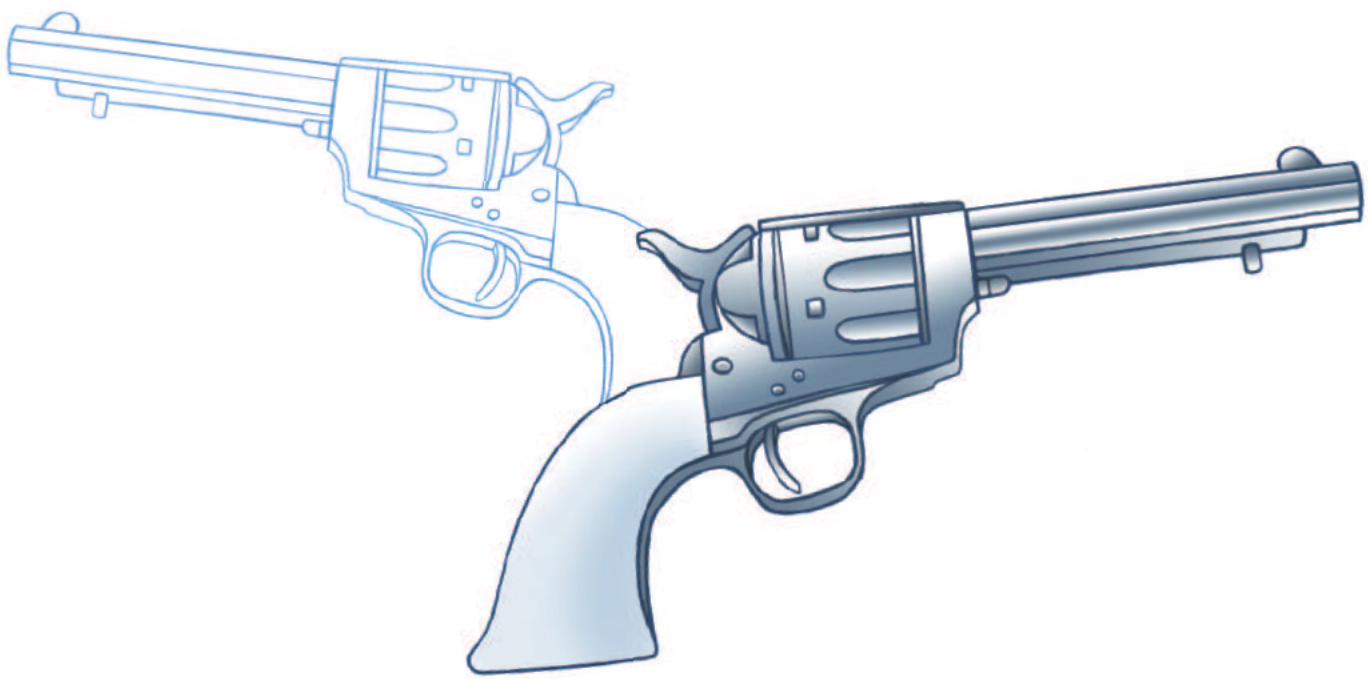
This toaster, should it remain in its static state, would be a part of the scene's background layout, and as such, would not need to be handled by the special effects animator. However, as soon as the toaster is moved in any way, it becomes an animating prop—as we see in the subsequent drawings—as the toaster springs into life, exploding violently as the toast is shot up out of it. In cases like this, a solid, inanimate object can become as volatile, stretchy, rubbery and animated as any animated character. This is very important to remember when modelling and rigging a prop in CGI, that may need to animate. The prop may need to be able to stretch and distort quite wildly, in order to give it the kind of life it needs, to be at home in an animated cartoon!



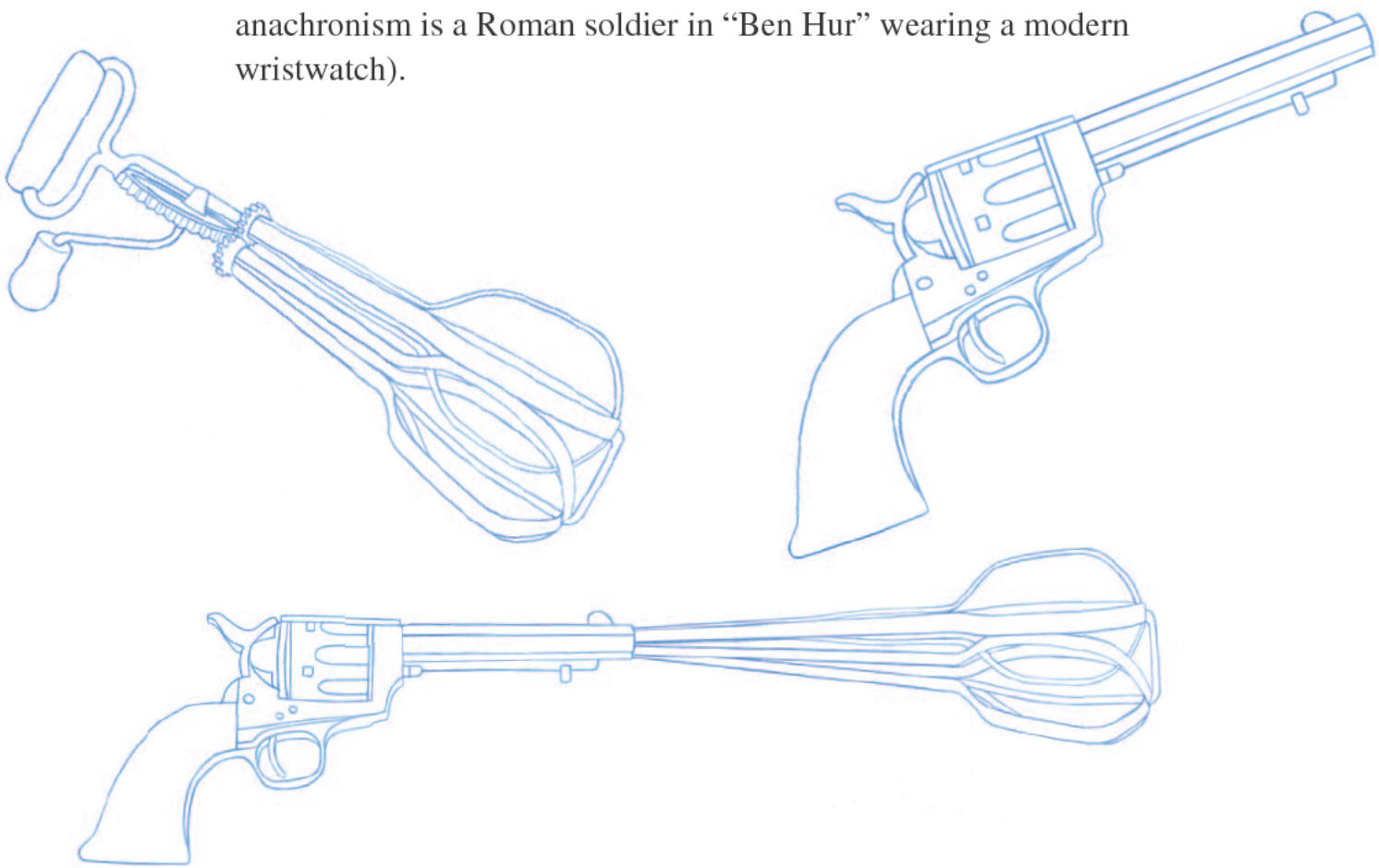
Combining organic elemental effects with our prop animation is frequently necessary. A smoking toaster, a breaking teapot full of tea, a gun firing, or two cars smashing into each other will require all kinds of organic effects. Explosions, liquids, sparks, dust, fire, or smoke can be used to add life to our solid, inorganic props.

The possibility of exaggerated physics is very important to consider when animating a usually inanimate object like this toaster, whether we are animating it by hand or as a digital 3D object.

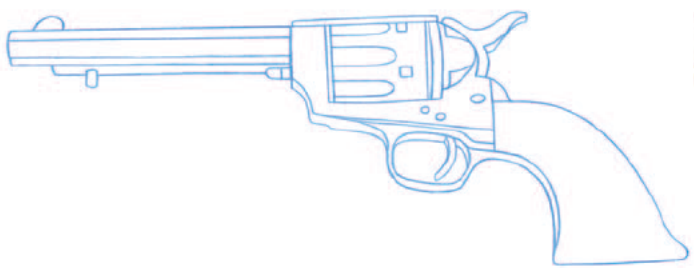
Let's talk a little more about a prop's function versus its utility. If you're a pacifist, like me, you probably wouldn't know what to do if someone gave you a Colt 45 handgun for a Christmas present. After all, its function is quite clear: It is designed to kill or injure other life forms. However, if I were a mad inventor, I could reconfigure the gun as a kitchen utensil by putting an egg beater in the barrel, then using the trigger to regulate the speed of the beater's rotation. I have thus altered the gun's original function by utilizing it to make a nice cake for my mother. Yet as artists, we have more to consider than function and utility we must also consider design and aesthetics through both line and shape.

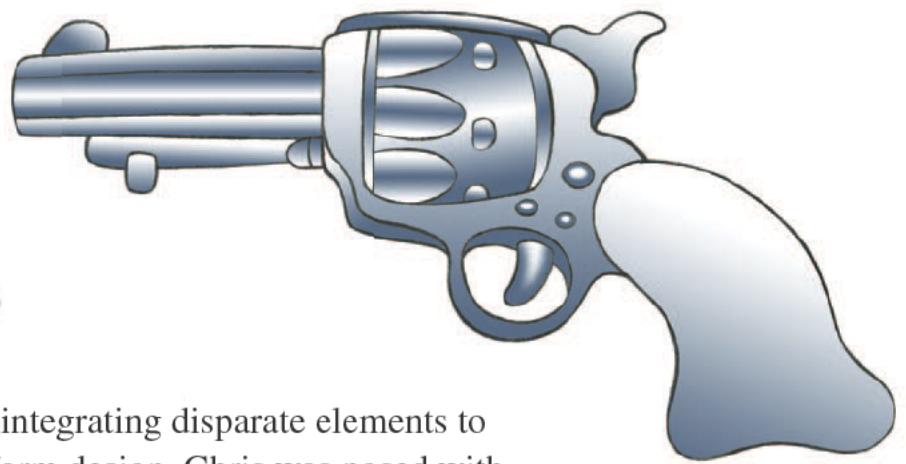


Let's imagine that the gun was an old-fashioned Colt six shooter, with a revolving barrel and a spur hammer. I might want to choose an eggbeater from that time period to match the Old West style, as opposed to a more modern version; otherwise, I run the risk of creating an "anachronistic" design. (A perfect example of an anachronism is a Roman soldier in "Ben Hur" wearing a modern wristwatch).



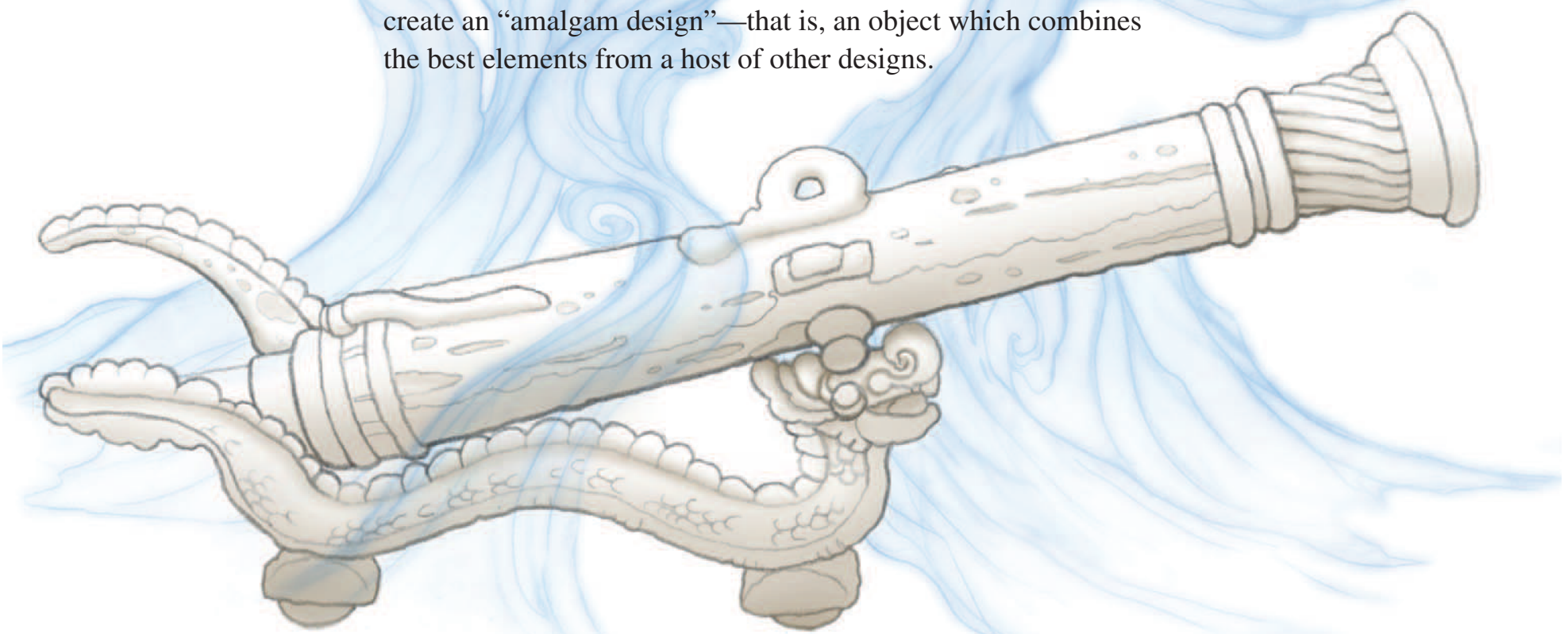
In Disney's "Lilo & Stitch," director Chris Sanders determined that the main style for all the props, characters, backgrounds, and natural effects, had to maintain a curvilinear simplicity. Therefore, all props with sharp angles and rectilinear lines had to be reconsidered as rounded off, "bowed-out" shapes, with thick-and-thin lines to match the character. He put it rather succinctly: "Imagine taking a bike pump and filling every prop with air."





This solved the first problem of integrating disparate elements to make them appear part of a uniform design. Chris was posed with an interesting challenge: How could he integrate the alien culture with Hawaii from the 1950s (replete with hot rods, Elvis, and Tiki bars)? He decided to look no further than the shores of Kauai, creating a stylistic substyle from the depths of the ocean (which is not unlike interstellar space) and its marine life. This is why the alien spaceships, weapons, and armor mimic the shapes of whales, manta rays, and crabs.

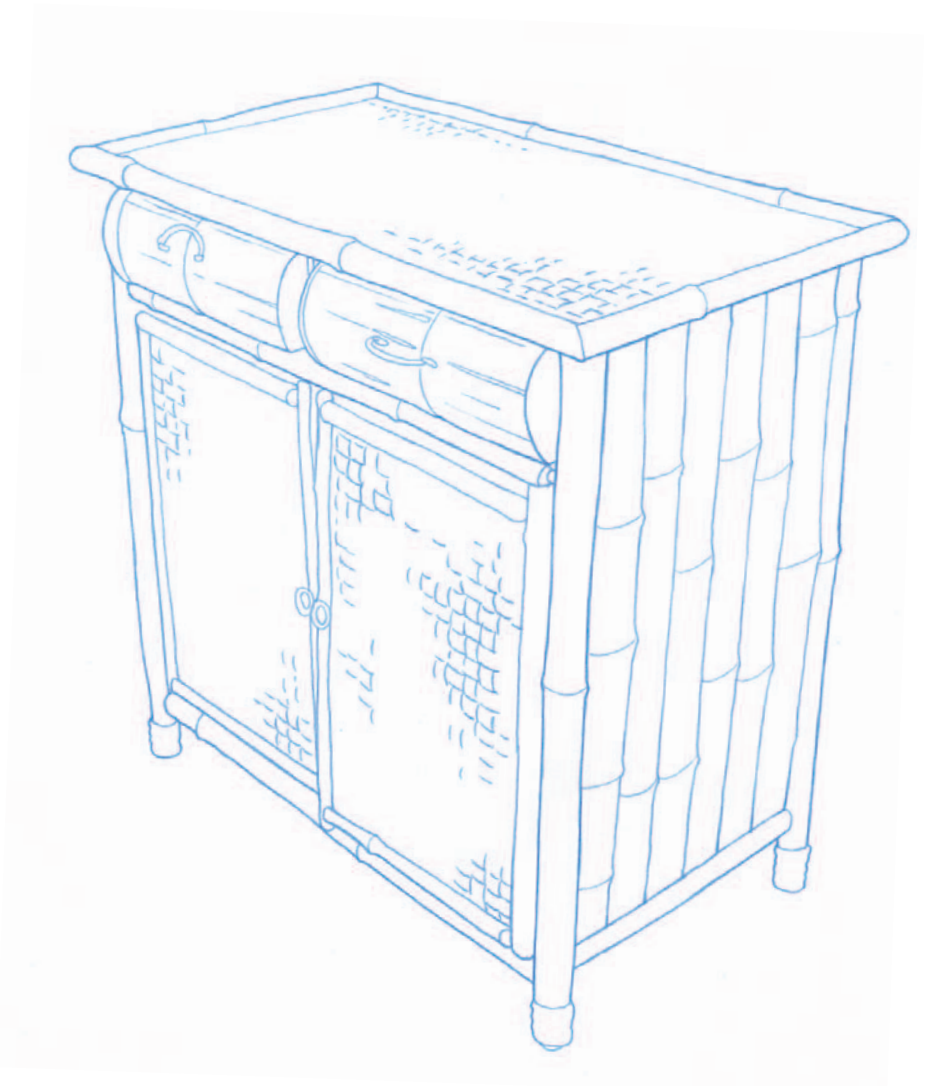
Props are so expansive as a group because they cover so many universes, cultures, and epochs, and therefore require copious amounts of research. Oftentimes, the best design has already been fully realized by a culture or by another artist, engineer, or architect (i.e., the cannon designs from both “Pocahontas” and “Mulan”). In most cases, the effects artist will research design motifs and commonalities from a particular culture, or a group of cultures, and create an “amalgam design”—that is, an object which combines the best elements from a host of other designs.



For instance, I may want to illustrate a Chinese vessel with the overall shape from a typical T'ang Dynasty piece, but complement it with patterns and illustrations from the Shang Dynasty. Please note that the more complex a pattern or texture, the more you should push to simplify. It's not necessary to draw every thread of a wicker basket or every link in a piece of chain mail. When designing props for animation, try pushing the detail until the design fails, then minimizing the detail until the design fails—the perfect iteration will lie somewhere in-between.

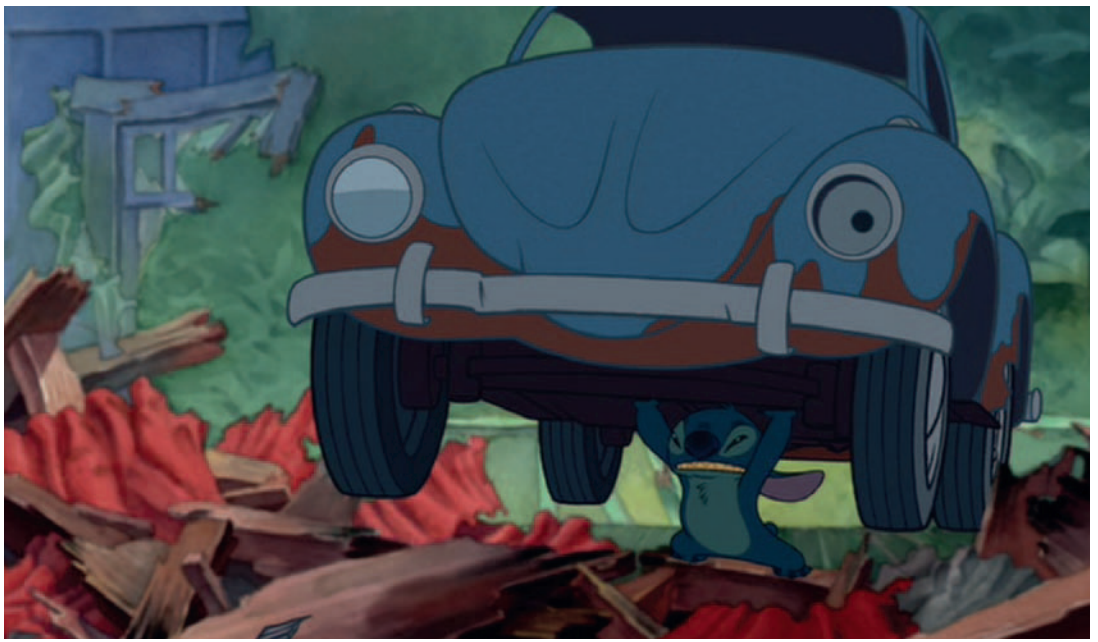
One technique I use to "see" a prop without so much detail, is to squint when looking at it. When we do this, we see less detail and more of the shadows and light that define the object. The photo on the right on the preceding page has that simplified look. From there we can design a concise, simple version of the object, as in these two drawn versions. Shading can then be applied to add lighting to the line drawing.

This bamboo and wicker chest has a great deal of detail, and it would be almost impossible to draw every tiny little line and shadow. The drawing on the right is vastly simplified, yet it still communicates clearly exactly what it is we are looking at. This technique can be very helpful when designing and modelling 3D props as well.



There are times when a prop's scale can give a story more metaphysical menace. For instance, what kind of threat would Darth Vader and the Emperor have posed without the Death Star? And would the Borg Queen from Star Trek have been as threatening without the Borg Planet? From H.G. Well's ominous tripods to John Henry's nemesis, the steam drill, oversized machines have become the modern equivalent of the bad guy par excellence. In "Howl's Moving Castle," the castle has both menacing and benevolent overtones, a force of nature and a personality at the same time. Once a prop becomes imbued with too many human attributes, however, it becomes a character in its own right.

Props with complex moving parts, particularly machines, are the most difficult to conceive and animate, which is why many enterprising directors will utilize computer graphics in a traditional picture. This saves the animator from copious amounts of pencil mileage and three-point perspective headaches—the key is to not let the CGI in these instances diverge stylistically from the rest of the picture. Because of these breakthroughs in visualization aids for complex man-made props, we will narrow our focus to organic props (and perhaps some simplified inorganic props, like bottles), both in terms of their design and structural variability.



Stitch wielding a Volkswagen Beetle, is a great example of a CGI prop, well designed to integrate with a 2D cartoon environment.

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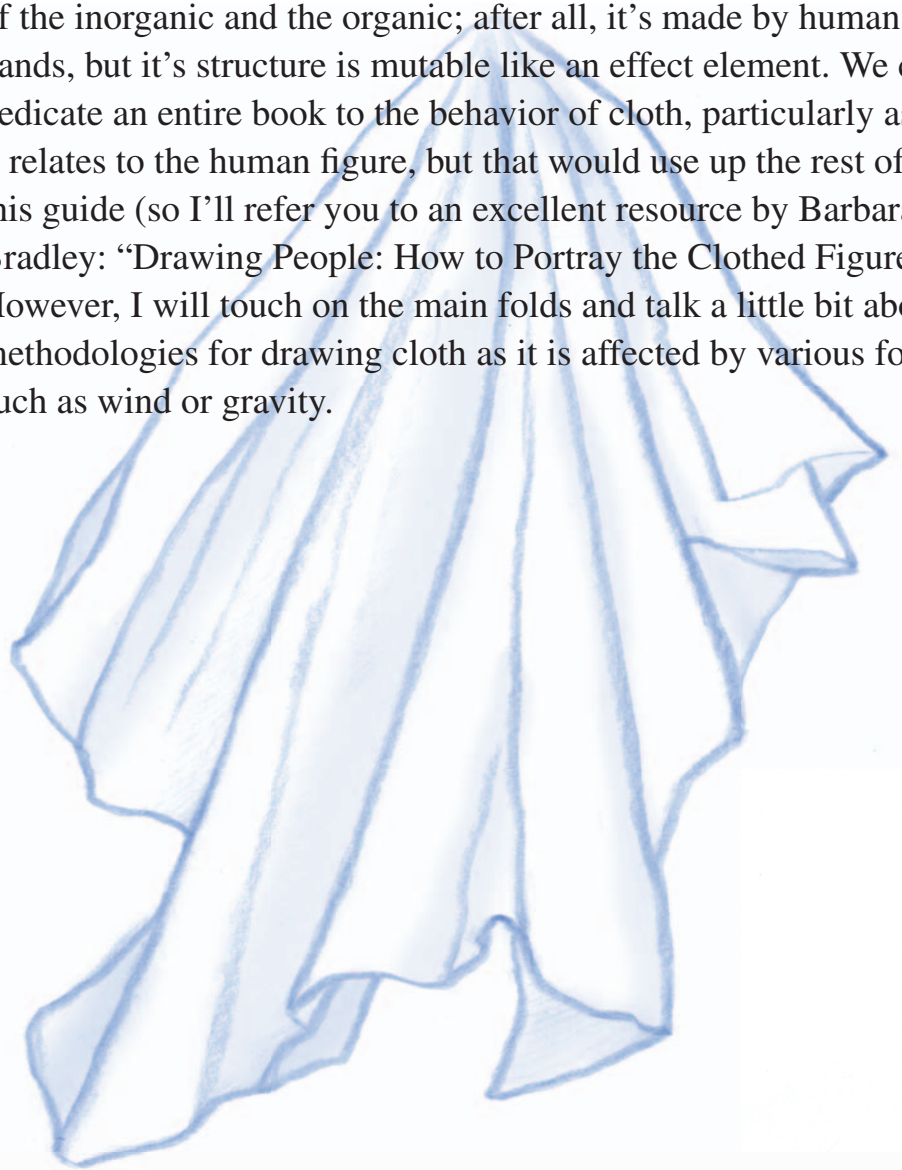
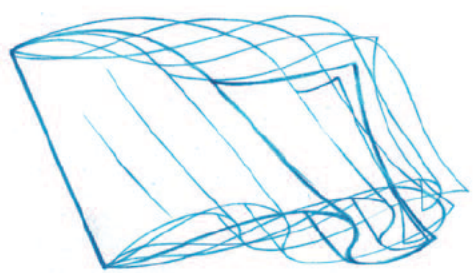
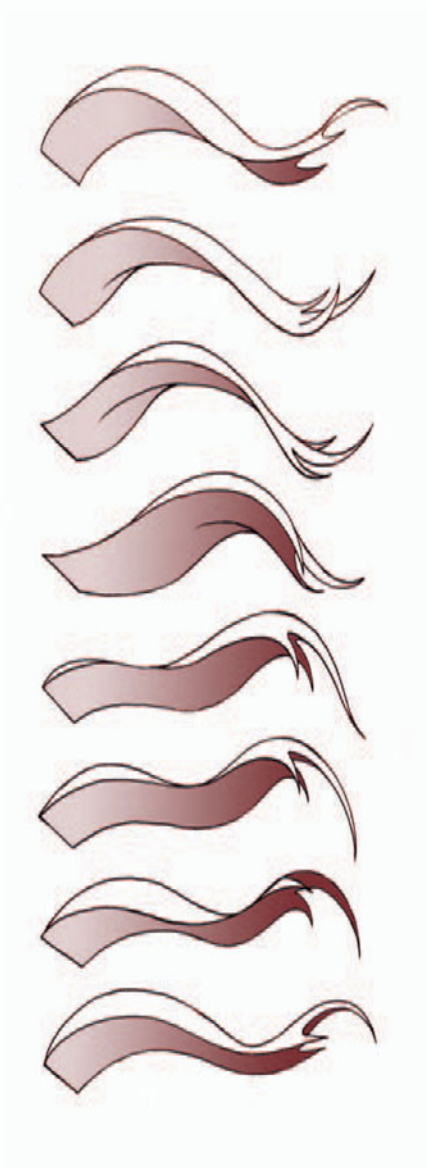
Perhaps the most complex prop is cloth, as it straddles the worlds of the inorganic and the organic; after all, it's made by human hands, but it's structure is mutable like an effect element. We could dedicate an entire book to the behavior of cloth, particularly as it relates to the human figure, but that would use up the rest of this guide (so I'll refer you to an excellent resource by Barbara Bradley: "Drawing People: How to Portray the Clothed Figure"). However, I will touch on the main folds and talk a little bit about methodologies for drawing cloth as it is affected by various forces, such as wind or gravity.



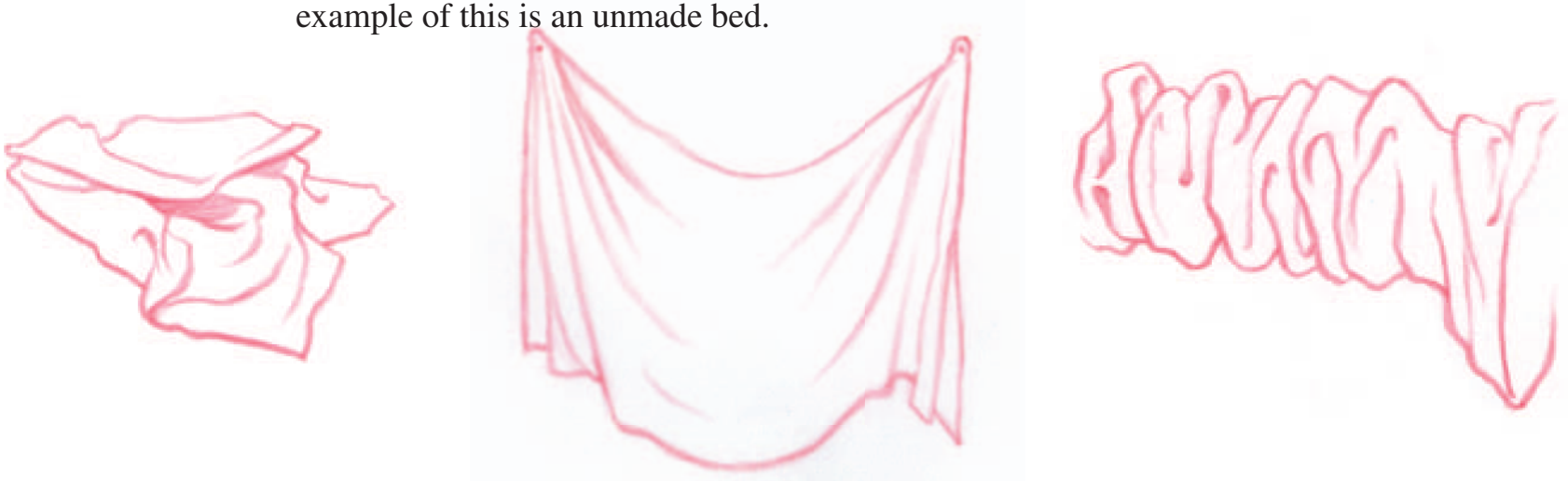
One of the most difficult problems from my early days as an animation student was the flag test. The goal is simple: Take a flag hanging from a flag pole and make it flap with a large gale of wind. Sounds simple, right? The first thing to consider is that the "shape" and direction of the wind must manifest in the interior folds and the exterior silhouette (imagine how wind might affect a wheat field). The trick is to maintain the sense of the flag's volume, while simultaneously collapsing and expanding convex and concave masses.



The most common types of folds are pipe folds, diaper folds, zigzag folds, spiral folds, half lock folds, falling folds, and inert folds. A pipe fold looks like it sounds, it's a long, cylindrical fold with a concave surface that begins with a pinched top and flares slightly outward toward the bottom. If you hang a robe on a hook, these will most likely be the types of folds you will see. A diaper fold is the result of cloth being suspended from two contact points. If you take that same robe and hang it from two hooks, the resulting pipe folds will meet in the middle, causing them to crease into one another. Zigzag folds are a series of tight, interconnected back-and-forth folds, creating a zigzag pattern.

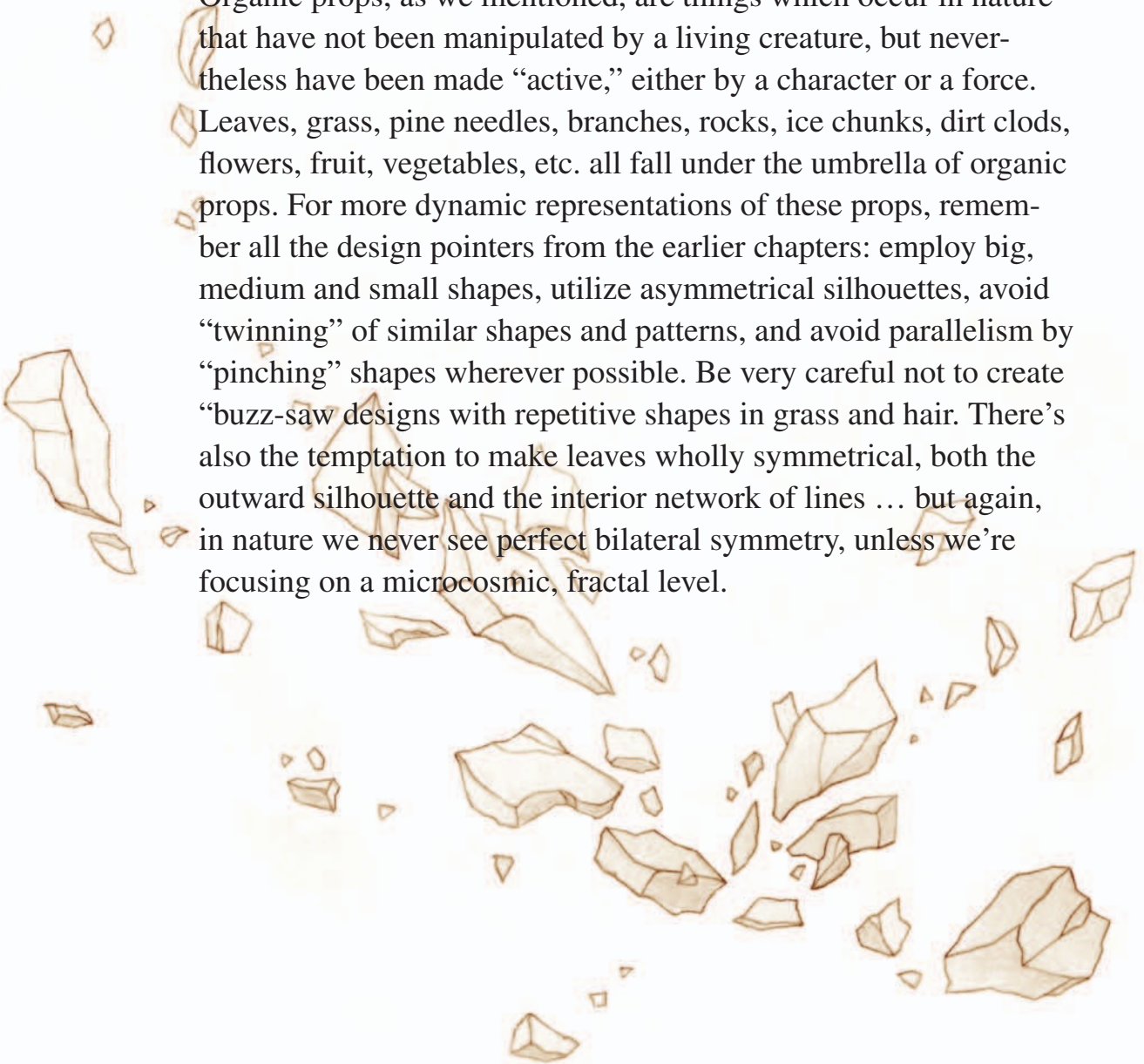


We typically see these types of folds in complex ribbon-work and in crimped dresses. Spiral folds are small, cylindrical convex folds that wrap around arms and torsos with lighter types of cloth. Most Greek sculptors employed an inordinate amount of spiral folds in their clothed figures. Half lock folds occur when a character bends a leg or an arm, thereby creating a series of interconnected folds at the locus of the joint. Falling folds occur when a hanging cloth begins to collide with a surface, causing the bottom part to bunch up in a complex array of folds. Lastly, an inert fold is just that—inert. There are no underlying forces or geometries affecting its makeup, it's just a random piling or bunching of folds. The best example of this is an unmade bed.

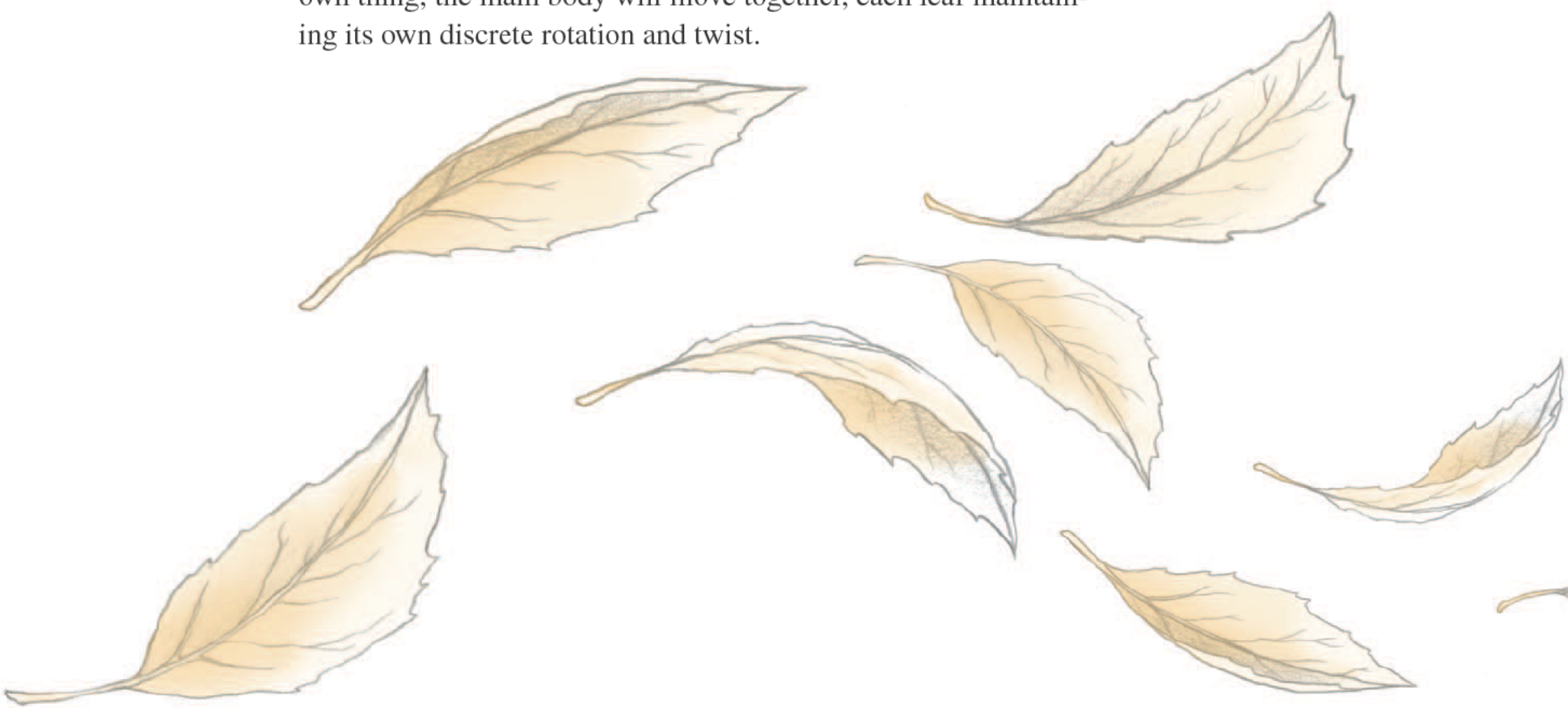


Drawing folds well is an art form unto itself. When animating any kind of cloth or fabric, clothing or hair, simply apply the same animation principles that apply to all effects. Wave and whipping actions, overlapping timing, follow-through, and weight. Keep in mind that the texture, softness, stiffness, or thickness of the fabric will affect its behavior as well.

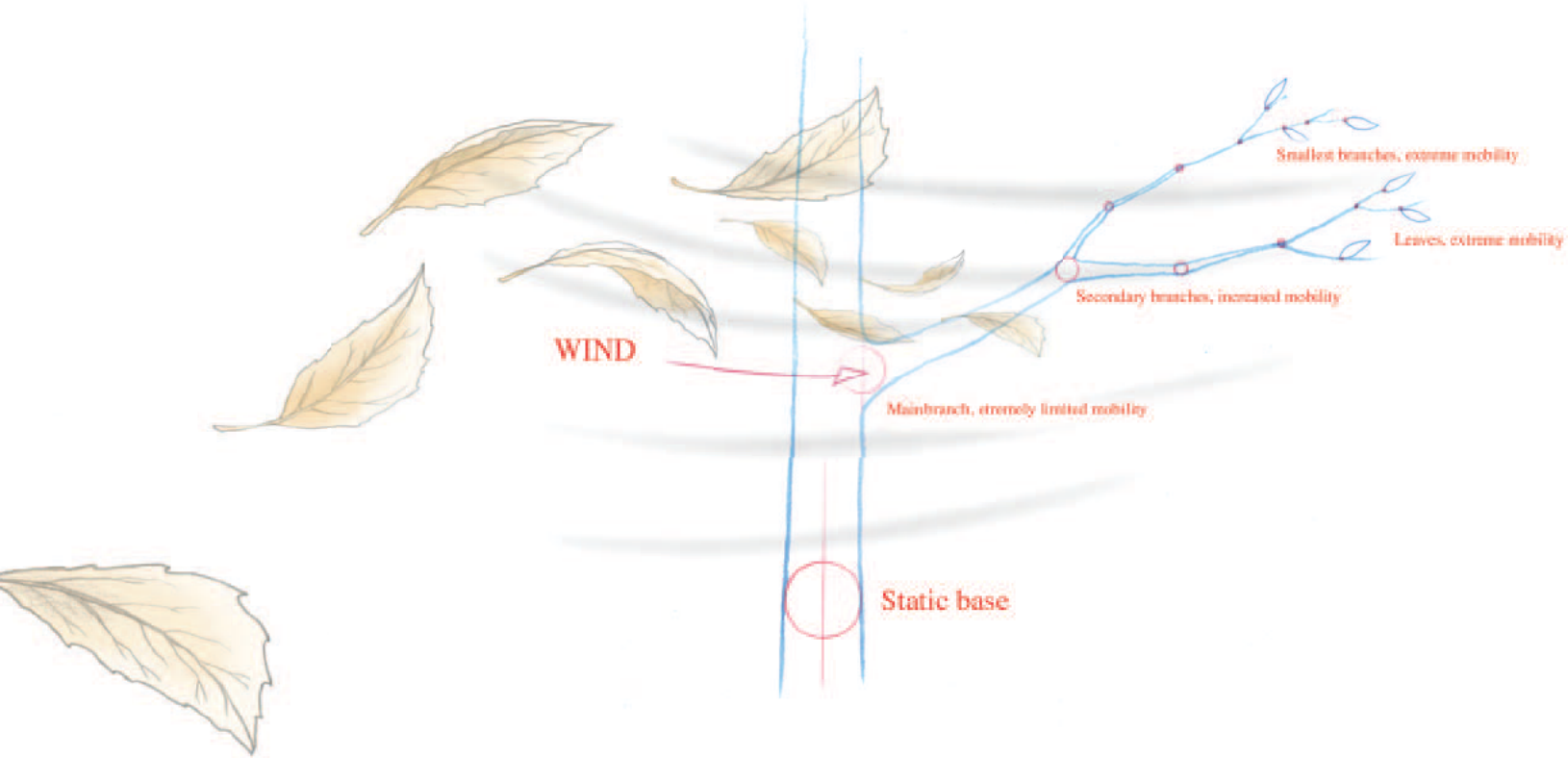
Organic props, as we mentioned, are things which occur in nature that have not been manipulated by a living creature, but nevertheless have been made "active," either by a character or a force. Leaves, grass, pine needles, branches, rocks, ice chunks, dirt clods, flowers, fruit, vegetables, etc. all fall under the umbrella of organic props. For more dynamic representations of these props, remember all the design pointers from the earlier chapters: employ big, medium and small shapes, utilize asymmetrical silhouettes, avoid "twinning" of similar shapes and patterns, and avoid parallelism by "pinching" shapes wherever possible. Be very careful not to create "buzz-saw designs with repetitive shapes in grass and hair. There's also the temptation to make leaves wholly symmetrical, both the outward silhouette and the interior network of lines … but again, in nature we never see perfect bilateral symmetry, unless we're focusing on a microcosmic, fractal level.

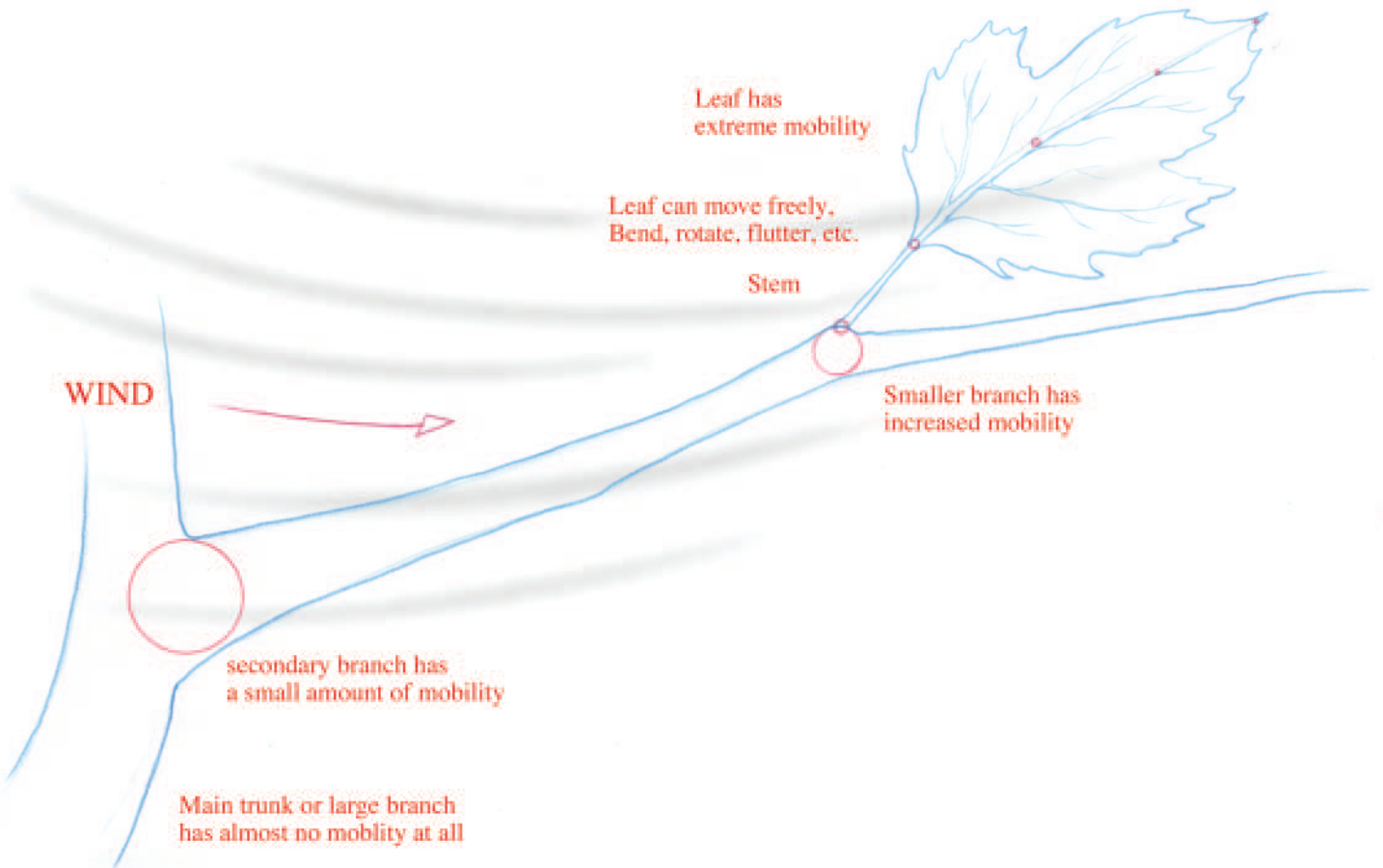


Animating leaves is similar to animating cloth, though a leaf has more tensile strength and does not form complex, interweaving folds. It will rotate, twist, and bend, however, when blowing in the breeze. When clumps of leaves move in the wind, the overall mass will follow a general path of action, just like a flock of birds. While some leaves will spin off from the main group and do their own thing, the main body will move together, each leaf maintaining its own discrete rotation and twist.



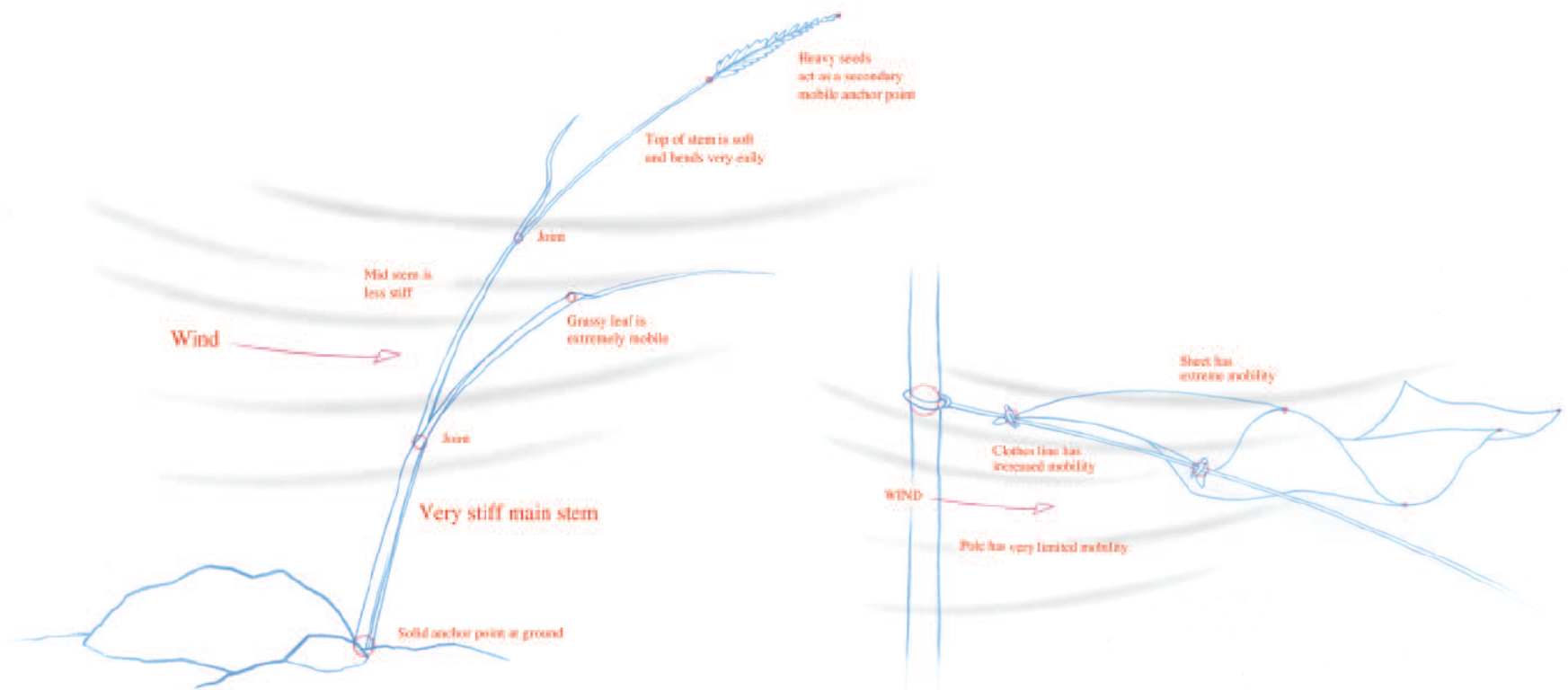
There are literally hundreds of overlapping actions in this type of phenomena. So what is the difference between leaves blowing freely in the air, versus leaves blowing in a bush or a tree? There are similarities, but in the former example, the leaves are connected by a common network of solid bodies—the branches. Just as the head is the unwitting passenger on the human body, a blowing leaf must follow the primary path of the branch first, regardless of how it reacts to the wind. This is why it is critical to animate branches before leaves!



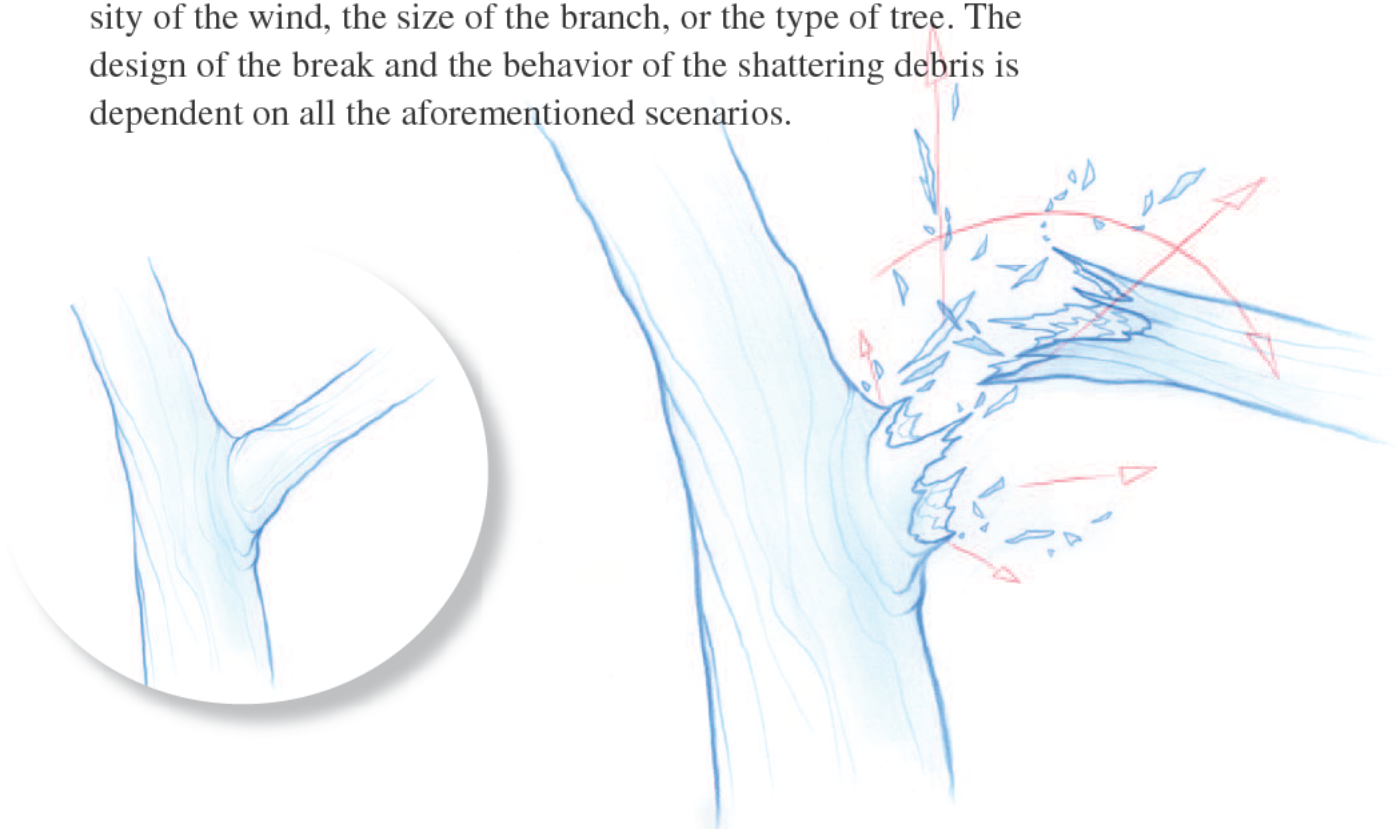


When animating branches and leaves moving in the wind, it is helpful to break down the structure of a tree into its main trunk, largest branches, smaller secondary branches, and so on down to the smallest leaves, flowers, fruits, seed pods, etc.

The same technique can be used when animating a wide variety of props and anything with an underlying structure that is being blown in the wind. Before beginning your animation, imagine you are engineering the element from its structural base. The more you understand the characteristics of its materials, the easier it will be to animate.



Sometimes a gust of wind is so strong that it actually manages to break a large branch off of the main body of a tree. There are dozens of parameters that determine how that break will occur. If it's a younger, more resilient tree, the branches may have more spring to them. The resulting break may look like this. If the branch is older and dryer, the shattering effect may be more explosive and contain far more smaller bits of wood debris. Yet there are other things to consider that may affect the outcome, such as the intensity of the wind, the size of the branch, or the type of tree. The design of the break and the behavior of the shattering debris is dependent on all the aforementioned scenarios.

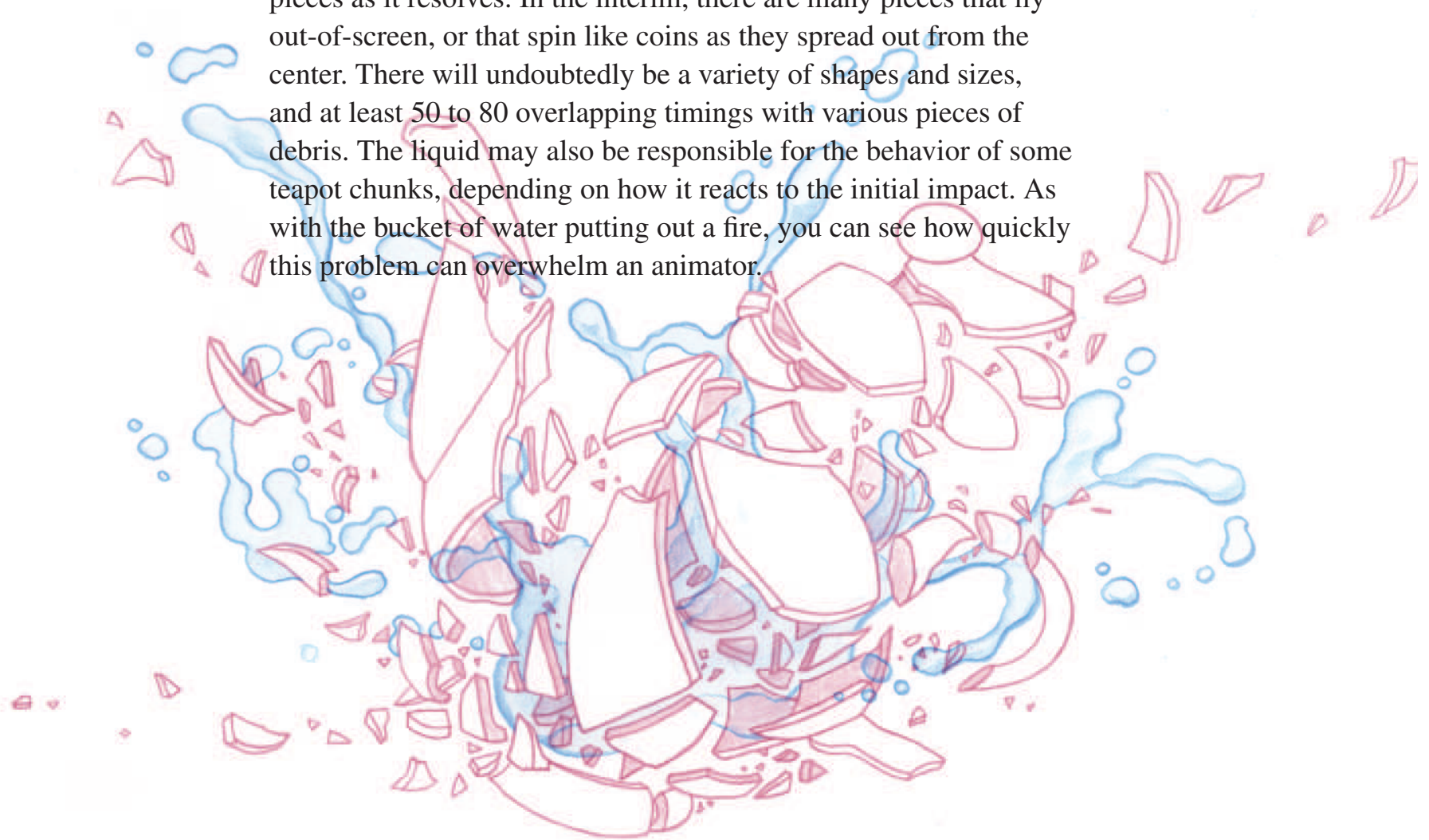


The wood debris will explode out from the center of the break much in the same way water splashes outward in proscribed paths of action. The larger pieces will have more hang time and generally rotate slower than the smaller pieces, and again, it's important to employ big, medium, and small chunks with varied timings.

Collisions of solid bodies and explosions are affected by gravity, wind, air pressure, the velocity of the impacting force, as well as the physical makeup of the object (or objects) being affected. In 3D dynamics, an animator will tweak these forces bit-by-bit and has the latitude to run many simulations of the colliding bodies. The 2D artist must use more planning and commit to a final choreographed scene before committing pencil to paper.

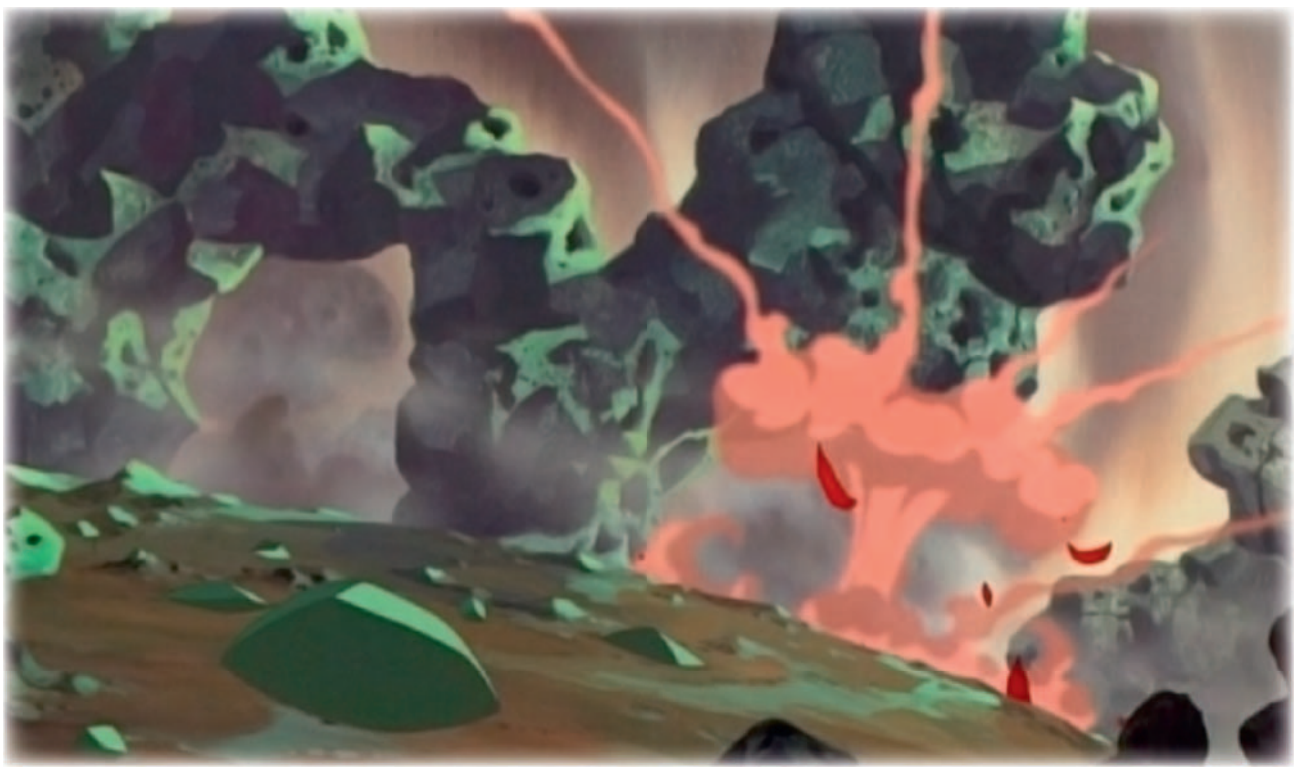
Perhaps the most dreaded visual effects problem an animation student can be assigned is the breaking teapot test. The parameters are simple: Animate a teapot half-filled with liquid dropping from 12 o'clock (as opposed to entering the scene at an angle) and impacting the ground in the center of the field. In this instance, gravity is the only physical force defining the outcome. After viewing hundreds of feet of visual reference, as well as researching photographic reference, I have observed the general similarities and created a basic formula for successfully depicting the collision of the teapot and its resolution. In most cases, an initial ring of debris quickly forms at the initial area of impact, with few pieces of the ring overlapping.

Meanwhile, one large piece will remain intact, moving at a slight angle away from the center; sometimes this piece will spin around as it moves, then will settle on the ground in a rocking type motion; other times this piece will break into two or three smaller pieces as it resolves. In the interim, there are many pieces that fly out-of-screen, or that spin like coins as they spread out from the center. There will undoubtedly be a variety of shapes and sizes, and at least 50 to 80 overlapping timings with various pieces of debris. The liquid may also be responsible for the behavior of some teapot chunks, depending on how it reacts to the initial impact. As with the bucket of water putting out a fire, you can see how quickly this problem can overwhelm an animator.

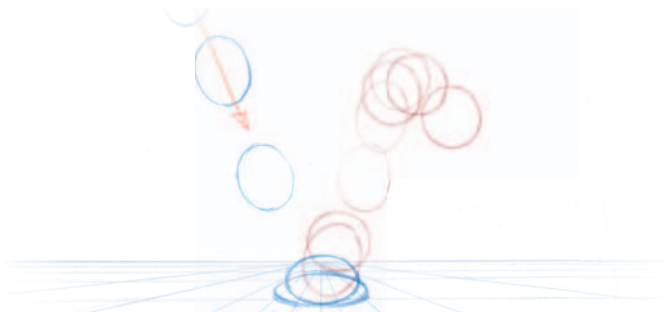


It's interesting to note the similarities between the dynamics of a water splash and an exploding teapot (even if the teapot is empty): Both actions have similar outward thrust, hang times, paths of action, overlapping actions, and concentric rings which form at the base. You can apply many of these principles to larger scale explosions or collisions. Remember that the most critical aspect of scale is the viewer's orientation to the scene. If an explosion of debris happens up close in the camera, the conventions for depicting depth vary. In Disney's "Mulan," the exploding munitions cart is a close-up shot, so the initial thrust of debris clears the field within four or five frames, which is a fraction of a second … this means that new debris has to generate from the center at the fifth frame, otherwise there will be nothing for the eye to hold onto.

In other words, there's simply not enough space in the field to show how some of the pieces resolve. The same explosion from the distance would look completely different, however, because we could see all the discrete paths of action for the resolving debris, and even watch where the pieces come to rest. Never forget that it's the forces in nature that create similarities between disparate actions: the teapot and the splash, the mountain range and the wave, the crack in the sidewalk and the bolt of lightning. Even animating a character moving through snow or dirt employs the same basic tenets of ocean wave timing, with some slight differences.



An excellent exercise for learning how to understand how different materials react differently to outside forces, is to take the standard bouncing ball animation exercise, and try it with a ball made of a different material other than a bouncy rubber ball. I have two examples, one is a ball made of ceramic. Imagine a hollow ball made out of china, or glass perhaps, falling from a height of one meter or so, and colliding with a solid, unmoving surface.



The second example on the following pages, is a water balloon falling from the same height. One of the things that jumps out at me when I consider these two balls made of entirely different materials, striking a solid surface, is that the trajectory path of action, timing and overall energy patterns of the two, are extremely similar. What does change is the shapes and patterns of the actual breaking object, dictated by its molecular makeup—the way its molecules hold themselves together, and the way they break apart when their ability to adhere to each other is challenged by an abrupt impact.

The ability to understand how various different materials behave under stress is a key element in understanding how to animate props well.

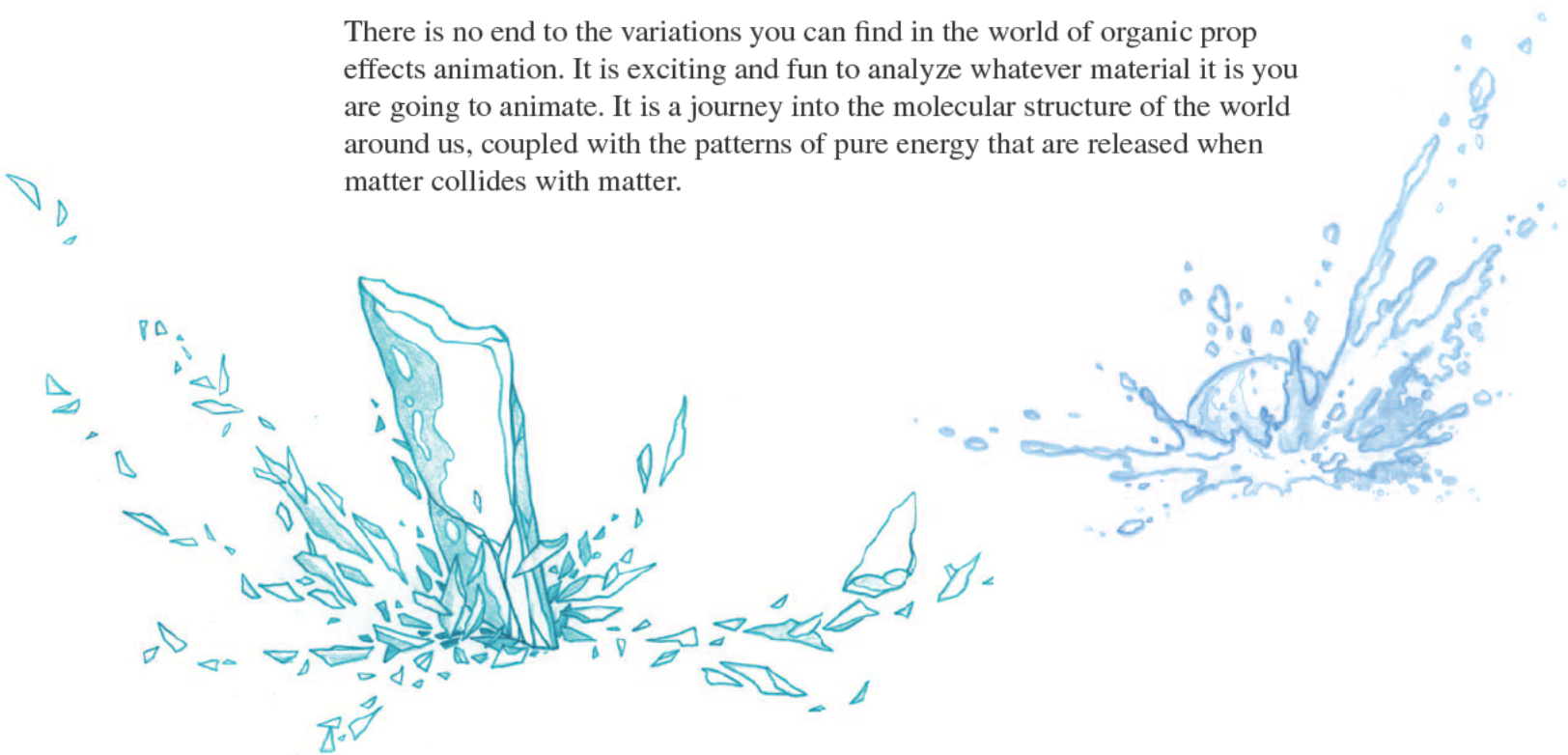
This exercise can be conducted using an infinite variety of different materials. Imagine a ball of mud or an icy snowball. A solid crystal ball or metal ball bearing. A ball made out of old, slightly decomposing wood, or a ball made out of fresh hardwood. A delicate Christmas tree ornament, (there's one that's quite easy to imagine breaking!) or a ball made out of putty. Each one of these examples will break apart in its own unique way. The cracks that form will have shapes that clearly indicate the underlying structure of the material.

If we move away from the ball, and experiment with objects of varying shapes and sizes as well as material, we will begin to see even more variables come into play. A randomly shaped chunk of dried cement impacting a solid surface will obviously break up in an entirely different way than a delicate ceramic teapot. The possibilities are endless. However, regardless of the material, the patterns of energy that underlie the physics of any collision of objects will contain similarities. As with a typical water splash, there is an initial impact, a shooting outward of energy and matter, a slowing down of that energy at the apex of its movement, and a resolution.

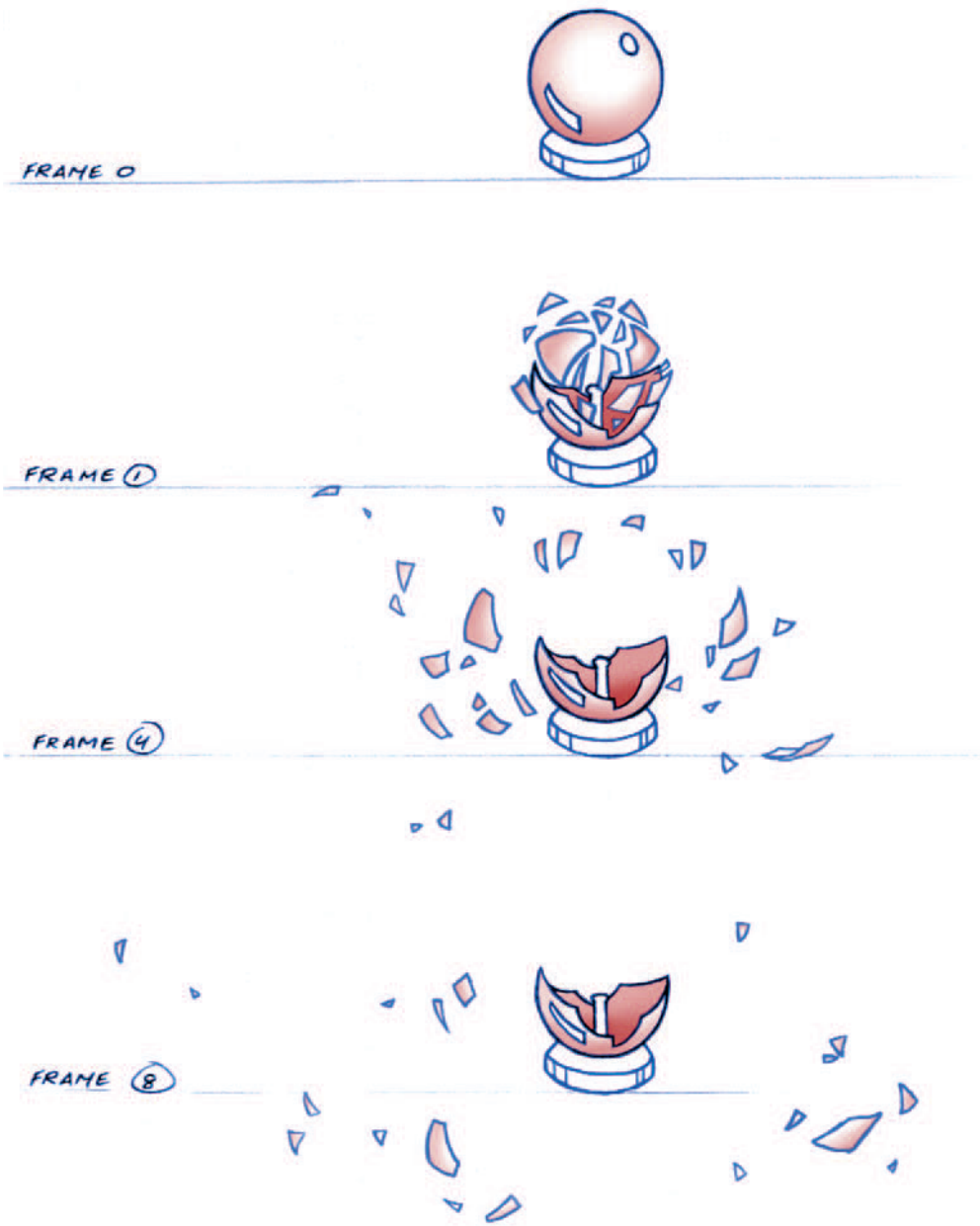
An old brick smashing to the ground breaks up in its own very unique way. There is a specific “brickness” to it, the way the chunks break up into smaller pieces, and the way some of it will be completely pulverized to a fine powder on impact. There is also a particular bounce to a piece of brick. It is dense and quite solid, but it has a little bit of give to it. Since it gives in and crumbles slightly on impact, its bounce is not as abrupt as something like glass, or ceramic. When the smaller pieces that break off on impact fall back to the ground, they will not skitter and slide as much either, due to the abrasive texture of the brick. Of course the density and texture of the surface that a given object collides with has a lot to do with the way the object bounces as well.

An icicle and a snowball, although they may be considered to be close relatives in the world of organic props, could not be more different in the way they react to an abrupt collision with an unforgiving surface. An icicle is extremely brittle, and if it lands on a very solid surface, it will really explode quite violently on impact, with many of the pieces breaking off flying quite far from the point of impact. The pieces break off in pointed shards and very sharp, angular pieces, due to ice's brittle, crystalline nature. A snowball on the other hand, will not bounce at all, but disintegrates into very small chunks of snowy powder on impact. Most of the pieces will not travel all that far, and when they land, they will stop abruptly. Depending on how wet the snowball is, there will be a certain amount of splatter to a snowball's impact as well.

There is no end to the variations you can find in the world of organic prop effects animation. It is exciting and fun to analyze whatever material it is you are going to animate. It is a journey into the molecular structure of the world around us, coupled with the patterns of pure energy that are released when matter collides with matter.



This is a simple rendition of what a small electric bulb might look like when it bursts. In the first frame we still see the shape of the bulb, as the pieces initially crack up into fragments. In the next two frames, the pieces shoot outwards away from the bulb in an arcing, splash-like manner. This type of glass is very thin and light, and the pieces are quickly slowed by the surrounding air resistance, falling and fluttering, down and away.



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One of the most stunning sequences of special effects animation ever created, was the avalanche in Disney's "Mulan." Taking over six months to animate, it was an extreme example of organic "prop" animation. The idea of animating snow took on completely new meaning, as our team of effects animators drew countless incredibly complex drawings of snow billowing, pouring, rolling, and exploding through the screen. This masterful scene, animated by effects animator Garrett Wren, is a wonderful example of just how complex much of the artwork actually was.

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The variety of shapes and sizes that a simple element as snow can take on, were quickly apparent when we we challenged to animate an avalanche catching up to, and overtaking, an army of Huns as they galloped down the side of a mountain. Rolling, billowing clouds of snow needed to be filled with massive sheets of compressed snow, that would explode and break into pieces as the avalanche engulfed everything in its path.

Here we see the leading edge of the avalanche as it catches up to Shan Yu, the evil leader of the Hun army. In this effects scene, the snow began as subtle and delicate puffs of snow and small chunks of snow being kicked up by the horse's feet as he ran full speed to escape the onslaught of the avalanche. (The sword in Shan Yu's hand also was animated by the effects department, but it was a small task compared to the intensity of the snow animation.)

As the rolling, boiling leading edge of the avalanche catches up with Shan Yu, we wanted to give the impression that within the clouds of snow there were massive sheets of snow crust. So as the billowing shapes overtake the horse and rider, we see the crust explode into chunks as it impacts them. Note the extreme variation in size of the various chunks, and the explosive silhouette of the shapes as they shoot outwards from the point of impact.
When we think of snow, the shapes we see in this scene definitely aren't the first images that come to mind. This is the challenge of animating organic props of all kinds. When objects we are familiar with, like teapots, trees, cars, and snow, are affected by outside forces and brought to their breaking point, fantastic and unexpected visual effects are frequently the result. It is a fascinating and endlessly challenging job to portray these amazing phenomena accurately.

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Matter Meets Computer

Of all the effects we have observed and discussed thus far, none is better suited to utilizing CGI animation technology than that of props, both man-made and organic. By their very nature, props lend themselves well to being created in virtual 3D space, much more simply and practically than water or fire.

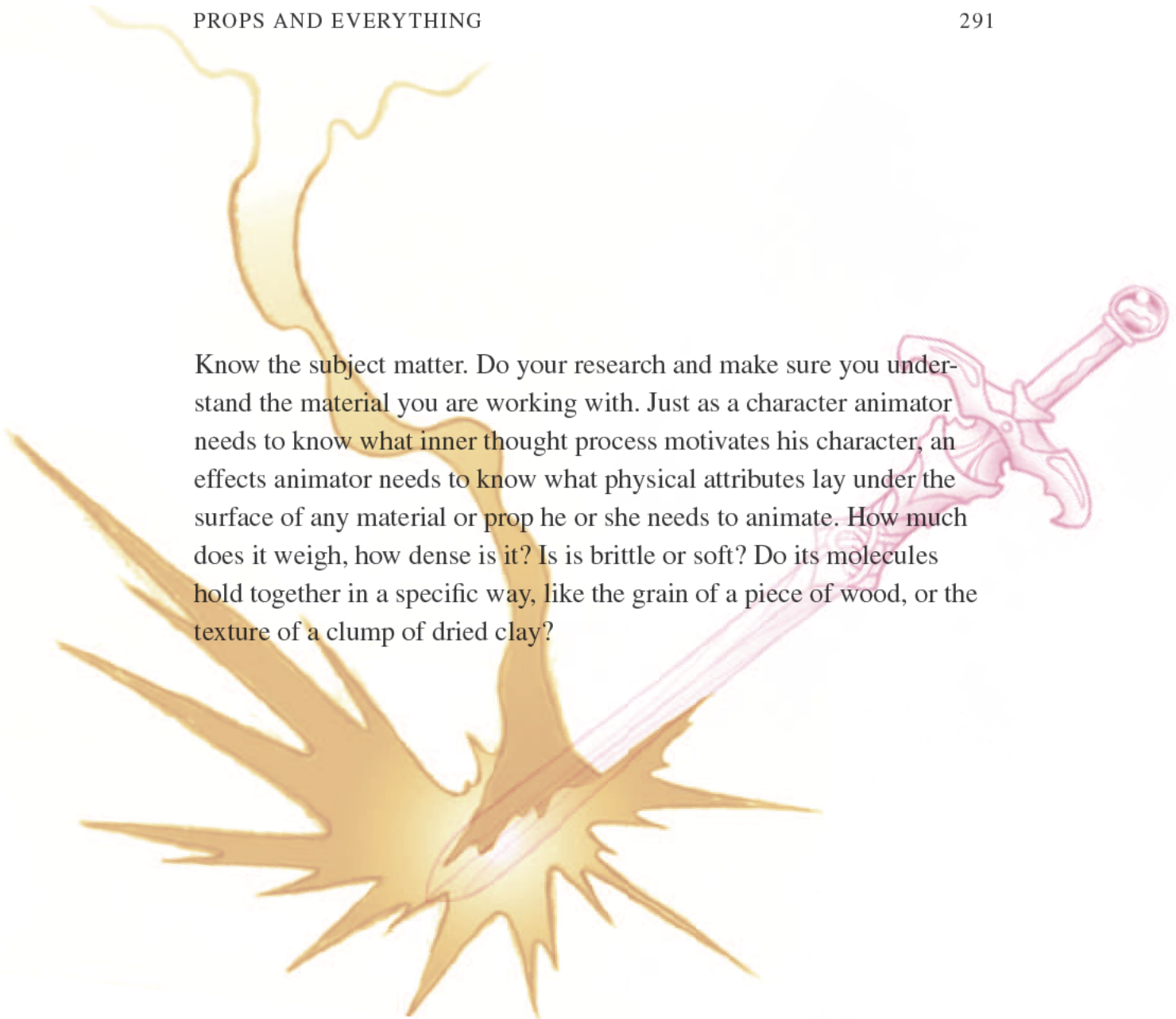
As far as special effects animation goes, things like doors, cars, pots and pans, trains, planes, bird cages, etc. are by far the most uninteresting and tedious to draw. This is technical drawing at best, or perhaps still life, if we want to put a positive spin on it. Animating rigid objects like this, is incredibly difficult to do, but not at all the same type of difficulty we encounter when animating organic effects like smoke and fire. Rather than "feeling" the energy inherent in an organic effect event, with man-made props we are limited to analyzing geometry, perspective, and minute measurements when we attempt to draw them moving. Only when these objects are broken, smashed, lit on fire or otherwise caused to interact with an outside energy force, do they become interesting to animate.

In the first ten or fifteen years of my career in animation, I was frequently assigned difficult-to-draw, man-made prop scenes, because I was a very accurate and experienced draftsman. My technical drawing abilities were pretty sharp, and knowing this, my supervisors would assign me the most mind-numbingly boring prop scenes to animate. Countless hours drawing door knobs, swords, vehicles and weapons of all description, and yes, bird cages too!

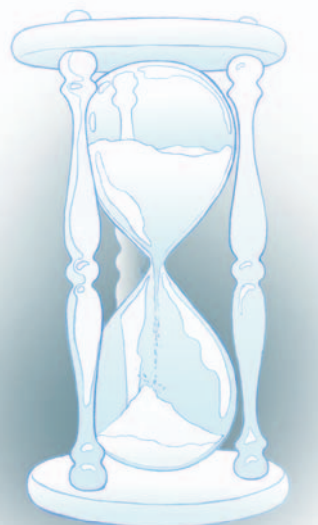
But computers finally came along, and soon I was saved from ever again needing to draw technical, rigid bodied props. Modeling and animating rigid geometric objects is a breeze with computer animation, compared to creating organic effects and characters. A door is such a simple piece of geometry, even a detailed old 17th century castle door can be modeled with relative ease. There are no difficult joints to deal with, no musculature, and no moving skin, hair or cloth to deal with either. And so, by the mid-nineties, the vast majority of rigid props that needed to be animated in big budget 2D animated feature films, was done using CGI technology.

Computer-animated cloth simulation, on the other hand, took much longer to become commonplace. Cloth is more difficult to manage with computers, especially when extremely broad movement is required, or a great deal of detailed overlapping folds. But simpler fabric effects elements are becoming easier and easier to manage, with increased computing power available, as well as more and more sophisticated software, that has specifically designed cloth capabilities.

Exploding, breaking, smashing and destroying props is becoming a simpler proposition for computers to handle extremely well. The only thing we need to always keep in mind, as with CGI character animation and with the organic elements discussed earlier in this book, is that if we are going to animate digitally, we need always to work with the classical principles of good quality animation.



Know the subject matter. Do your research and make sure you understand the material you are working with. Just as a character animator needs to know what inner thought process motivates his character, an effects animator needs to know what physical attributes lay under the surface of any material or prop he or she needs to animate. How much does it weigh, how dense is it? Is is brittle or soft? Do its molecules hold together in a specific way, like the grain of a piece of wood, or the texture of a clump of dried clay?



Then, once the material properties are understood,, we know how it needs to react. We can push the object around, or break it, throw it, crush it, whatever needs to be done. Dynamic CGI effects are complex, and very technically demanding, but if we have done our homework, there is an enormous advantage from the get-go.

We are then more free to focus on the important principles of all good quality effects animation—Energy. Dynamic, flowing, natural, pure energy. Always keep that at the center of your CGI animation. Add to it your understanding of weight, exaggeration, anticipation, overlapping action and dynamic timing, and your effects animation will sparkle.

Chapter 8

The Last Element

Writing and illustrating a book about special effects animation turned out to be a far greater undertaking than I had ever anticipated. Now, in retrospect, I can only see how much ground I did not manage to cover, although I was, to the best of my knowledge, as thorough as I could possibly be. What has now come to light for me is that this topic, this fantastic and utterly unique art form, is truly one of the most vast and varied disciplines in the animation world as we know it.

One has only to go back and read the first couple of paragraphs of Chapter 3 of this book to see what I mean. In that chapter, I ran through an extensive list of special effects elements that I have had to animate through the years. And now that I am finished with this book, how many of those effects did I manage to cover in detail? About half of them.

My experience with animating has made it abundantly clear to me that the art of animating the elements carries with it a learning curve that never ends. Every single time I animate a special effect, no matter how many times I have animated it before, I learn something new about the element I am attempting to create.

I have animated bubbles enough times to tell you that I could easily write an entire book just about animating bubbles. I have animated smoke enough times to tell that a complete book could easily be dedicated to the topic. For all of the elements, without exception, volumes could and should be written.

I wish I had another three or four hundred pages to fill, so that I could finally say to myself, "Now *that* is a complete book on special effects animation!" But as the nature of time and practical conventions would have it, I am out of time, and out of paper to put it all on. I am driven, by some inexplicable reason, to keep at it. Special effects animation has not even begun to tire me out. I return each day to the drawing board, or the computer, as excited about creating elemental magic as I was as a young man in art school over 33 years ago. And I will continue to compile effects drawings, photos, scribbles, notes and the like, as long as I am walking and breathing.

Using the incredible new tools we now have before us, we are only limited by our imaginations. Animation is now an art form where *literally* anything is possible. If we can dream it up in our mind's eye, we can create it. We can even go farther than that, with just a little *more* imagination, a little *more* energy, and just a little more love!

Until next time, keep drawing, keep looking, keep imagining, keep animating, and spill your elemental magic out into the world! Keep asking the questions, keep pushing the envelope, and keep raising the bar on what we can do with our imaginations.

Joseph Gilland, September 2, 2008

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